

Worker Wealth and the Option of Self-Employment

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Abstract

Self-employment is widespread in Brazil, where informal businesses rarely borrow. We use Brazilian panel data to estimate a labor-market model in which workers save out of wages to finance business entry, and wages respond to their self-employment options. The estimates imply that two-thirds of labor-market entrants cannot finance entry, while wage adjustment retains employees who can. Targeted grants expand self-employment and raise welfare. Better savings opportunities raise welfare while reducing self-employment, as financial assets become more attractive than business ownership. Relaxing firms' collateral constraints has little effect. Financial development need not promote entrepreneurship to improve workers' welfare.

JEL codes: E21, J46, J64, L26, O17.

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I. Introduction

Nearly a third of Brazil’s private labor force works on own account (IBGE, 2012–2019a). These businesses rarely use credit: among urban own-account businesses and employers with at most five employees, 94 percent use no credit at all, half of their owners started the business with own resources, a third needed no starting capital, and 1.5 percent used a bank loan (IBGE, 2005). When workers must finance a business from their own resources, wage employment can provide a means of accumulating the wealth needed to enter self-employment. The prospect of self-employment can, in turn, affect the compensation workers receive while employed. This paper studies how the interaction among worker saving, firm asset accumulation, and wage determination shapes the effects of financial development. We develop and estimate a model of the Brazilian labor market to examine how workers’ and firms’ financing opportunities affect entry into self-employment and worker welfare.

We begin with empirical evidence from Brazil, where the financing of own-account businesses relies largely on owners’ resources. Formal employment also provides an institutional channel for accumulating these resources: the severance fund FGTS requires employers to deposit eight percent of wages into workers’ accounts, with balances available upon dismissal without just cause. Using the PNADC panel for 2012–2019 (IBGE, 2012–2019a), we document that own-account work becomes more prevalent with age and that older formal employees are more likely to enter it. Earlier studies document age variation in self-employment and labor-market transitions in Brazil and other Latin American countries (Perry et al., 2007; Bosch and Maloney, 2010). Within manufacturing, the increase in entry with age is more pronounced in industries with lower investment per worker. We complement these profiles with ECINF evidence linking owners’ previous employment to startup finance: compensation payments are a more frequent principal source of capital following formal employment, especially among owners who report having been dismissed. The evidence motivates studying worker accumulation alongside the resources of employers; the employer differences provide additional heterogeneity against which to assess the scope of the quantitative model.

In the model, workers save out of wages and must meet a wealth requirement to enter self-employment. Firms retain earnings, which provide collateral for financing capital and wage payments. Wages are determined through proportional bargaining, subject to firms’ financing constraints (Kalai, 1977). Workers who meet the wealth requirement can enter self-employment from unemployment or following a job separation. This option affects their bargaining position while employed, so wages both finance access to self-employment and respond to the opportunity that saving creates. Because firms must finance wages in

advance, higher compensation can reduce the capital they can finance and, when their assets are insufficient, make an employment relationship unviable.

We estimate the model using labor-market stocks, earnings, and quarterly transitions in the PNADC panel. At the estimated parameters, worker wealth limits access to self-employment, while employed workers who meet the startup requirement prefer to remain in their current jobs at the bargained wage. Sixty-six percent of labor-market entrants begin with wealth below the startup requirement. Among entrants initially below the requirement who subsequently meet it, the average time to eligibility is nearly twelve years, with the necessary resources accumulated during formal employment. At the same time, 70 percent of employed workers meet the requirement and prefer self-employment to unemployment. Self-employment therefore provides a valuable alternative following job loss even when workers prefer to remain in their current jobs.

Our counterfactuals show that workers' financial opportunities shape both access to self-employment and its attractiveness relative to other uses of their wealth. Removing the startup wealth requirement raises the self-employment share by about eleven percentage points but leaves welfare at labor-market entry nearly unchanged. Providing entrants below the requirement with enough resources to meet it raises entrant welfare by 19.8 percent in consumption-equivalent terms, before accounting for financing costs. Yet better savings opportunities can raise welfare while reducing self-employment, even as they help workers accumulate the resources needed for entry. A 25-basis-point increase in returns on wealth held during employment and unemployment, with self-employment returns unchanged, produces this result: the improved alternative to investing in a business outweighs the easier access to entry. Self-employment thus partly reflects workers' limited financial alternatives in the estimated economy. Expanding firms' borrowing capacity against collateral, by comparison, has little effect on self-employment or entrant welfare over the range examined.

The paper relates first to research on entrepreneurship and financial frictions. Studies of firm dynamics examine how internal funds and access to external finance shape capital accumulation and allocation ([Buera and Shin, 2013](#); [Moll, 2014](#); [Midrigan and Xu, 2014](#)). Related work studies how household wealth and borrowing constraints affect business entry ([Evans and Jovanovic, 1989](#); [Banerjee and Newman, 1993](#); [Quadrini, 2000](#); [Cagetti and De Nardi, 2006](#); [Buera, 2009](#)). We connect worker saving and firm asset accumulation through wage bargaining in a search labor market. Wages finance workers' access to self-employment and respond to the outside option that this access creates, while firms' accumulated assets finance capital and wage payments. This interaction links the financing of business entry to the compensation and viability of continuing employment relationships.

[Itskhoki and Moll \(2019\)](#) show that lower wages early in development can accelerate entrepreneurial accumulation and thereby raise future productivity and wages. Our model includes an additional role for wages: they provide the resources workers accumulate before entering self-employment. Wage payments therefore contribute to worker accumulation while reducing the earnings firms can retain. The interaction between these financing needs links workers' access to self-employment to firms' ability to sustain employment. Our analysis focuses on entry into own-account businesses and does not model their subsequent growth into employers.

[Michelacci and Quadrini \(2009\)](#) and [Guiso, Pistaferri, and Schivardi \(2013\)](#) study how firms obtain financing from workers by paying lower wages initially and higher wages later. Our analysis examines how workers' own financing opportunities affect wage bargaining through their access to self-employment.

The second literature studies search with precautionary wealth ([Krusell, Mukoyama, and Şahin, 2010](#); [Bils, Chang, and Kim, 2011](#); [Eeckhout and Sepahsalari, 2024](#)). Relative to this literature, the worker's asset serves not only as insurance but also as the entry requirement for self-employment, so the saving decision is also an occupational choice.

The paper also relates to research on why workers and firms operate outside the formal sector and how these decisions respond to labor-market conditions and policies ([La Porta and Shleifer, 2014](#); [Meghir, Narita, and Robin, 2015](#); [Ulyssea, 2018](#); [Gerard and Gonzaga, 2021](#)). [Narita \(2020\)](#) studies self-employment over the working life in a search model for Brazil, with transition shocks that vary exogenously with age. Our focus is on workers' accumulation of startup resources and firms' accumulation of assets, connected through bargained wages. We emphasize that own-account self-employment is also an investment decision: workers commit their own wealth to a business as well as supply their labor. Its attractiveness therefore depends on the financial alternatives available to workers alongside their employment opportunities. This perspective helps interpret our savings-return result. Self-employment can be preferable under existing financial conditions, while better opportunities to save outside a business can raise worker welfare and reduce self-employment.

Section [II](#) presents the motivating evidence, Sections [III–III.D](#) the model and its mechanics, Section [IV](#) the estimation, Section [V](#) the estimates, their fit and identification, and the estimated economy's four organizing properties, and Section [VI](#) the counterfactuals. Section [VII](#) concludes. The online appendix contains model details, the bargaining protocols, proofs, the entry closure, the empirical and estimation apparatus, and the counterfactual conventions.

II. Motivating Evidence

This section examines how self-employment in Brazil varies with indicators of the resources available to workers and their employers, and how workers finance business entry. Participation in self-employment and entry from formal employment rise with age, while entry profiles also differ with employers' industry investment and establishment size. Business owners' employment histories provide a more direct connection to financing: compensation payments are a more common source of startup capital following formal employment, especially among workers who report being dismissed. Section II.A describes the data and samples, Section II.B presents the age profiles across employment environments, and Section II.C examines the connection between previous employment and startup finance.

A. Data and sample

Data Sources. The Pesquisa Nacional por Amostra de Domicílios Contínua (PNADC) is a nationally representative quarterly household survey conducted by the Instituto Brasileiro de Geografia e Estatística (IBGE) (IBGE, 2012–2019a). Its rotating design interviews sampled households for five consecutive quarters. Industry investment comes from IBGE's Pesquisa Industrial Anual–Empresa (PIA–Empresa), which reports fixed-asset acquisitions and employment by manufacturing division (IBGE, 2012–2019b). For information on startup finance and owners' previous jobs, we use IBGE's 2003 Economia Informal Urbana (ECINF), which covers nonagricultural businesses operated by urban residents working on own account or employing at most five workers (IBGE, 2005). The 1997 ECINF wave provides an earlier comparison of the financing patterns (IBGE, 1999).

Sample. We use PNADC interviews from 2012 through 2019 and link individuals across consecutive quarters. The sample covers people aged 18–65 in formal private employment, own-account work, or unemployment. Formal private employees have a signed work card; own-account workers operate a business without employees; unemployment follows the ILO definition.¹ The age profiles begin at age 20. Both employer comparisons restrict origin jobs to manufacturing, matching the industry coverage of PIA–Empresa, and retain the same destination states.

The 2003 ECINF comparison retains own-account businesses whose owners became proprietors less than five years before the survey and previously held a private wage job. The five-year restriction follows the questionnaire's coverage of previous employment. We

¹Own-account work is the closest PNADC counterpart to the nonemployer self-employment state modeled below. Appendices E.10 and E.5 examine outcome definitions and next-quarter observation in the broader sample used for the appendix employer comparisons. Appendix E.11 compares age profiles by origin-job formality.

distinguish previous formal and informal jobs by signed-card status and compare reported dismissal with other reasons for leaving.

Variables. The own-account share is measured among formal private employees, own-account workers, and unemployed people. The quarterly entry rate is the share of formal private employees who report own-account work in the next interview, restricting the denominator to those next observed in one of these three categories. Other employment categories and people outside the labor force are excluded from these denominators. Shares use the interview’s sampling weight, and transitions use the origin interview’s weight.

Age provides a worker-side indicator of the time available to accumulate resources. On the firm side, industry investment measures resources devoted to acquiring productive assets. We rank the 24 two-digit manufacturing divisions covered by PIA-Empresa by average log fixed-asset acquisitions per worker over 2012–2019 and divide them at the median. Each employee is assigned the investment group of the origin job’s industry. This measure describes an industry investment flow, rather than the assets of the individual employer. Establishment size provides a second employer characteristic, distinguishing workplaces with 1–10 workers from those with at least 11. Appendix E.9 gives the investment construction and division ranking.

ECINF records the principal source of startup capital, including compensation received, previous saving or asset sales, and loans, as well as whether capital was needed. The financing comparisons use survey weights. Appendices E.1 and E.2 provide the construction and inference details.

B. Self-employment and entry over the working life

Figure 1 presents two age profiles: the share in own-account work and the quarterly rate of entry from formal employment. Together they describe participation at each age and new entry among those still in formal employment. The profiles compare workers observed at different ages and combine accumulation, cohort differences, and selection.

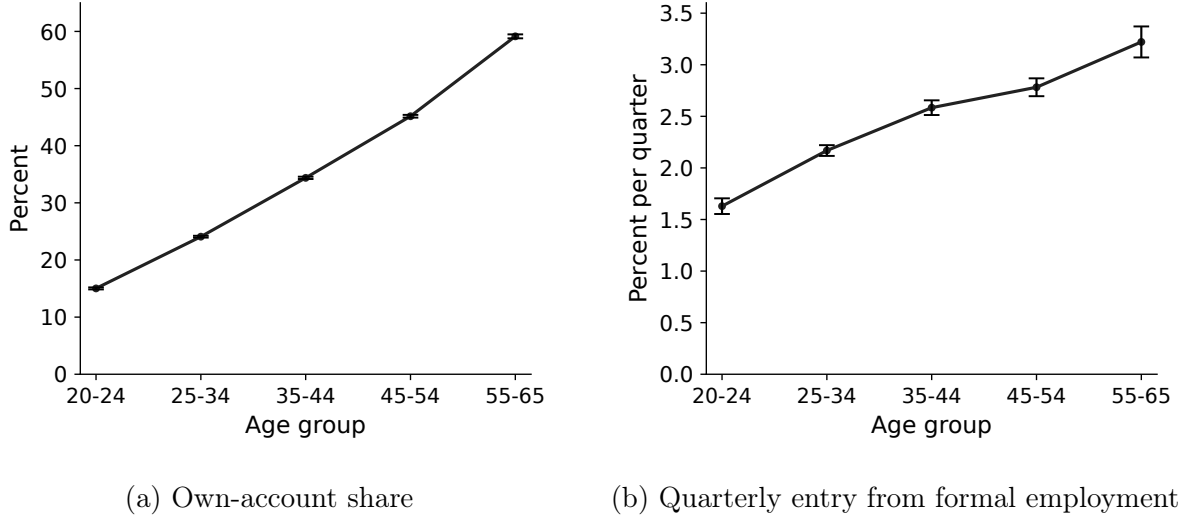


Figure 1: Own-account participation and entry by age

Notes: Panel (a) reports the own-account share among formal private employees, own-account workers, and ILO-unemployed persons in each age group. Panel (b) reports quarterly transitions from formal private employment to own-account work, conditional on the destination remaining in these three states. Transitions are grouped by origin age; both interviews lie in 2012–2019 and satisfy the age restriction of 18–65. All estimates use PNADC sampling weights. Vertical bars are pointwise 95 percent confidence intervals based on 200 bootstrap resamples of individuals. Appendix E.1 provides details. Source: PNADC, 2012–2019.

Own-account work is more prevalent at older ages: its share rises from 15.0 percent at ages 20–24 to 59.1 percent at ages 55–65. Similar participation profiles for Brazil are documented by [Perry et al. \(2007\)](#) and [Narita \(2020\)](#).

Higher own-account participation at older ages is accompanied by more frequent entry from formal employment. The quarterly entry rate rises from 1.63 percent at ages 20–24 to 3.22 percent at ages 55–65 (panel (b)), consistent with the age pattern of entry propensities reported by [Bosch and Maloney \(2010\)](#).²

Own-account work also becomes a more frequent destination relative to unemployment. Among formal employees next observed either unemployed or working on their own account, the own-account share increases from 25.6 percent at ages 20–24 to 67.2 percent at ages 55–65. Alongside the increase in entry, the quarterly transition rate into unemployment falls from 4.72 to 1.57 percent between these age groups.³

We next ask whether the age profile of entry differs across employers’ investment environments and scale. Figure 2 restricts the comparison to manufacturing. Panel (a)

²The entry propensities in [Bosch and Maloney \(2010\)](#) condition on separation; our quarterly probabilities condition on the destination remaining in the three states defined in Section II.A. Participation and entry need not have the same age profile: [Evans and Leighton \(1989\)](#) report rising U.S. self-employment shares alongside broadly age-invariant entry over much of working life.

³These comparisons describe states in consecutive interviews; they do not identify involuntary dismissals or immediate destinations at separation. Moves into other jobs and nonparticipation are outside the two-destination comparison.

distinguishes industries below and above the investment median; panel (b) compares establishments with 1–10 workers and those with at least 11. The industry classification refers to the worker’s origin job.⁴ Both panels use the transition definition and origin ages of Figure 1.

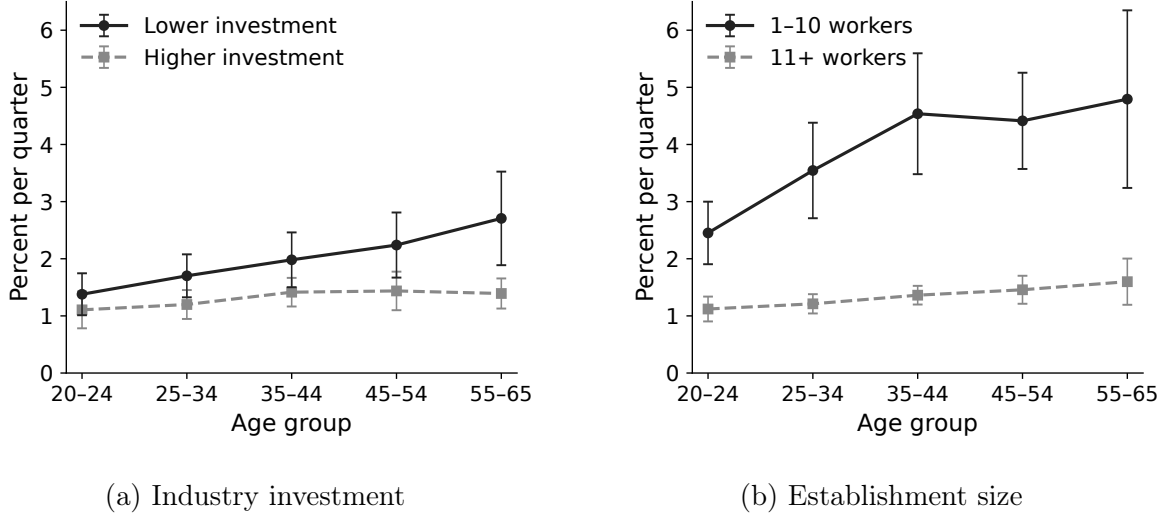


Figure 2: Entry into own-account work by age and employer characteristics

Notes: The figure reports sampling-weighted quarterly transitions from formal private employment in manufacturing to own-account work, conditional on the destination being formal private employment, own-account work, or unemployment. Age, industry, and size are measured in the origin interview. Industries are divided at the median of average log fixed-asset acquisitions per worker over 2012–2019. Panel (a) contains 312,448 transitions. Panel (b) excludes 29,536 observations with unknown establishment size, leaving 282,912. Both interviews lie in 2012–2019; origin ages are 20–65 and destination ages are 18–65. Vertical bars are pointwise 95 percent confidence intervals based on standard errors clustered by primary sampling unit and two-digit industry. Appendix E.1 provides inference details, and Appendix E.9 gives the industry ranking. Source: PNADC and PIA-Empresa, 2012–2019.

The increase in entry with age is larger in lower-investment industries. Between ages 20–24 and 55–65, quarterly entry rises from 1.38 to 2.71 percent in these industries, compared with 1.11 to 1.39 percent in higher-investment industries (panel (a)).

Entry is higher from smaller establishments at every age, and the gap is larger among older workers. Between the youngest and oldest groups, entry rises from 2.45 to 4.79 percent in smaller establishments, compared with 1.12 to 1.60 percent in larger establishments (panel (b)).⁵

⁴Studies of startup requirements instead classify the business entered. [Hurst and Lusardi \(2004\)](#) find no evidence of a stronger wealth–entry relationship in high-startup-capital industries, while [Zissimopoulos, Karoly, and Gu \(2010\)](#) report that entry into such industries continues to rise at high wealth among older workers.

⁵Related work examines entry in relation to prior-employer size ([Parker, 2009](#); [Elfenbein, Hamilton, and Zenger, 2010](#)) and employer productivity and R&D intensity ([Hyytinen and Maliranta, 2008](#)). These employer measures differ from our industry-level fixed-asset investment measure. Appendices E.1, E.9, and E.6 report common-sample checks and broader comparisons, including adjustments for worker and job characteristics.

These profiles show how entry varies with age and the employment environment.⁶ We next examine a financial connection between prior employment and business entry: the resources that owners report using to start their businesses.

C. Previous employment and startup finance

The end of a formal job can change both a worker’s employment options and the resources available to pursue them. Brazil’s Fundo de Garantia do Tempo de Serviço (FGTS) provides employees in formal jobs with individual restricted savings accounts, into which employers must deposit eight percent of covered remuneration. Dismissal without just cause permits withdrawal of the balance associated with the job and requires an additional employer payment equal to 40 percent of the accumulated deposits, including their statutory adjustment and interest (Brazil, 1990). This institution connects past earnings to resources available following job loss.⁷

Using ECINF, we relate the principal source of startup capital to the formality of the owner’s previous job, the reason for leaving it, and its duration.⁸ The survey records compensation payments but does not identify FGTS receipts separately. Table 1 compares four groups defined by the formality of the previous private wage job and whether the owner reports having been dismissed. Other exits include contract expiry, retirement, personal or family reasons, and other reported reasons. Each column reports the distribution of the principal financing source, excluding owners who report that no startup capital was needed. Borrowing combines finance from banks, relatives or friends, and other lenders.

⁶The entry-rate gaps between employer groups are also larger at ten or more years of tenure than at three to less than ten years (Appendix Figure 5). In the broader sample including informal private employees, own-account entry rises at long tenure while transitions to a new wage job and unemployment decline (Appendix Figures 6 and 7).

⁷Related evidence shows that severance liquidity affects job search in Austria (Card, Chetty, and Weber, 2007) and that unemployment-benefit protection supports business creation in France (Hombert, Schoar, Sraer, and Thesmar, 2020). Appendix H describes aggregate withdrawals and other withdrawal channels.

⁸Earlier ECINF studies document the use of personal resources, including compensation payments, to finance small businesses (IETS, 2006; Fontes and Pero, 2011). Pinto, Chein, and Pinto (2013) use own-resource startup financing as part of a wealth measure in a study of credit constraints and investment. Da Mata, Mattos, and Vilarouca (2025) study job loss and registered-business creation in Brazil. Our comparison concerns financing among own-account businesses operating at the survey date.

Table 1: Startup finance and previous employment

Financing source	Previous formal job		Previous informal job	
	Dismissed	Other exit	Dismissed	Other exit
Compensation received	37.7 (2.3)	24.5 (2.0)	7.5 (1.9)	7.7 (1.3)
Prior saving or asset sales	6.2 (1.2)	9.0 (1.1)	8.7 (2.7)	12.3 (1.6)
Other own resources	38.5 (2.3)	40.7 (2.1)	49.9 (4.5)	46.3 (2.5)
Borrowing	12.7 (1.4)	17.6 (1.8)	26.1 (4.4)	25.0 (2.2)
<i>of which:</i> Bank loans	1.9 (0.5)	3.5 (0.7)	2.0 (0.9)	2.9 (0.7)
<i>of which:</i> Relatives or friends	8.6 (1.2)	11.5 (1.5)	20.3 (4.4)	19.3 (2.1)
<i>of which:</i> Other lenders	2.2 (0.6)	2.7 (0.7)	3.8 (1.5)	2.9 (0.9)
Other sources or unreported	5.0 (0.7)	8.2 (1.4)	7.8 (1.8)	8.7 (1.7)
Observations	1,307	1,744	374	1,176

Notes: Entries are percentages, with standard errors in parentheses. The sample consists of own-account owner–business observations in ECINF 2003 whose owners became proprietors less than five years before the survey and previously held a private wage job. Owners reporting no startup capital needed are excluded. Entries describe principal sources, not shares of capital invested. Borrowing combines banks, relatives or friends, and other lenders. Indented rows provide the full breakdown of borrowing, using the same denominator as the total. Components may not sum to totals because of rounding. Formal jobs have a signed work card. Dismissal is self-reported and does not distinguish dismissal with and without just cause. “Compensation received” is the survey category *indenização recebida*; the survey does not identify FGTS receipts separately or link the payment to the reported previous job. Estimates use survey weights. Standard errors account for survey stratification and clustering within census sectors. Appendix E.2 reports the unconditional distribution, construction details, and alternative samples and weights. Source: Author’s calculations from the ECINF 2003 microdata (IBGE, 2005).

Among owners previously in formal jobs, 37.7 percent name compensation as their principal startup source following dismissal, compared with 24.5 percent following other exits. The corresponding shares after informal employment are 7.5 and 7.7 percent. The difference between these dismissal contrasts is 13.4 percentage points, with a 95 percent confidence interval of [6.7, 20.1]. The greater reliance on compensation following dismissal from

formal employment is consistent with a role for FGTS withdrawals in financing business entry.

Borrowing is more often named following informal employment, and bank loans are the principal startup source for only 1.9–3.5 percent of owners across the four groups. The 1997 survey shows a similar dismissal contrast: compensation shares are 41.4 percent after dismissal and 28.6 percent after other exits from formal jobs; the corresponding informal-job shares are 10.6 and 8.7 percent (Appendix E.2).

Compensation financing also varies with the duration of previous employment. Within the same 2003 sample, we compare owners whose previous jobs lasted at least five years with those whose jobs lasted less than one year, adjusting for age and whether the owner reports dismissal. The difference in the share naming compensation as the principal startup source is 23.5 percentage points following formal employment and 2.2 points following informal employment. The difference between these gradients is 21.3 points, with a 95 percent confidence interval of [11.9, 30.6]. Longer formal employment is thus associated with greater reliance on compensation to finance business entry.⁹

The entry profiles document differences across employment environments, while the financing comparisons connect prior employment to the resources used to start a business. The model below examines how worker wealth shapes access to self-employment, how this opportunity affects wages while employed, and how wage setting feeds back into worker saving and firm asset accumulation.¹⁰

III. The Model

The model connects two uses of internal funds: workers save toward business entry, while firms retain earnings to finance capital and wage payments. Bargaining links these accumulation decisions because compensation responds to the opportunities that worker wealth makes available.

Section III.A introduces the environment and financing arrangements. Section III.B describes saving and wage determination; Section III.C characterizes participation and equilibrium; and Section III.D develops the implications for compensation and match viability. Appendix A states the complete dynamic problems, dividend policy, and

⁹Appendix E.2 reports the profiles by departure reason and alternative samples. With the same exclusion of owners who needed no startup capital, the corresponding formal–informal tenure contrast is smaller and less precisely estimated in 1997.

¹⁰Empirical evidence on the wage effects of workers’ outside options is mixed across settings. [Castellaneta, Conti, Kacperczyk, and Mamadehussene \(2025\)](#) find wage gains among high earners in incumbent firms following a reduction in business-entry costs in Portugal, with patterns consistent with an entrepreneurial outside option. [Jäger, Schoefer, Young, and Zweimüller \(2020\)](#), however, find little wage response to increases in unemployment benefits in Austria.

boundary conditions; Appendix B derives the wage equation and compares bargaining protocols; and Appendix C proves the formal results.

A. Environment

Time is continuous. Workers have constant-relative-risk-aversion (CRRA) flow utility $u(c)$ over consumption c , with coefficient $\gamma > 0$. The discount rate and the return on financial assets are $r > 0$. A worker is unemployed, self-employed, or matched with a firm. Worker wealth is $b \geq 0$, and the internal assets of a matched firm are $a \geq 0$. Workers exit the economy at rate ν and are replaced by unemployed entrants whose initial wealth is drawn from distribution ψ . This turnover permits a stationary cross-sectional distribution.

Starting a self-employed business requires wealth of at least $b_{SE} \geq 0$. Define the entry-eligibility indicator

$$I_{SE}(b) = \mathbf{1}\{b \geq b_{SE}\}. \quad (1)$$

The threshold governs entry; incumbent businesses continue operating below it. With $b_{SE} = 0$, all workers are eligible.

Self-employment opportunities arrive to eligible unemployed workers at rate ρ , and are taken when the worker prefers a business to continued unemployment. Job offers arrive to unemployed workers at rate p , with firm assets a' drawn from distribution F_A , and are accepted when both parties gain from the match. A self-employed business fails at rate ξ_{SE} , returning its owner to unemployment. Self-employed workers receive no job offers and return to wage employment through unemployment following business failure. Matches end exogenously at rate s , or endogenously when either party's gain from continuation falls to zero. At an exogenous separation, $\lambda \in [0, 1]$ is the probability that an eligible worker is routed into self-employment when the worker prefers it to unemployment. All other separations lead to unemployment. Business opportunities do not arrive during employment in the baseline; Appendix F.5 considers an extension that allows them to do so. Figure 3 summarizes these transitions.

Production and financing. A firm produces zk^α with capital k , productivity $z > 0$, and capital elasticity $\alpha \in (0, 1)$. Capital depreciates at rate $\delta > 0$. Firm assets carry retained earnings across time and support rented capital at each instant. This separation between accumulated financial wealth and current capital demand follows the rental-market formulation in Moll (2014) and Itskhoki and Moll (2019). Capital adjusts with current financing capacity, while a records the firm's accumulation history.

Firms operate at full collateralized capacity. Let $\phi \geq 1$ be the collateral multiplier and $\chi \geq 0$ the duration, in years, for which the annual wage flow w must be financed in

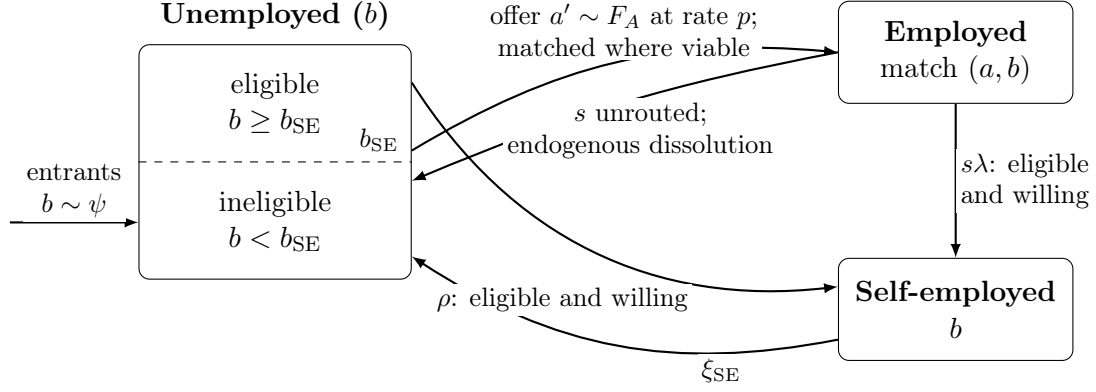


Figure 3: States and transitions

Notes: The dashed line marks entry eligibility; incumbent businesses continue operating below the threshold. Willing workers prefer self-employment to unemployment. Workers exit every state at rate ν and are replaced by unemployed entrants with wealth drawn from ψ . Saving and dissaving change wealth within each state.

advance. Capital and the prefunded wage stock together equal ϕa , so

$$k(a, w) = \phi a - \chi w. \quad (2)$$

Equation (2) is a maintained restriction on the financing technology: all collateral capacity is committed to production and wage finance. With advance wage finance, a higher wage absorbs financing that supports capital at given assets; retained earnings expand the resources available for both uses.

The firm borrows funds in excess of its internal assets at rate r . Its flow resources before dividends are

$$\pi(a, w) = zk^\alpha - w - \delta k - r(k + \chi w - a). \quad (3)$$

Substituting (2) gives

$$\pi(a, w) = zk(a, w)^\alpha - (1 - \delta\chi)w - [(r + \delta)\phi - r]a. \quad (4)$$

The user cost $r + \delta$ applies to capital, while internal funds reduce borrowing costs at rate r .¹¹ These resources can be retained to expand the firm's asset base or distributed to its owner.

Self-employment and unemployment. An unemployed worker receives flow income z_u . A self-employed worker uses their own wealth to operate capital $k_{SE}(b) = \phi_{SE}b$, where $\phi_{SE} \geq 0$ is the capital intensity of wealth. Let $z_{SE} > 0$, $\alpha_{SE} \in (0, 1)$, and $\delta_{SE} > 0$ denote

¹¹The prefunded wage stock incurs interest at rate r ; depreciation applies to physical capital. Raising the wage by one unit reduces capital by χ , saving $(r + \delta)\chi$ in capital costs while adding $r\chi$ in wage-financing costs. Together with the wage payment, this gives the coefficient $1 - \delta\chi$.

self-employment productivity, capital elasticity, and depreciation. Self-employment flow resources, net of the user cost of capital, are

$$\Pi^{\text{SE}}(b) = z_{\text{SE}} k_{\text{SE}}(b)^{\alpha_{\text{SE}}} - (r + \delta_{\text{SE}}) k_{\text{SE}}(b). \quad (5)$$

A self-employed worker's wealth evolves according to $\dot{b} = rb + \Pi^{\text{SE}}(b) - c$. Worker wealth therefore determines both access to a business through the startup requirement and the scale of production once the business operates.

The contact rate p and the offer distribution F_A are held fixed in the baseline. Appendix D makes the contact rate endogenous through free entry, calibrated to reproduce the estimated equilibrium; the counterfactuals compare the fixed-contact and free-entry responses.

B. Saving and wage determination

Let $U(b)$ and $V^{\text{SE}}(b)$ denote the values of unemployment and self-employment. Let $W(a, b)$ denote the worker's value in a match and $J(a, b)$ the value of the matched firm, inclusive of its internal assets. Bargaining breakdown gives the worker $U(b)$ and the firm its liquidation value a . Writing $[x]^+ = \max\{x, 0\}$, the expected worker gain from an acceptable offer is

$$\Lambda(b) = \int_{\mathbb{R}_+} [W(a', b) - U(b)]^+ \mathbf{1}\{J(a', b) > a'\} dF_A(a'). \quad (6)$$

Unemployment provides access to both business opportunities and employment offers. Their combined contribution to its value is the option flow

$$\mathcal{O}(b) = \rho I_{\text{SE}}(b) [V^{\text{SE}}(b) - U(b)]^+ + p\Lambda(b). \quad (7)$$

The baseline uses a proportional bargaining protocol (Kalai, 1977). Define worker and firm surplus by $S_W = W - U$ and $S_J = J - a$. Whenever the proportional allocation is feasible, it satisfies

$$S_W(a, b) = \kappa S_J(a, b), \quad \kappa = \frac{\eta}{1 - \eta}, \quad (8)$$

where $\eta \in (0, 1)$ is the worker's bargaining weight. The rule assigns the worker a fixed proportion of the firm's gain from continuation. Appendix B derives the resulting wage equation and compares the allocation with Nash protocols used in models with worker savings (Krusell, Mukoyama, and Şahin, 2010; Bils, Chang, and Kim, 2011). Worker surplus is measured in utility-value units and firm surplus in monetary units, so the interpretation of η uses the CRRA normalization specified in Appendix A.

Within an active match, the firm chooses dividends d , while the worker chooses consumption c . Their asset positions evolve according to

$$\dot{a} = \pi(a, w) - d, \quad \dot{b} = rb + w - c. \quad (9)$$

Wages finance worker consumption and saving, while the resources left after wage payments support firm accumulation. Let $c_U(b)$ denote consumption in unemployment, and let subscripts denote value derivatives. At an interior consumption choice, the proportional rule and the continuation problems imply

$$u'(c(a, b)) = U_b(b) + \kappa J_b(a, b). \quad (10)$$

$$\begin{aligned} \kappa[\pi(a, w) - ra] &= u(c) - u(c_U) - \mathcal{O}(b) + U_b(b)[w - z_u - c + c_U] \\ &\quad + s\lambda I_{SE}(b)(V^{SE}(b) - U(b))^+. \end{aligned} \quad (11)$$

The first equation equates the marginal utility of consumption to the worker's marginal continuation value of wealth. The second determines the interior wage and displays two distinct contributions of self-employment. Opportunities available during unemployment enter through $\mathcal{O}(b)$ and strengthen the worker's bargaining position. The final term values access to self-employment at exogenous separation, which is part of the return to holding the job. Section III.D separates their effects on compensation. Appendix B derives these equations from the worker and firm continuation problems.

C. Participation and equilibrium

A match operates only when both parties receive strictly more than their separation values. Define the bilateral participation margin

$$m(a, b) = \min\{W(a, b) - U(b), J(a, b) - a\}. \quad (12)$$

The active set is

$$\mathcal{A}(b) = \{a \geq 0 : m(a, b) > 0\}. \quad (13)$$

Wages must also be feasible under the financing technology. When $\chi > 0$, the largest wage consistent with nonnegative capital is $\bar{w}(a) = \phi a / \chi$; set $\bar{w}(a) = +\infty$ when $\chi = 0$. The wage lies between zero and this cap. Where a bound binds, participation is determined by each party's continuation value at the feasible wage. Contact with either separation value ends the match. Appendix A states the value problem at the collateral cap, where the proportional allocation gives way to the two separate participation conditions.

The minimum firm assets required for formation are

$$a_c(b) = \inf\{a \geq 0 : m(a, b) > 0\}. \quad (14)$$

Set $a_c(b) = +\infty$ when the active set is empty.

Retained earnings have value because they support future capital and wage payments. The firm retains available profits when the marginal value of assets, $J_a(a, b)$, exceeds one, and distributes them when that value is below one. Among active matches, the first contact with the dividend region is

$$a^*(b) = \inf\{a \in \mathcal{A}(b) : J_a(a, b) \leq 1\}. \quad (15)$$

Thus $a_c(b)$ describes the assets required to form a match, while $a^*(b)$ describes the firm's incentive to retain earnings within one. Both depend on worker wealth through wages and continuation values. Appendix A.4 characterizes retention and participation even when the active set consists of several intervals, and Appendix A.2 derives the dividend policy from the marginal value of assets.

Equilibrium.

Definition 1 (Equilibrium). *An equilibrium consists of value functions (U, V^{SE}, W, J) , a wage policy $w(a, b)$, consumption and dividend policies, participation sets $\{\mathcal{A}(b)\}$, and an offer surplus Λ such that (i) U and V^{SE} solve the unemployment and self-employment problems given Λ and the match values; (ii) on the active set, W and J solve the match problem under the proportional rule (8), with the capped and zero-wage regimes where the interior root is infeasible; (iii) the participation sets are generated by the resulting values through (13); and (iv) Λ is generated by the values through (6). A stationary equilibrium adds a distribution over worker and match states that is invariant under the implied transitions and the entry–exit process.*

The equilibrium links bargaining to future employment opportunities: the value of unemployment affects wages and participation, and the resulting acceptable offers determine $\Lambda(b)$ and hence the value of unemployment. This feedback permits multiple joint equilibria even when the wage is unique conditional on continuation values. Appendix B establishes conditional wage uniqueness from the monotonicity of the wage equation. At parameter vectors with a low startup requirement and high unemployment income, numerical solutions have recovered two joint equilibria with different participation sets and entry flows. Their coexistence makes equilibrium determination consequential for the predicted allocation of workers across employment states.

The quantitative analysis restricts attention to parameter vectors for which independently initialized solutions agree. Candidate estimates with detected multiplicity are excluded. The reported moments agree across the initializations examined. Counterfactuals follow the equilibrium from the estimated economy and are checked against independent solutions along each path. These checks recover the same solution at the estimate and at the reported counterfactual points. They establish robustness to the initializations examined; uniqueness of the full equilibrium remains an open theoretical question. Appendix A.5 distinguishes the fixed-point selection convention from this restriction on admissible estimates, and Appendix G states the agreement and perturbation checks applied to counterfactuals.

The stationary distribution records how saving and labor-market transitions distribute workers and firms across these states. Appendix F derives its flow-balance equations and the transition probabilities used to match the quarterly data.

D. Economic implications

The model links workers' access to self-employment to the compensation and financing of employment relationships. An eligible worker may prefer to remain employed when the wage responds to the business option. Firm assets matter for that response because wage payments and capital share the same financing capacity. The following results distinguish adjustment within continuing matches, contraction of the set of viable employers, and conditions under which that set becomes empty.

Self-employment contributes to compensation through two channels. An opportunity available during unemployment raises the value of the worker's alternative to the job. Access to a business upon exogenous separation, in turn, adds to the value of holding the job and can substitute for current wages. The first proposition isolates these two contributions, holding the other determinants of the wage fixed.

Proposition 1 (Self-employment opportunities and compensation). *Hold fixed (a, b) , the value derivatives, and the consumption policies. Suppose the interior wage solves (11), $U_b(b) > 0$, and $M(a, w) \equiv -\pi_w(a, w) > 0$. Treating the unemployment option $\mathcal{O}$ and the reallocation premium $R_\lambda = s\lambda I_{SE}(b)(V^{SE} - U)^+$ as independent flow objects,*

$$\frac{\partial w}{\partial \mathcal{O}} = \frac{1}{\kappa M(a, w) + U_b(b)} > 0, \quad \frac{\partial w}{\partial R_\lambda} = -\frac{1}{\kappa M(a, w) + U_b(b)} < 0. \quad (16)$$

Let $\Delta_{SE} = (V^{SE} - U)^+$. Holding $p\Lambda$ fixed, the direct effect of Δ_{SE} is

$$\frac{\partial w}{\partial \Delta_{SE}} = \frac{I_{SE}(b)(\rho - s\lambda)}{\kappa M(a, w) + U_b(b)}, \quad \frac{\partial k}{\partial \Delta_{SE}} = -\chi \frac{\partial w}{\partial \Delta_{SE}}. \quad (17)$$

Thus the direct wage effect is positive when $\rho > s\lambda$ and entry is feasible.

The net direct effect depends on where the worker can exercise the business option. When opportunities arrive more frequently during unemployment than through separation from employment, $\rho > s\lambda$, a higher self-employment surplus raises the wage at eligible states. With advance wage finance, the increase also reduces operated capital at given firm assets. When $\rho < s\lambda$, the separation channel dominates and the direct wage effect reverses. These derivatives describe a local compensation response; an equilibrium change in self-employment technology or access also changes saving, continuation values, and future acceptable offers.

The next question is which firms can sustain a mutually acceptable match after the outside option changes. An asset-poor firm has little capacity to finance compensation alongside productive capital. If the change reduces the bilateral participation margin, some previously viable matches may end and a higher asset base may be required for formation. Let q index the strength of the worker's outside option, and write the equilibrium margin and active set as functions of q .

Proposition 2 (Outside options and the assets required for employment). *Suppose $m(a, b; q)$ is weakly decreasing in q at every (a, b) . Then*

$$\mathcal{A}(b; q') \subseteq \mathcal{A}(b; q) \quad \text{and} \quad a_c(b; q') \geq a_c(b; q) \quad \text{for } q' > q.$$

Take the infimum of an empty active set to be $+\infty$. If inactive firms have no source of asset accumulation, the states $a < a_c(b; q)$ form a capital trap for matches with workers at wealth b . A stronger outside option weakly expands that region.

The proposition connects a decline in match surplus to the financing requirement for employment. Firms below the formation boundary need additional assets to operate a viable match, yet operating profits are available only after formation. Retained earnings help firms already in viable matches expand their financing capacity. The assumption that the equilibrium margin falls is central: it incorporates the response of wages, worker saving, and future opportunities. The quantitative analysis determines whether a particular change in the outside option generates that response and how far the formation boundary moves.

Contraction can also reach the extensive margin of employment as a whole: at some worker wealth levels, the greatest attainable gains from employment may fall to zero. To characterize this possibility, consider a worker-wealth interval $[b, \bar{b}]$ and a firm-asset bound $\bar{a}$ containing every active match on that interval. Define the highest bilateral participation margin and its first nonpositive point by

$$H(b) = \max_{a \in [0, \bar{a}]} m(a, b), \quad b_{\text{trap}} = \inf \{b \in [b, \bar{b}] : H(b) \leq 0\}. \quad (18)$$

$H(b)$ is the highest participation margin attainable at worker wealth b . Its sign therefore distinguishes loss of some employers from closure of the entire viable set.

Proposition 3 (Closure of the viable match set). *Suppose $H(b)$ is weakly decreasing on $[\underline{b}, \bar{b}]$, $H(\underline{b}) > 0$, and $H(\bar{b}) \leq 0$. Then the threshold in (18) is finite, $\mathcal{A}(b) \neq \emptyset$ for $b \in [\underline{b}, b_{\text{trap}})$, and $\mathcal{A}(b) = \emptyset$ for $b \in (b_{\text{trap}}, \bar{b}]$. At the threshold itself, a match is viable exactly when $H(b_{\text{trap}}) > 0$.*

Closure requires the best attainable participation margin to fall far enough to reach zero. Worker wealth can improve consumption possibilities outside employment and raise the value of self-employment, while firm accumulation expands the resources available to sustain a match. The conditions of the proposition describe equilibria in which the first force eventually dominates over the stated wealth interval. They allow a discrete change at startup eligibility as well as a gradual decline. Appendix C derives the wage responses and proves the set-contraction and closure results under their respective monotonicity conditions.

The three thresholds have different economic meanings. The startup requirement b_{SE} governs the worker's access to a business. The formation floor $a_c(b)$ governs the firm's ability to sustain employment. Under the closure conditions, b_{trap} marks the loss of every viable match on the relevant wealth interval. An economy can have a substantial startup barrier while wages sustain employment for eligible workers; another can also lose matches as outside opportunities strengthen. Estimation and the stationary distribution establish which margins matter for Brazil.

IV. Estimation Strategy

We internally estimate thirteen parameters using Brazilian labor-market data and externally calibrate the remaining parameters. Section IV.A separates these two sets, Section IV.B explains the targets and their identifying information, and Section IV.C presents the estimator.

A. Estimated and assigned parameters

The thirteen internally estimated parameters are

$$(z_{\text{SE}}, \alpha_{\text{SE}}, z_u, \bar{b}_0, \beta_1, \beta_2, \eta, \lambda, b_{\text{SE}}, s, \rho, p, \xi_{\text{SE}}).$$

These govern self-employment returns, unemployment income, the startup requirement, bargaining, transitions, and the distributions of initial worker wealth and offered firm assets. Initial worker wealth follows a truncated exponential distribution ψ with scale $\bar{b}_0$, and offered firm assets follow a scaled beta distribution with shape parameters (β_1, β_2) .

The externally assigned parameters use evidence from the literature or maintained assumptions. Relative risk aversion is fixed at two, following [Quadrini \(2000\)](#). Self-employed capital equals own wealth ($\phi_{SE} = 1$). Wage prefinancing is set to one quarter ($\chi = 0.25$). Collateral capacity is set to $\phi = 2.6$ times internal assets ([Midrigan and Xu, 2014](#); [Assunção, Benmelech, and Silva, 2014](#)). Appendix A reports the full set of assignments in Table 8, their rationale, and the distributional supports.

B. Targets and identification

We use PNADC interviews from 2012–2019 for people aged 18–65 in formal private employment, own-account work, or unemployment as defined by the ILO. Quarterly transitions are measured among those next observed in one of these three categories, which correspond to the model’s employment states.

Table 2 reports seventeen targets. We match stationary model shares to shares implied by the empirical transition matrix, keeping aggregate stocks consistent with measured flows. Age-specific shares are observed directly. Relative earnings compare cross-sectional medians of self-employment and formal earnings, and tenure is measured in years with the current employer.

We interpret the observed age profiles through a model cohort entering unemployment at age 20 with wealth drawn from ψ . Preferences, technologies, and arrival rates are common across ages and cohorts, so the model’s age profiles reflect accumulation and selection. Appendix F.1 describes the model counterparts.

Table 2: Targeted moments

Moment	Data
<i>A. Aggregate levels and flows</i>	
Implied stationary self-employment share	0.316
Implied stationary unemployment share	0.101
Median self-employment/formal earnings	0.769
Formal to self-employment, aggregate	0.0232
Unemployment to formal	0.163
Unemployment to self-employment	0.138
Self-employment to unemployment	0.042
<i>B. Age and tenure profiles</i>	
Formal to self-employment, tenure 10+	0.0236
Formal to unemployment, tenure 10+	0.0085
Formal to self-employment, age 35–44	0.0258
Formal to self-employment, age 45–54	0.0278
Formal to self-employment, age 55–65	0.0322
Self-employment share, age 20–24	0.1502
Self-employment share, age 25–34	0.2408
Self-employment share, age 35–44	0.3437
Self-employment share, age 45–54	0.4516
Self-employment share, age 55–65	0.5914

Notes: Stocks and transitions refer to formal private employment, own-account self-employment, and ILO unemployment. Aggregate shares are implied by the empirical conditional transition matrix; age-specific shares are directly observed, sampling-weighted stocks. Age-specific targets begin at age 20; aggregate targets use ages 18–65. Transitions are quarterly, conditional on remaining in these three states. Tenure is measured in years with the current employer. The earnings target is the ratio of the two cross-sectional medians. Source: PNADC, 2012–2019.

The age and tenure transition targets emphasize later accumulation and mature matches. Younger entry rates and short-tenure separations remain untargeted diagnostics of the common initial condition and early-match turnover.¹²

Young workers’ self-employment shares constrain initial wealth relative to the startup requirement, while subsequent age profiles constrain changes in access through accumulation and selection. Wealthier entrants can partly offset a higher requirement, limiting the precision of their separate estimates.

Relative earnings provide information about business returns conditional on financial access. Self-employment productivity and capital elasticity determine returns at a given wealth level and incentives to save toward entry. Their estimation also uses stocks and transitions because the earnings ratio depends on wealth composition and formal wage bargaining.

¹²The exogenous separation rate is constant; tenure dependence arises through worker and firm accumulation and selection. Appendix Figure 9 reports the empirical tenure profiles.

Unemployment-to-formal transitions inform the offer-arrival rate p , and exits from self-employment inform the failure rate ξ_{SE} . Entry from unemployment depends on both the opportunity rate ρ and the fraction eligible and willing to start a business. The two separation destinations among long-tenure employees provide information about the separation rate s and routing probability λ , conditional on the resources of surviving matches.

Relative earnings and employment histories inform bargaining and the distribution of offered firm assets through wages, accumulation, and match survival. Identification is conditional on the assigned financing technology. Section V.B evaluates identification at the estimates.

C. Estimation and inference

For computation, let θ collect the eleven parameters other than p and ξ_{SE} . At each trial value of θ , we recover these two rates from the formal job-finding and business-exit targets (Appendix F.2). All thirteen parameters are therefore estimated from the data, with the numerical minimization conducted over eleven. Let t_k denote empirical target k , $m_k(\theta)$ its model counterpart including the conditionally recovered rates, and $\hat{\sigma}_k$ its bootstrap standard error. With scale $s_k = \max\{\hat{\sigma}_k, 0.05|t_k|\}$ and admissible parameter set Θ_{adm} defined in Appendix F.3, the diagonally weighted minimum-distance estimator is

$$\hat{\theta} = \arg \min_{\theta \in \Theta_{adm}} \sum_{k=1}^{17} h\left(\frac{m_k(\theta) - t_k}{s_k}\right), \quad (19)$$

where h is the quadratic-linear loss of Huber (1964), specified in Appendix F.2. The five-percent floor binds for every target, and all fitted deviations lie in the quadratic region. Equal percentage errors therefore receive equal weight locally; profiles represented by more cells contribute more terms.

Standard errors use the minimum-distance sandwich formula and a joint individual-level bootstrap covariance of the empirical moments, preserving their cross-moment correlations. Appendix F.3 gives the covariance formula and inference qualifications.

V. The Estimated Economy

A. Parameter estimates

Table 3 reports the parameter estimates and their standard errors.

Table 3: Point estimates

Parameter		Estimate
Self-employment productivity	z_{SE}	0.316 (0.055)
Self-employment curvature	α_{SE}	0.386 (0.022)
Unemployment income	z_u	0.100 (—)
Mean entrant wealth (scale)	$\bar{b}_0$	0.594 (0.189)
Offer-distribution shapes	β_1, β_2	0.507, 0.637 (0.020, 0.013)
Worker bargaining weight	η	0.510 (0.020)
Reallocation routing	λ	0.860 (0.025)
Opportunity arrival	ρ	0.995 (0.016)
Startup-wealth threshold	b_{SE}	0.686 (0.191)
Exogenous separation	s	0.139 (0.003)
Business failure (inverted)	ξ_{SE}	0.213
Offer arrival (inverted)	p	0.848
Wage prefinancing (calibrated)	χ	0.25
Collateral multiplier (calibrated)	ϕ	2.6

Notes: Unemployment income is estimated at an imposed lower bound: the loss improves slowly as z_u falls through implausible values, and 0.10 is a stated plausibility floor rather than an interior optimum (Appendix F.3). b_{SE} is identified up to one wealth-grid cell and reported at its effective grid node. ϕ is calibrated to the micro collateral evidence (anchors at 2.58–2.66; Appendix A), with sensitivity band [1.35, 3.3]. Standard errors, in parentheses, are minimum-distance delta-method errors: the sandwich formula with the moment Jacobian of Appendix F.3 and the joint bootstrap covariance of the empirical moments (the scale floors make the weighting non-efficient, so the sandwich form is required). No standard error is reported for z_u , estimated at its imposed lower bound. $\bar{b}_0$ and b_{SE} are individually imprecise because they trade along the least-identified direction (correlation 1.00); their gap $b_{SE} - \bar{b}_0 = 0.105$ carries a standard error of 0.005. The inverted rates inherit the uncertainty of their targeted flows and are reported without standard errors.

In the technology block, self-employment productivity and curvature govern the level and concavity of output in self-employment. Unemployment income, at $z_u = 0.100$, replaces about 40 percent of the equilibrium average wage. In the mobility block, formal offers reach the unemployed at rate $p = 0.848$ and self-employment opportunities at rate $\rho = 0.995$, so an unemployed worker waits on average about a year for each option to arrive. Matches dissolve exogenously at rate $s = 0.139$; businesses fail at rate $\xi_{SE} = 0.213$, an expected business life of close to five years. A worker whose match ends moves directly into self-employment with probability $\lambda = 0.860$ where willing and eligible. At the estimates, $\rho = 0.995$ exceeds $s\lambda \approx 0.120$, so the direct wage response to a larger self-employment surplus in Proposition 1 is positive for eligible workers at an interior solution. In the wealth block, entrant wealth averages $\bar{b}_0 = 0.594$, below the startup threshold of 0.686, which equals about 35 months of average earnings. In 2003 prices, the threshold is about R\$33,000, more than the equipment stock of 97 percent of urban own-account businesses (IBGE, 2005). The threshold represents a wealth requirement rather than an equipment requirement. It bundles the capital a business uses with the working balances and reserves its owner must hold. The offer-distribution shapes, both below one, imply offered firm capital is concentrated at the ends of its support. The estimated bargaining parameter is $\eta = 0.510$ under the maintained utility normalization. The calibrated financing parameters hold the working-capital duration at one quarter and the collateral multiplier at 2.6.

B. Fit and identification

Figure 4 revisits the age profiles introduced in Figure 1, adding their model counterparts on the same axes. Panel (a) shows self-employment shares, and panel (b) shows quarterly entry from formal employment. All five self-employment shares are targeted in estimation, together with entry rates at ages 35–44, 45–54, and 55–65. Entry rates in the two younger groups are untargeted. The comparison therefore shows both the fit to targeted levels and the model’s predictions for younger workers.

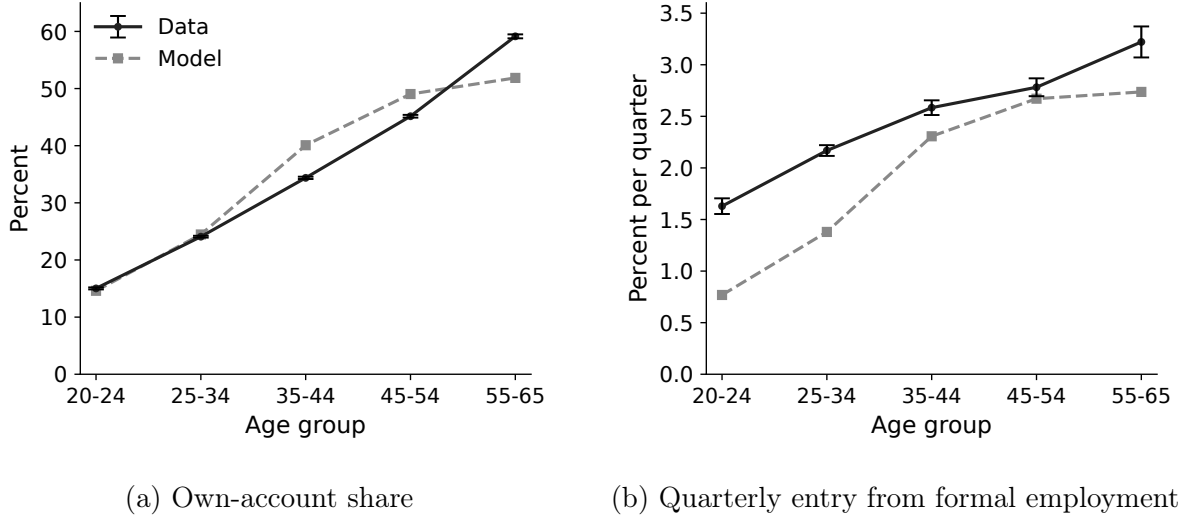


Figure 4: Age profiles of self-employment: data and model

Notes: The data, age groups, and axis scales are the same as in Figure 1. Panel (a) reports own-account workers as a share of formal private employees, own-account workers, and unemployed persons within each age group. Panel (b) reports quarterly transitions from formal private employment to own-account work, conditional on the next-quarter state remaining in this three-state sample. Model values are evaluated at the estimates in Table 3. All five stock shares and the three entry rates at ages 35–65 are targeted; the two younger entry rates are not. Vertical bars show pointwise 95 percent confidence intervals for the data, based on 200 bootstrap draws resampling individuals. Source: PNADC, 2012–2019, and model calculations.

The model reproduces the rise in self-employment with age, but the increase in the stock is too small at older ages. It closely matches the shares below age 35, overpredicts the share at ages 35–44 (40.1 versus 34.4 percent), and underpredicts it at ages 55–65 (51.9 versus 59.1 percent). Entry from formal employment also rises with age. The model understates entry in the two younger groups: 0.77 versus 1.63 percent at ages 20–24 and 1.38 versus 2.17 percent at ages 25–34. At ages 55–65, the model predicts 2.74 percent against 3.22 percent in the data.

Appendix Table 28 reports each data and model moment and its scaled deviation: the gap divided by five percent of the empirical target (equation F4). The model matches the unemployment block tightly (scaled deviations of +0.8 on the implied stationary unemployment share and 0.0 on the job-finding transition), the margins that identify the mobility rates (−0.2 on self-employment exit and −0.6 on long-tenure separations), and the targeted formal-to-self-employment levels within about three scaled units. The largest discrepancies concern the aggregate self-employment share and the unemployment-to-self-employment flow: the model predicts a share of 40.9 percent against a target of 31.6 percent (+5.9 scaled units), and a flow of 10.5 percent against 13.8 percent (−4.8). Section VI.A examines how lowering the startup requirement changes this flow within the model. The age-share discrepancies shown in Figure 4 correspond to +3.3 scaled units at

ages 35–44 and -2.5 at 55–65. Relative self-employment earnings are 11 percent above the empirical median ratio ($+2.3$).

The estimated model generates little of the employer heterogeneity in tenure profiles documented in Appendix Figure 5 and Appendix E.6. These comparisons are untargeted. Splitting matches by firm assets at formation, the low-minus-high asset difference in the increase in entry from three-to-seven years of tenure to ten or more years is 0.01 percentage points. In the broader-sample conditional comparisons of Appendix E.6, the corresponding differences are 2.49 percentage points across establishment sizes and 0.98 across industry-investment groups. This comparison is imperfect: initial firm assets in the model do not map directly to establishment size or industry investment at the origin interview. Moreover, model firms’ assets converge to a common target within surviving matches, limiting the persistence of employer differences (Appendix E.12).

The untargeted moments show which features of the data the model abstracts from. In the model, separations at short tenure are not elevated (a quarterly separation rate of 0.018 against 0.052 in the data) because the model has no match-quality learning (Jovanovic, 1979); quarterly transitions from self-employment to formal employment are nearly absent (0.004 against 0.045) because they require an intervening unemployment spell in the baseline; and within-state earnings dispersion is far below the data (75th-to-25th percentile ratios of 1.26 against 2.00 in formal work, 1.01 against 3.24 in self-employment) because a single worker type accumulates through a single wealth channel. Each discrepancy corresponds to a deliberately excluded mechanism rather than to a parameter that failed to fit. The model also has no margin along which own-account businesses grow into employers. In the PNADC panel that margin is small and unstable: employers are 4 percent of the employed against 23 percent own-account workers; 3.5 percent of own-account workers report employing someone in the following quarter, but 19 percent of employers revert to own-account work each quarter, and the one-year own-account-to-employer transition rate, 4.2 percent, is barely above the quarterly one. These moves therefore do not accumulate into durable firms, consistent with the evidence that informal businesses rarely grow (La Porta and Shleifer, 2014).

Identification is examined directly. At the estimates, the Jacobian of the seventeen scaled deviations with respect to the eleven estimated parameters has full rank. Its condition number is 1,162 and its minimum singular value is 0.97. Movement along even the least-informed direction of the parameter space, at steps of one wealth-grid cell or more, shifts the moment vector by essentially a full detectable unit. Appendix Table 29 reports the three most responsive moments for each parameter, and the pattern matches the design of Section IV: the startup threshold and the entrant-wealth scale act through the young self-employment shares, the bargaining weight through the long-tenure margins and relative earnings, and the offer-distribution shapes through the long-tenure cells and

the unemployment-to-self-employment flow. These diagnostics condition on the assigned financing technology; the targets respond weakly to the collateral multiplier.

The least-identified direction is economically interpretable: it trades a higher startup threshold against wealthier entrants on the young-worker margin, and because the threshold enters through wealth-grid nodes, the loss is locally flat within one grid cell along this pair. The threshold is accordingly reported at its effective grid node, identified up to the cell width. Its imprint on inference is direct: the standard errors of $\bar{b}_0$ and b_{SE} are individually wide (0.19 each, correlation 1.00), while their gap, the distance a typical entrant must save, is estimated at 0.105 with a standard error of 0.005. The second-weakest direction mixes the offer-distribution shapes with the bargaining weight and the self-employment curvature. Appendix F.3 reports the full sensitivity analysis and the global search behind the estimate.

With the estimates, their fit, and their identification established, four computed properties of the estimated economy organize the remainder of the section: entry into self-employment is rationed by wealth (Section V.C), the bargained wage exceeds the self-employment value on every surviving match (Section V.D), the region of parameter space in which match viability collapses lies far from the data (Section V.E), and a large part of the self-employed stock is a constrained portfolio (Section V.F). Each carries a policy consequence in Section VI.

C. Entry is rationed by wealth

The estimated startup threshold exceeds mean entrant wealth. Sixty-six percent of labor-market entrants begin working life ineligible for self-employment, and 40 percent of the unemployed at any moment search without the self-employment option. Eligibility arrives slowly: among entrants who ever cross the threshold, the mean time to first crossing is nearly twelve years, and sixteen percent of a cohort never crosses it.

Workers who begin below the startup threshold acquire eligibility through saving during employment. At the threshold, unemployed wealth falls on both sides, so unemployment spells can erode access to business entry. Consumption is lower just above the threshold than just below it, slowing the loss of wealth while the worker remains eligible. This eligibility-protection response preserves access to arriving business opportunities. Appendix A.1 derives the associated one-sided consumption policies and value derivatives. Wages finance the saving through which workers initially acquire this option.

Prior wages provide a limited empirical measure of this accumulation process. In the formal-employment sample underlying Figure 1, quarterly entry rises from 2.24 percent in the bottom third of the prior-wage distribution to 2.42 percent in the top third, with wage groups defined within calendar year. After adjustment for age, tenure, and observed worker

and job characteristics, the top-minus-bottom difference is instead -0.23 percentage points (Appendix E.3). The positive raw association therefore does not provide independent corroboration of the accumulation channel. A wage describes the return to employment as well as an opportunity to save; it does not measure accumulated wealth.

The threshold has a firm side, and it is quantitatively neutral. In the cross-section, matches with below-threshold workers carry about eight percent more firm surplus than matches with eligible workers, which might suggest that employers profit from their workers' exclusion from self-employment. The model rejects this reading: granting the option to the excluded, the threshold-removal experiment of Section VI.A, moves firm values at the affected matches by half of one percent, and upward. Two facts explain the neutrality. Under proportional bargaining the wage adjusts to the outside option state by state, so the employer cannot keep a margin that depends on the worker lacking it. And much of what workers gain from eligibility arrives at separation, through the routing margin, rather than in the wage: when the excluded receive the option, the flow wage at their matches falls by about three percent while their value rises. In the estimated economy, employers therefore hold no rents from workers' exclusion from self-employment. This does not contradict the evidence that firms borrow from their workers through the wage path (Michelacci and Quadrini, 2009; Guiso, Pistaferri, and Schivardi, 2013): that channel requires the firm's collateral constraint to bind, so that delayed wages relax it, and at the estimates no active match is at the constraint. The channel is present in the model but is inactive at the estimates.

D. Retention under the bargained wage

At the estimates, 70 percent of employed workers satisfy the startup requirement and prefer self-employment to unemployment. These workers value access to own-account work if their job ends, while all employed workers prefer their current match to self-employment at the bargained wage. Under proportional bargaining, the worker's gain from remaining in the match depends on its surplus. In the estimated economy, matches cease to be viable before quitting into self-employment becomes attractive.¹³

The routing counterfactual in Appendix G.4 measures the importance of direct access to self-employment following separation within the model. This exercise describes how the estimated economy generates transitions; the quarterly data do not identify whether a worker left a continuing match voluntarily. The retention result is therefore a property of the estimated model: self-employment provides an alternative following job loss even when its value is below that of current employment.

¹³Appendix F.5 discusses an extension with opportunities to enter self-employment while employed. An inactive quit margin leaves the opportunity-arrival rate unidentified; it does not establish the frequency of voluntary quits in the data.

E. Distance from the closure frontier

The theory of Section III.D exhibits a starker regime in which a strengthening self-employment option raises wages until the set of mutually acceptable matches collapses. The estimated economy lies well outside that regime. The joint-surplus band remains open at every wealth level we solve for, out to twice the wealth range used in estimation. Raising self-employment productivity closes the band only at roughly four times its estimated value.

At the estimates, self-employment is therefore limited by the wealth requirement rather than by the destruction of matches through the outside option. Formal employment, in turn, is sustained by a wage that compensates workers for the option, which four-fifths of the employed hold. The counterfactuals of Section VI accordingly operate on the entry margin rather than on match viability.

F. Part of self-employment is a constrained portfolio

The self-employed at the estimate divide into those who would choose the business over search at any plausible return on savings and those who hold it because no alternative asset pays more than the market return. The division is measured by offering workers a return above the market rate on wealth held outside a business, holding the return on business capital fixed, and asking who would leave; Section VI.C reports the experiment. At the market rate every self-employed worker prefers the business to search. At a return two percentage points higher, 4 percent would leave. The share rises to 23 percent at four points higher, 44 percent at six, and 92 percent at eight (Table 6). The preference of eligible entrants for self-employment falls in step, from 31 percent to zero. Self-employment in Brazil therefore takes three distinct roles. The startup requirement rations entry by wealth for the entrants it excludes. The bargained wage exceeds the value of self-employment for the eligible employed. A large part of the stock represents a portfolio choice, constrained by the absence of an alternative asset that pays more than the market return. Between a fifth and a half of current businesses, depending on the return offered, exist because their owners lack an alternative asset that pays a comparable return on their savings.

VI. Counterfactuals: Financial Development in the Estimated Economy

The estimated economy restricts entry into self-employment through a startup-wealth requirement. This section quantifies how financial development moves that economy on the three margins the term describes in practice: the cost of starting a business, workers' access to savings instruments, and firms' collateral capacity. Two further exercises compare

the targeted grant with a flow-matched capital grant to firms, and evaluate the transfer Brazil already operates, the statutory severance account. Every experiment is evaluated at the estimates of Table 3 under fixed conventions. Parameters other than the policy lever are held at their estimated values. Results are stated under two closures, the partial-equilibrium benchmark with the contact rate and offer distribution fixed, and the free-entry closure of Appendix D, in which vacancy creation transmits the policy to market tightness. Welfare is measured as the consumption-equivalent value of labor-market entry at the entrant wealth distribution. Appendix G records the conventions, the structural ceiling on the savings-return experiment (worker returns cannot approach the discount rate), and the sensitivity of the entry closure to the location of the offer distribution.

A. Deregulating entry and subsidizing startups

A long literature measures the cost of starting a business and asks what removing it would achieve (Djankov, La Porta, López-de-Silanes, and Shleifer, 2002); an experimental literature asks the same of capital grants to microenterprises (de Mel, McKenzie, and Woodruff, 2008). We evaluate both policies against the same estimated startup threshold, and the model implies very different effects for the two. Removing the startup threshold raises the self-employment share by 11 percentage points and lowers the unemployment rate by nearly 2 percentage points, but leaves entrant welfare essentially unchanged. A targeted grant that completes eligibility, bringing an entrant’s wealth up to the threshold, raises entrant welfare by 19.8 percent at 45 percent of the cost of doubling the entrant endowment, and delivers three and a half times the welfare per unit of transfer of the most generous uniform scheme. The two policies therefore operate on different margins: deregulation expands participation, while targeted completion raises entrant welfare.

Table 4 lowers the threshold from its estimated value to zero along the wealth distribution, holding all other parameters fixed.

The structural $U \rightarrow SE$ flow, which the model leaves 24 percent below its empirical counterpart at the estimate, rises through the data value of 0.138 when the threshold falls to roughly half its estimated level, and overshoots it by 29 percent under full removal. The wedge between the model’s unemployment-to-self-employment flow and the data measures the entry the threshold suppresses. Lowering the threshold to about half its estimated value closes the wedge. The size of the participation effect under the entry closure depends on the location of the offer distribution, a margin the data do not discipline. Tilting it across its plausible range moves the full-removal self-employment share between 0.37 and 0.56 around the closure’s 0.52, a range that includes the baseline share (Appendix G.3).

Entrant welfare, by contrast, barely moves: full removal raises the consumption-equivalent value of labor-market entry by one-tenth of one percent. The workers whom the threshold

Table 4: Removing the startup threshold

Startup threshold b_{SE}	0.686	0.51	0.34	0.17	0.05
Self-employment share	0.409	0.446	0.479	0.508	0.516
Unemployment rate	0.105	0.098	0.092	0.087	0.086
U→SE transition rate	0.105	0.125	0.146	0.171	0.178
Ineligible share of unemployed	0.396	0.292	0.183	0.057	0.000
Entrant welfare (CE)	0.189	0.189	0.189	0.189	0.189

Notes: Partial equilibrium; each column moves only b_{SE} . The empirical U→SE transition rate is 0.138. Entrant welfare is the consumption-equivalent value of labor-market entry (Appendix G); the change from the first to the last column is +0.1 percent. Propagating the parameter uncertainty of Table 3 through the experiment by the delta method, the first-to-last change in the self-employment share is +10.7 percentage points (standard error 2.0) and in the U→SE rate +7.3 points (standard error 1.8).

excludes are those whose self-employment surplus is smallest, so the marginal entrant enters a business worth little more than continued search. This is the quantitative counterpart, on the entry margin, of the retention result of Section V.D: at the estimated parameters, self-employment functions in substantial part as an option that workers retain rather than as an occupation they actively prefer.

The comparison suggests that welfare gains come from targeting rather than from deregulation. Table 5 compares two transfers to labor-market entrants: a uniform increase in the entrant wealth endowment, and a targeted grant that tops up every below-threshold entrant exactly to the threshold, the cheapest eligibility-completing transfer.

Table 5: Uniform endowments versus the targeted eligibility grant

	Uniform endowment increase			Targeted grant to the threshold
	+25%	+50%	+100%	
Cost per entrant	0.148	0.297	0.594	0.266
Self-employment share	0.417	0.421	0.424	0.484
Entrant welfare gain (%)	+4.1	+7.5	+12.5	+19.8
Welfare gain per unit cost	27.8	25.1	21.0	74.4

Notes: Partial equilibrium at the estimates. Cost is the mean transfer per labor-market entrant in wealth units; the targeted grant treats the 66 percent of entrants below the threshold. Welfare gains are consumption-equivalent percent changes for a newborn entrant; the last row divides the percent gain by the cost. Delta-method standard errors at the estimated parameter covariance: the targeted grant’s welfare gain of 19.8 percent carries a standard error of 1.1 points, and its advantage over the most generous uniform scheme, 3.5 to one per unit of transfer, a standard error of 0.8.

The targeted grant dominates on every margin because welfare and entry both turn on crossing a threshold. Transfers are valuable exactly to the extent that they complete eligibility, and a uniform grant spends most of its budget moving wealth where no threshold

binds. In this economy, the effectiveness of financial-inclusion policy therefore depends primarily on how transfers are targeted.

A natural objection comes from the optimal-development-policy literature: when firms are the constrained party, a planner may hold wages down while entrepreneurial wealth accumulates, because capital deepening is then the only route to higher wages (Itskhoki and Moll, 2019). In the estimated economy that route is nearly shut, within matches and through entry alike (Section VI.D). Moving across the interdecile range of firm assets shifts the wage by half of one percent of its mean, holding worker wealth fixed; the same movement across worker wealth shifts it by 49 percent. At the estimated parameters, wages therefore respond to worker wealth rather than to firm assets.

The matched-flow experiment makes the comparison direct: a capital grant tops up the initial assets of newly forming matches, at the same fiscal flow as the targeted grant. The worker-side grant is more effective by an order of magnitude. Per unit of flow, it raises mean worker earnings thirteen to twenty-three times as much, and it raises entrant welfare by 19.8 percent where the firm-side grant cannot move it.¹⁴ The firm-side grant operates on no active margin: every offer with positive initial assets is already accepted, whatever the worker’s wealth, so the grant rescues no match, and the assets it adds barely move the wage. At the estimate, capitalizing workers is the policy that raises wages. Whether an optimal sequence would ever favor firms is a transition question outside this stationary comparison.

B. The statutory severance account

Brazil’s labor code mandates the saving flow at the center of this paper’s mechanism. Under the Fundo de Garantia do Tempo de Serviço (FGTS), employers deposit eight percent of every formal wage into a locked account in the worker’s name. A dismissed worker receives the balance plus a forty percent fine paid by the firm; a worker who quits receives nothing (Lei 8.036/1990). The average separator receives about seven months of pay. We add the balance as a third state of the match and re-estimate the model; Appendix H reports the design, the estimates, and the transfer comparison. No parameter moves by more than one standard error: the estimates are robust to the institution.

We find that the account changes little in equilibrium. Workers treat it as their own wealth and cut their liquid saving by its full expected payout: the employed hold 12 percent less liquid wealth than they would without the institution. The share of employed workers below the startup threshold rises from 24 to 30 percent, and the average wait for eligibility lengthens by 2.4 years. The release at dismissal then restores the eligibility the

¹⁴The range spans earnings measured over the employed, over the employed and self-employed, and over the whole population. Entrant welfare is computed from worker states under the fixed value layer, so the firm-side grant leaves it unchanged by construction; gains to firm owners are not evaluated (Appendix G).

deposits crowded out; the ineligible share of the unemployed barely moves (37.8 against 37.1 percent). On net, entry into self-employment is almost unchanged.

Deposits and fines transfer 0.9 percent of workers' liquid wealth to separating workers each year. Redirected to the eligibility margin through a grant of the form of Section VI.A, the same flow raises entry into self-employment twenty-two times as much per unit and raises entrant welfare by 29 percent; the statutory transfer leaves welfare unchanged.¹⁵ The forced-saving account finances little entry, because workers reduce their voluntary saving by the amount mandated and because the balance is released only after the separation decision. The same flow at the eligibility margin finances substantially more.

C. Savings accounts: workers' asset-market participation

Financial underdevelopment restricts workers' access to asset markets as well as the formation of firms. Access to basic savings instruments is a policy margin with a substantial empirical literature of its own (Dupas and Robinson, 2013; Karlan, Ratan, and Zinman, 2014). The second experiment gives workers that access. Wealth held while employed or unemployed earns $r_k > r$. Capital committed to a business continues to earn r . This return structure corresponds to financial development that reaches wage earners before it reaches informal firms.

The design of the experiment is constrained by the model itself, so we state it first. The baseline has $r_k = r = 0.05$: all wealth earns the market return. Raising r_k is bounded above, because workers discount at $r + \nu$ and a stationary wealth distribution exists only while the savings return stays below the discount rate: the entire policy space is a corridor of 2.2 percentage points, and no reform of this type can exceed it. Within the corridor, wealth accumulation accelerates quickly. Only for small spreads can a stationary equilibrium be solved and verified. The experiment is therefore reported in two layers. A 25 basis point spread is solved and verified as a complete stationary equilibrium. Larger spreads, up to $r_k = 0.15$, are reported through the worker's value problem alone, which is well defined at any r_k because it involves no stationary distribution (Table 6). Appendix G.2 states the boundary and shows that the stationary-point results are conservative bounds.

¹⁵The grant lifts below-threshold entrants to a common wealth target and exhausts the matched flow exactly. The comparison is partial-equilibrium; the fine is a firing cost whose effects on dismissals and job creation are not evaluated. The instruments are also financed differently: the statutory flow is deferred wages plus a firm-paid fine, the grant an outside transfer, so its welfare gain is an upper bound relative to any financed implementation; the per-unit entry comparison is free of this asymmetry, the welfare comparison is not. Delta-method standard errors at the extended model's parameter covariance: the statutory effect on the self-employment share is 0.45 percentage points (standard error 1.5) and the grant's 9.8 (0.2); the welfare effects are -0.02 percent (0.005) and 28.7 percent (2.1). The ratio of the two entry effects is therefore not separately precise; what the comparison establishes is a near-zero effect against a large one.

Better outside saving contracts self-employment, and it raises welfare. The self-employment premium $V^{\text{SE}}(b) - U(b)$ falls at every wealth level as r_k rises. At $r_k = 0.15$, the wealth frontier below which self-employment is preferred contracts to less than a third of its level at the estimate. It falls below the startup threshold itself (Table 6). The share of eligible entrants who would choose self-employment falls from 31 percent to zero. The incumbent stock responds only beyond the corridor’s edge: at $r_k = 0.07$, 4 percent of the self-employed would leave their businesses for search, at 0.09 nearly a quarter, and at 0.13 nine in ten (Section V.F). In the stationary equilibrium at a 25 basis point spread, the self-employment share falls by 0.59 percentage points, the ineligible share of the unemployed falls by 1.7 points, mean time to eligibility shortens by about two months, and entrant welfare rises by 0.19 percent.

Table 6: Savings access and the value of self-employment

r_k	0.05	0.07	0.09	0.11	0.13	0.15
$V^{\text{SE}} - U$ at the threshold	+1.34	+1.06	+0.78	+0.54	+0.41	−0.63
$V^{\text{SE}} - U$ at $b = 1.5$	+0.41	+0.13	−0.33	−2.01	−3.58	−4.79
SE-preference frontier $b^\dagger$	2.23	1.68	1.30	1.10	0.83	0.62
Eligible entrants preferring SE	0.31	0.28	0.22	0.17	0.08	0.00
Self-employed preferring exit	0.00	0.04	0.23	0.44	0.92	1.00

Notes: Value-side objects from the solved worker problem; no stationary distribution is involved, so the entries are free of the wealth-accumulation boundary discussed in Appendix G. The baseline is $r_k = r = 0.05$. $b^\dagger$ is the largest wealth at which $V^{\text{SE}} > U$. The last row is the share of the self-employed, at the estimated self-employed wealth distribution, for whom $U > V^{\text{SE}}$ at the stated return.

Both channels of the model are active. Accumulation toward the threshold accelerates, shortening the eligibility wait, but the price channel dominates from the first basis point: a better outside asset raises the opportunity cost of tying wealth up in a business. The sign of the welfare change helps interpret this result. Workers made better off by an instrument that reduces their self-employment reveal that part of the observed self-employment stock was a constrained portfolio choice rather than a preferred occupation. On this margin, improved savings access is associated with exit from self-employment rather than with entry.

D. Interactions: complements or substitutes?

Whether credit-market and labor-market frictions compound one another is a classic question: where they are complements, partial reform disappoints and joint reform is amplified (Wasmer and Weil, 2004). In the estimated economy the question has a structural answer, because free entry (Appendix D) converts financial levers into tightness and contact-rate responses. The experiments point to substitution: the threshold reform has its largest effect where matching efficiency is lowest, the firm-side collateral margin

has essentially no effect, and no amplification emerges through entry. Table 7 computes each financial lever at three levels of matching efficiency under the entry closure and reports the cross-partial.

Table 7: Financial levers and matching efficiency

Lever	Effect on SE share at matching efficiency			Cross-partial
	0.85×	1.00×	1.15×	
Startup threshold halved	+0.086	+0.073	+0.063	−0.024
Savings access (+25bp)	−0.002	−0.004	−0.002	−0.000
Collateral capacity (φ 2.6→3.2)	+0.000	−0.000	−0.001	−0.001

Notes: Free-entry closure; each cell a converged tightness equilibrium at the estimates with the stated lever and matching-efficiency multiplier. The cross-partial is the difference in the lever’s effect between high and low matching efficiency. Welfare cross-partials are zero to reported precision. Appendix G reports the full grid, including unemployment, and the closure convention.

Worker-side financial development and labor-market efficiency are substitutes on the self-employment margin. The threshold reform raises self-employment by 8.6 percentage points where matching is poor but only 6.3 where it is good; the savings-access lever contracts self-employment at every matching level, and its welfare gain does not vary with matching efficiency at reported precision. Each reform reduces the group of workers for whom the other is valuable: self-employment entry is most valuable, and the threshold most costly, where formal jobs are hardest to find. The policy corollary extends the targeting statement of Section VI.A: financial-inclusion reforms have their largest effects in economies, or regions, with the least efficient labor markets.

The firm-side collateral margin has little effect at the estimates. Deepening collateral capacity from 2.6 to 3.2 moves the self-employment share by at most one-tenth of a percentage point at any matching efficiency, and every other reported margin by similar amounts. At the estimates, the binding financial friction is the worker’s startup wealth, not the firm’s borrowing capacity.

No complementarity emerges through entry. Tightness stays within three percent of baseline under every lever at estimated matching efficiency, and partial-equilibrium and entry-closure results differ by at most 0.6 percentage points on any reported share. The partial-equilibrium tables of this section are, to first order, the general-equilibrium tables.

E. Summary

Two margins govern self-employment in the estimated economy: a wealth threshold in front of entry, and a wage that retains those who pass it. The policy experiments divide along these two margins. Transfers that complete eligibility deliver three and a half times the welfare per unit of uniform transfers; removing the threshold expands participation without

raising welfare; better savings instruments raise welfare while shrinking self-employment; firm-side collateral deepening moves nothing; and the threshold reform earns most where the labor market matches worst. Brazil’s statutory severance account leaves the estimates unchanged and finances little entry: workers save less, one for one, and the release comes after the separation decision. The same flow at the eligibility margin buys twenty-two times the entry, and a matched capital grant to firms a small fraction of the worker-side gain, because wages respond to worker wealth rather than to firm assets. In the estimated economy, financial development has little effect on the level of entrepreneurship. Its main effect is to make self-employment a choice rather than a constrained outcome.

VII. Conclusion

In Brazil, the capital that finances self-employment is the owner’s own rather than borrowed. We model its accumulation as saving out of wages and build a labor-market equilibrium on that basis. In the model, workers save the wealth a startup requires while employed, and, once they have saved it, the wage responds to their option of leaving. Two-thirds of labor-market entrants begin below the startup requirement, and those who meet it take over a decade to do so. No employed worker who has met it prefers self-employment to the current job. We call these workers eligible.

A grant that lifts a worker’s wealth to the startup requirement delivers three and a half times the welfare per unit of transfer that a uniform transfer does. Removing the requirement expands self-employment but adds little to welfare, because the workers it admits value a business at little more than continued search. Better savings instruments raise welfare while shrinking self-employment, because for many workers the business is mainly a place to hold savings rather than a preferred occupation: at a return four percentage points above the market rate, nearly a quarter of the self-employed would leave their businesses. Of the model’s two financial frictions, only the worker’s startup requirement constrains outcomes. Firm-side collateral deepening leaves every outcome the paper measures unchanged, whereas each worker-side instrument changes the self-employment rate, welfare, or both, and the threshold reform’s gains are largest where labor-market matching is weakest. In this economy, financial development has a larger effect on whether workers are self-employed by choice than on the number of workers who are self-employed.

The model omits learning about the quality of a job match, differences in worker skill, permanent differences in employer size, and the growth of own-account businesses into employers; the untargeted moments show where each matters, and none overturns the wealth mechanism. The last omission matters most for the comparison with the literature in which entrepreneurs self-finance out of retained profits. In the data the margin is small

and unstable: employers are 4 percent of the employed, and nearly a fifth of employers revert to own-account work each quarter. Were some businesses able to grow and hire, entering self-employment would be worth more, the startup requirement would bind more tightly, and the return to meeting it would be larger, so the worker-side estimates are conservative on that margin. The firm's collateral constraint, which does not bind in the estimated economy, would again matter for the businesses that grew. A companion paper supplies heterogeneous firms and asks whether deferred compensation or equity-like claims could pay workers the return on savings that households cannot obtain from available assets.

The model admits two qualitatively different regimes. The comparison between them raises a question that extends beyond Brazil and that we are pursuing in ongoing work. In one, Brazil's case, the startup requirement limits entry and, at the estimated wages, eligible employees prefer their current job to self-employment. In another, workers' option of leaving raises the wage even in jobs that continue, ends matches at asset-poor employers, and at the extreme leaves no match mutually acceptable. Which applies depends on two ratios: what a worker earns on own account relative to what a job produces, and the startup requirement relative to household wealth. Both move against the poorest economies: own-account work yields at least subsistence while what a job produces falls with a country's income, and the capital that own-account work requires is small, so most households can meet it. That pushes the poorest economies toward the second regime, consistent with their small and short-lived formal sectors; in rich economies the second regime should appear first among wealthy, high-wage workers, where entrepreneurial entry concentrates ([Hurst and Lusardi, 2004](#)).

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Online Appendix

Worker Wealth and the Option of Self-Employment

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This appendix contains the theoretical, empirical, and computational material supporting the results in the paper.

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A. Model Details

This appendix states the functional forms, policy functions, and boundary conditions used to solve the model. It also states the solution concept and the numerical domain.

Table 8 records the preference, technology, demographic, and financing assignments used in the quantitative analysis. The turnover rate $\nu = 1/45$ implies an expected working life of 45 years. Section A.2 below describes the evidence for the financing calibration and its sensitivity ranges.

Table 8: Assigned parameters

Parameter	Value
Relative risk aversion, γ	2
Discount rate and asset return, r	0.05
Worker turnover rate, ν	1/45
Formal productivity, z	1
Formal capital elasticity, α	0.30
Capital depreciation, $\delta = \delta_{SE}$	0.06
Self-employed capital per unit of wealth, ϕ_{SE}	1
Wage-prefinancing duration, χ	0.25
Formal collateral multiplier, ϕ	2.6
Cohort entry age	20

Notes: Rates are annual and durations are in years. The productivity normalization fixes the model's goods unit; income is a flow per year and wealth is a stock in the same goods unit. Wealth amounts are interpreted relative to the model's annual mean formal wage. Preference, technology, and demographic values are maintained assignments. The risk-aversion coefficient follows [Quadriini \(2000\)](#); Section A.2 describes the financing parameters and their evidence.

A.1 Preferences and worker policies

Preferences are CRRA:

$$u(c) = \begin{cases} \frac{c^{1-\gamma} - 1}{1-\gamma}, & \gamma \neq 1, \\ \log c, & \gamma = 1. \end{cases} \quad (\text{A1})$$

The startup-wealth indicator $I_{SE}(b)$ is defined in (1). It governs transitions into self-employment, while incumbents continue operating on the full wealth domain, including $b < b_{SE}$.

The unemployment and self-employment values solve

$$(r + \nu + \xi_{SE})V^{SE}(b) = \max_{c \geq 0} \{u(c) + V_b^{SE}(b)[rb + \Pi^{SE}(b) - c]\} + \xi_{SE}U(b), \quad (\text{A2})$$

$$(r + \nu)U(b) = \max_{c \geq 0} \{u(c) + U_b(b)[rb + z_u - c]\} + \mathcal{O}(b). \quad (\text{A3})$$

The unemployment and self-employment consumption policies satisfy

$$u'(c_U(b)) = U_b(b), \quad \dot{b}_U = rb + z_u - c_U(b), \quad (\text{A4})$$

$$u'(c_{SE}(b)) = V_b^{SE}(b), \quad \dot{b}_{SE} = rb + \Pi^{SE}(b) - c_{SE}(b). \quad (\text{A5})$$

Regularity at the startup threshold. The option flow in (A3) jumps by $\Delta_{\mathcal{O}} = \rho [V^{SE}(b_{SE}) - U(b_{SE})]^+$ at b_{SE} . The value U remains continuous. The option is a flow arriving at a finite rate, so its value accrues over occupation time. The derivative U_b , however, is generically one-sided at b_{SE} . Writing (A3) as $(r + \nu)U = \mathcal{H}(b, U_b) + \mathcal{O}(b)$ with

$$\mathcal{H}(b, q) = u((u')^{-1}(q)) + q[rb + z_u - (u')^{-1}(q)], \quad (\text{A6})$$

continuity of U replaces smooth pasting by the H-continuity condition

$$\mathcal{H}(b_{SE}, U_b^+) - \mathcal{H}(b_{SE}, U_b^-) = -\Delta_{\mathcal{O}}, \quad (\text{A7})$$

where $U_b^{\pm}$ denote the one-sided derivatives. Because $\mathcal{H}_q(b, q) = rb + z_u - (u')^{-1}(q)$ equals the wealth drift, the sign of $\dot{b}_U$ at the threshold determines the kink. With positive drift, U_b falls across b_{SE} and c_U jumps up: the marginal value of wealth is elevated just below the threshold, which gives unemployed workers a motive to save toward eligibility; if the induced consumption jump reverses the drift sign, the threshold is an attractor and the stationary distribution places an atom of unemployed workers at b_{SE} who await the ρ arrival. With negative drift on both sides, unemployed workers cannot self-finance eligibility. The threshold is crossed upward only through saving inside formal matches. The incentive then inverts into eligibility protection just above the threshold, where $U_b^+ > U_b^-$. Consumption drops as wealth crosses b_{SE} from below. In every case c_U , the interior wage $w(a, \cdot)$, and the unemployed density g_U are one-sided at b_{SE} ; W and J remain continuous with kinks in b ; and all value equations hold with one-sided derivatives at the threshold. At the estimated parameters the drift is negative on both sides ($\dot{b}_U \approx -0.06$ at the threshold), so the eligibility-protection case obtains and no atom forms.

A.2 Formal production and dividends

Formal production is summarized by

$$k(a, w) = \phi a - \chi w, \quad (\text{A8})$$

$$y(a, w) = zk(a, w)^{\alpha}, \quad (\text{A9})$$

$$\pi(a, w) = y(a, w) - (1 - \delta\chi)w - [(r + \delta)\phi - r]a. \quad (\text{A10})$$

The first line imposes full use of collateralized capacity for capital and advance wage finance. The wage coefficient $1 - \delta\chi$ in the third line follows from the financing primitive (3). The firm borrows $k + \chi w - a$ at rate r ; interest applies to all borrowing and depreciation to physical capital. The marginal resource cost of an additional unit of wages is

$$M(a, w) \equiv -\pi_w(a, w) = 1 - \delta\chi + \chi\alpha z k(a, w)^{\alpha-1}. \quad (\text{A11})$$

When $\chi > 0$, nonnegative operated capital requires

$$w \leq \bar{w}(a) \equiv \frac{\phi a}{\chi}. \quad (\text{A12})$$

There is no wage cap from working capital when $\chi = 0$. Because w is an annual flow, χ is measured in years of wage-bill prefinancing. The baseline calibration sets $\chi = 0.25$; sensitivity exercises use $\chi \in [0.125, 0.50]$.

For reference, flow resources at a zero wage satisfy $\pi(a, 0) = z(\phi a)^\alpha - [(r + \delta)\phi - r]a$. Their marginal return to assets is $z\alpha\phi(\phi a)^{\alpha-1} - [(r + \delta)\phi - r]$, which decreases with assets and vanishes at

$$\hat{a} = \frac{1}{\phi} \left[\frac{z\alpha\phi}{(r + \delta)\phi - r} \right]^{\frac{1}{1-\alpha}}. \quad (\text{A13})$$

Thus $\hat{a}$ maximizes the zero-wage resource flow. This benchmark scales the offer distribution and the numerical asset domain below. The dividend boundary $a^*(b)$ instead depends on the marginal continuation value J_a , including the effects of wages and worker wealth.

The collateral multiplier is calibrated at $\phi = 2.6$ from micro evidence on secured lending. [Midrigan and Xu \(2014\)](#) estimate $\lambda = 2.58$ by indirect inference on plant-level moments under a constraint of the identical form, working capital and wage bill included; Brazilian secured auto lending after the 2004 repossession reform financed 62.4 percent of asset value, implying a per-asset multiplier of $1/(1 - 0.624) = 2.66$ ([Assunção, Benmelech, and Silva, 2014](#)); and the World Bank Enterprise Survey for Brazil reports collateral requirements of 114 percent of loan value for small firms ([World Bank, 2010](#)), which maps to $\phi \approx 2.6$ at a pledgeable capital share of about 0.7. When ϕ is instead estimated, the point estimate falls within this cluster. The loss is nearly flat in ϕ , so the moments do not pin it down. The external calibration disciplines it instead. Sensitivity exercises use $\phi \in [1.35, 3.3]$, whose lower end corresponds to the external-finance-to-GDP tradition ([Buera and Shin, 2013](#); [Moll, 2014](#)).

Inside an active match,

$$\dot{a} = \pi(a, w) - d, \quad \dot{b} = rb + w - c. \quad (\text{A14})$$

The firm value satisfies the variational inequality

$$\min \{(r + s + \nu)J - d - (s + \nu)a - J_a[\pi(a, w) - d] - J_b[rb + w - c], J - a\} = 0. \quad (\text{A15})$$

The dividend policy solves

$$d(a, b) \in \arg \max_{\tilde{d} \in [0, \pi(a, w)^+]} \{\tilde{d} + J_a(a, b)[\pi(a, w) - \tilde{d}]\}. \quad (\text{A16})$$

Consequently,

$$d(a, b) = \begin{cases} 0, & J_a(a, b) > 1, \\ \pi(a, w)^+, & J_a(a, b) < 1, \\ \text{any value in } [0, \pi(a, w)^+], & J_a(a, b) = 1. \end{cases} \quad (\text{A17})$$

Substitution of these policies into (A15) gives the continuation equation used in the active region and the obstacle $J = a$ outside it.

On uncapped active states, the proportional identity determines W from J . If the interior wage root exceeds $\bar{w}(a)$, the wage is fixed at the collateral cap and the proportional identity is suspended. On those states the worker value satisfies

$$0 = \min \left\{ (r + s + \nu)W - u(c) - W_a[\pi(a, \bar{w}) - d] - W_b[rb + \bar{w} - c] - s [U + \lambda I_{\text{SE}}(b)(V^{\text{SE}} - U)^+] , W - U \right\}, \quad (\text{A18})$$

with $u'(c) = W_b$. Equality with either participation obstacle places the state outside the strict active set.

A.3 Formal offers

The asset level in a formal offer is drawn from F_A . In the quantitative model, F_A is a beta distribution with shape parameters (β_1, β_2) scaled to the support $[0, \hat{a}]$, where $\hat{a}$ is the zero-wage resource-maximizing scale in (A13). The firm-asset search bound $\bar{a} > \hat{a}$ in (18) sets the range over which match viability is evaluated, while $\hat{a}$ sets the support of offers. The expected contribution of an offer to unemployment is

$$\Lambda(b) = \int_{\mathbb{R}_+} [W(a', b) - U(b)]^+ \mathbf{1}\{J(a', b) > a'\} dF_A(a'). \quad (\text{A19})$$

Equivalently,

$$\Lambda(b) = \int_{\mathcal{A}(b)} [W(a', b) - U(b)] dF_A(a'). \quad (\text{A20})$$

Under an unconstrained proportional allocation, Lemma 1 permits either participation inequality to characterize the offer set. The bilateral definition (13) remains valid at wage caps.

A.4 Participation and boundaries

The bilateral participation margin $m(a, b)$ is defined in (12). The central boundaries are

$$a_c(b) = \inf\{a \geq 0 : m(a, b) > 0\}, \quad (\text{A21})$$

$$a^*(b) = \inf\{a \in \mathcal{A}(b) : J_a(a, b) \leq 1\}, \quad (\text{A22})$$

$$b_{\text{trap}} = \inf\{b \in [\underline{b}, \bar{b}] : \max_{a \in [0, \bar{a}]} m(a, b) \leq 0\}. \quad (\text{A23})$$

Here $[\underline{b}, \bar{b}]$ is the interval considered in Proposition 3, and $\bar{a}$ contains its active firm-asset states. The infimum defining $a^*(b)$ is restricted to the active set because $J_a \equiv 1$ on the liquidation region, where $J = a$. This restriction isolates the payout decision within continuing matches. The earnings-retention and low-asset regions are, respectively,

$$\mathcal{G} = \{(a, b) : m(a, b) > 0, J_a(a, b) > 1\}, \quad (\text{A24})$$

$$\mathcal{T} = \{(a, b) : 0 \leq a < a_c(b)\}. \quad (\text{A25})$$

The set $\mathcal{G}$ describes retention incentives; positive asset growth additionally requires $\pi(a, w) > 0$. Both sets remain well defined when $\mathcal{A}(b)$ consists of several intervals in firm assets. The startup threshold b_{SE} governs entry into self-employment, while b_{trap} governs the existence of formal matches over the stated wealth interval.

A.5 Solution concept and equilibrium selection

The equilibrium of Definition 1 is a fixed point in the expected surplus from an acceptable offer, Λ . Given this surplus, the unemployment and self-employment problems determine outside values. These values enter the match problem and wage equation, whose solutions determine participation and hence the next offer surplus. The implementation solves the match problem as a linear complementarity problem, updates Λ , and uses damped iteration to convergence.

Conditional on continuation values, the wage equation has at most one interior root under the monotonicity conditions derived in Appendix B. The joint equilibrium additionally requires consistency between outside values and acceptable offers. Numerical searches with a low startup threshold and high unemployment income have produced two distinct fixed points at the same parameter vector. Each satisfies the residual and complementarity checks, while participation sets and entry flows differ. The primitive offer distribution F_A is the same at both solutions; the values and set of acceptable offers entering Λ differ.

The numerical selection convention is the solution reached by damped iteration from the baseline initialization, using continuation in the separation rate where prescribed by the solution procedure. A fixed selection convention assigns a moment vector, and hence an estimation loss, to a parameter vector whenever that procedure converges. The additional admission rule excludes candidate estimates for which independent initializations recover different equilibria. This restriction makes the reported fit stable across the equilibrium searches performed, while avoiding reliance on a particular branch at parameter vectors with detected multiplicity.

At the estimated parameters, the independently initialized solutions agree to numerical tolerance. Counterfactual solutions are continued from the estimated equilibrium and compared with independent solutions at the reported points. Appendix G specifies their agreement and local-perturbation checks. Agreement across these searches establishes robustness to the initializations examined. A global uniqueness result would require control of the joint fixed-point map over all admissible continuation values.

A.6 Numerical domain and boundary conditions

The worker problems are solved on a truncated wealth grid $[b_{\min}, b_{\max}]$ with state-constraint boundary conditions at both edges: consumption is capped at cash on hand at the borrowing edge, and the outward drift is suppressed at the top, so wealth cannot leave the domain. The match problem is solved on $[0, \bar{a}] \times [b_{\min}, b_{\max}]$, where $\bar{a}$ is a fixed multiple of the zero-wage resource-maximizing scale $\hat{a}$; the top row of the firm-asset grid lies in the payout region and imposes $J_a = 1$, and the wage is zero on the $a = 0$ row, where no capital can be operated. The discretized Kolmogorov system uses conservative upwinding with no-flux edges, with one exception: an active match whose worker wealth reaches the top of the grid is treated as a dissolution into unemployment, so no formal mass accumulates on that edge.

B. Bargaining Protocols

B.1 Derivation of the proportional wage

The proportional protocol fixes the ratio of worker to firm surplus:

$$W(a, b) - U(b) = \kappa[J(a, b) - a], \quad \kappa = \frac{\eta}{1 - \eta}. \quad (\text{B1})$$

This restriction is imposed in levels on the region where the proportional allocation is feasible: uncapped active states away from the participation obstacles. On the interior of

that region, differentiation with respect to worker wealth gives

$$W_b(a, b) = U_b(b) + \kappa J_b(a, b). \quad (\text{B2})$$

The worker's consumption first-order condition, $u'(c) = W_b$, therefore yields (10). Because the two surpluses are measured in different units, κ and the associated η are interpreted conditional on the stated CRRA normalization. A money-metric protocol would be invariant to rescaling worker utility but would introduce additional U_{bb} terms; it is not the protocol implemented in the baseline.

In an active match, the worker value satisfies

$$(r + s + \nu)W = u(c) + W_a \dot{a} + W_b \dot{b} + sU + s\lambda I_{\text{SE}}(b)(V^{\text{SE}} - U)^+, \quad (\text{B3})$$

where $\dot{a} = \pi(a, w) - d$ and $\dot{b} = rb + w - c$. Equation (B1) implies

$$W_a = \kappa(J_a - 1), \quad W_b = U_b + \kappa J_b. \quad (\text{B4})$$

At the boundaries of the feasible region these derivatives are one-sided. On the continuation region, the firm equation is

$$(r + s + \nu)J = d + (s + \nu)a + J_a \dot{a} + J_b \dot{b}. \quad (\text{B5})$$

The unemployment equation evaluated at its consumption policy is

$$(r + \nu)U = u(c_U) + U_b(rb + z_u - c_U) + \mathcal{O}(b). \quad (\text{B6})$$

Worker exit at rate ν is an exit from the economy, not a transition to unemployment. It therefore enters the discount rate in (B3) but has no continuation-value source. Firm assets return to the firm owner at either worker exit or separation, which accounts for $(s + \nu)a$ in (B5). Substituting (B1) into (B3), using (B5), and subtracting (B6) gives

$$\kappa[\pi(a, w) - ra] = u(c) - u(c_U) + U_b(b)[w - z_u - c + c_U] - \mathcal{O}(b) + s\lambda I_{\text{SE}}(b)(V^{\text{SE}} - U)^+, \quad (\text{B7})$$

which is (11); the reallocation term has no firm-side counterpart, so it survives the cancellation, and it is constant in w , so uniqueness and monotonicity of the wage root are unaffected.

Conditional on continuation-value derivatives and consumption policies, the residual defined in (C1) has derivative

$$G_w = -\kappa M(a, w) - U_b(b) < 0, \quad (\text{B8})$$

where M is the marginal resource cost of wages (A11); $M > 0$ under the maintained calibration ($\delta\chi < 1$) and $U_b = u'(c_U) > 0$. The interior wage is therefore unique whenever a root exists.

B.2 Feasibility

Let $w^0(a, b)$ be the root of (B7). If $0 \leq w^0(a, b) \leq \bar{w}(a)$, the proportional allocation is implemented. If $w^0(a, b) > \bar{w}(a)$, the wage is fixed at the collateral cap and W solves (A18); the proportional identity need not hold on those states. If $w^0(a, b) < 0$, the wage is fixed at zero. The worker and firm obstacles are then evaluated under the resulting policy. A match continues only where $m(a, b) > 0$; equality with either obstacle is dissolution. Thus the implemented model does not retain matches at participation corners.

B.3 Alternative Nash protocols

A spot Nash protocol that holds continuation values fixed gives

$$\eta S_J u'(c) = (1 - \eta) S_W M. \quad (\text{B9})$$

This protocol is common in search models with worker savings; see [Krusell, Mukoyama, and Şahin \(2010\)](#) and [Bils, Chang, and Kim \(2011\)](#). A one-sided dynamic protocol that accounts for the effect of wages on worker savings gives

$$\eta S_J u'(c) = (1 - \eta) S_W [M - J_b]. \quad (\text{B10})$$

A directional two-sided protocol that also accounts for retained earnings gives

$$\eta S_J [u'(c) - q M W_a] + (1 - \eta) S_W \{J_b - M[1 + q(J_a - 1)]\} = 0, \quad (\text{B11})$$

where $q \in [0, 1]$ weights the local retention response along the bargaining direction; $q = 0$ recovers (B10). Equation (B11) applies in the interior. At the lower wealth edge, the condition blends $u'(c)$ with the marginal utility of cash on hand. These protocols do not replace the proportional protocol in the baseline model.

At the estimated parameters none of these conditions can be implemented state by state. On the active set the proportional identity fixes $W - U = \kappa(J - a)$, so the spot condition (B9) reduces to $u'(c) = M(a, w)$; the bargaining weight and both surpluses cancel. Because employed workers save toward the startup threshold, $u'(c)$ typically exceeds M at every wage below the collateral cap. The unique root of each condition therefore lies against the financing constraint $k + \chi w \leq \phi a$ on 98.5 percent of employment mass (98.4 percent for (B10), 97.5 for (B11)), with operated capital below one one-thousandth of pledgeable capacity and flow profit below the owner's outside return; the worker-side participation

and consumption requirements never bind. A feasible root exists on 1.5 to 2.3 percent of employment mass, low-asset firms matched with wealthy workers, but it carries no option value: the reallocation term enters none of the conditions, so the wage response to the self-employment option is zero state by state, against $-s\lambda/(\kappa M + U_b)$ under the proportional protocol, and the implied wealth gradient of the wage has the wrong sign (-1.4 to -1.8 against $+0.08$).

C. Proofs

C.1 Common participation

Lemma 1 (Common participation). *At every state where the proportional allocation is feasible,*

$$W(a, b) - U(b) > 0 \iff J(a, b) - a > 0.$$

At a wage cap, participation is determined by the two inequalities in (13).

At a state where the proportional allocation is feasible, $W - U = \kappa(J - a)$ and $\kappa = \eta/(1 - \eta) > 0$. Multiplication by a strictly positive constant preserves signs. Hence $W - U > 0$ if and only if $J - a > 0$. If a wage cap prevents implementation of the proportional allocation, equation (8) need not hold; both participation inequalities must then be checked separately. $\square$

C.2 Proof of Proposition 1

Move the right-hand side of (11) to the left and define

$$\begin{aligned} G(w, \mathcal{O}, R_\lambda) &= \kappa[\pi(a, w) - ra] - u(c) + u(c_U) \\ &\quad - U_b(b)[w - z_u - c + c_U] + \mathcal{O} - R_\lambda, \\ R_\lambda &= s\lambda I_{SE}(b)(V^{SE} - U)^+. \end{aligned} \tag{C1}$$

At fixed continuation-value derivatives and consumption policies,

$$G_w = \kappa\pi_w - U_b = -\kappa M - U_b < 0, \quad G_{\mathcal{O}} = 1, \quad G_{R_\lambda} = -1. \tag{C2}$$

The implicit-function theorem therefore gives

$$\frac{\partial w}{\partial \mathcal{O}} = \frac{1}{\kappa M + U_b} > 0, \quad \frac{\partial w}{\partial R_\lambda} = -\frac{1}{\kappa M + U_b} < 0. \tag{C3}$$

For $\Delta_{SE} = (V^{SE} - U)^+$, holding $p\Lambda$ fixed gives $\partial \mathcal{O} / \partial \Delta_{SE} = \rho I_{SE}(b)$ and $\partial R_\lambda / \partial \Delta_{SE} = s\lambda I_{SE}(b)$. Combining the two derivatives proves (17). Since $k = \phi a - \chi w$, the capital derivative is $-\chi$ times the wage derivative. $\square$

C.3 Proof of Proposition 2

For $q' > q$, the assumption gives $m(a, b; q') \leq m(a, b; q)$ at every (a, b) . Hence $m(a, b; q') > 0$ implies $m(a, b; q) > 0$, which proves $\mathcal{A}(b; q') \subseteq \mathcal{A}(b; q)$. Taking infima, with $\inf \emptyset = +\infty$, gives $a_c(b; q') \geq a_c(b; q)$. At a given worker wealth b , a firm below the formation boundary requires additional assets to form a match. Its assets remain below that boundary in the absence of a source of accumulation or an external transfer. $\square$

C.4 Proof of Proposition 3

Let $B_0 = \{b \in [\underline{b}, \bar{b}] : H(b) \leq 0\}$. The endpoint conditions make B_0 nonempty and imply $b_{\text{trap}} = \inf B_0 \in [\underline{b}, \bar{b}]$. For $b \in [\underline{b}, b_{\text{trap}})$, $b \notin B_0$, so $H(b) > 0$ and the active set is nonempty. For any $b \in (b_{\text{trap}}, \bar{b}]$, there exists $\tilde{b} \in B_0$ with $\tilde{b} < b$. Weak monotonicity gives $H(b) \leq H(\tilde{b}) \leq 0$, so the active set is empty. No conclusion about membership of b_{trap} in B_0 is required. $\square$

D. The Entry Closure

The estimated model is general-equilibrium in every margin it determines. Wages are bargained against outside values that themselves reflect search and entry options. Acceptance and dissolution sets respond to the whole value system, and the offer-surplus field is a fixed point over all of it. Two objects, however, are held fixed in estimation: the rate at which unemployed workers meet firms, and the distribution of capital behind the offers they meet. For counterfactual analysis this margin matters, because a policy that changes firm values should change how many firms post vacancies, and hence how often workers are met. This appendix closes that margin with the minimal entry layer the counterfactuals require, constructed so that the estimated equilibrium is exactly undisturbed.

Free entry. Let Θ denote market tightness, with a constant-returns matching technology: vacancies fill at $q(\Theta) = \bar{m} \Theta^{-\alpha_m}$ and unemployed workers meet offers at $p = \Theta q(\Theta)$, with benchmark elasticity $\alpha_m = 0.5$ and efficiency $\bar{m}$ normalized so that $\Theta = 1$ at the estimated contact rate. Potential entrants post vacancies with capital drawn from the offer distribution at flow cost κ , meeting searchers drawn from the equilibrium unemployed composition and forming a match where jointly viable:

$$q(\Theta) \mathbb{E}_{a \sim F_A} \mathbb{E}_{b \sim g_U} [(J(a, b) - a)^+ \mathbf{1}\{\text{viable}\}] = \kappa. \quad (\text{D1})$$

The firm's surplus from matching is $J - a$, not J net of a vacancy value: free entry competes the vacancy's option value down to its liquidation value, so a dissolved firm is

indifferent between re-posting and liquidating. The closure therefore delivers the estimated model’s dissolution outside option as an arbitrage outcome rather than contradicting it.¹⁶

Calibration and baseline invariance. The entry cost κ , together with the marginal cost that normalizes the offer-composition condition below, is backed out so that equation (D1) holds exactly at the estimated equilibrium with $\Theta = 1$. The baseline is therefore unchanged by construction: no parameter is reinterpreted and nothing is re-estimated. The layer activates only in counterfactuals, where Θ adjusts until entry again breaks even and the contact rate moves with $p = \Theta q(\Theta)$, feeding back into values, wages, acceptance sets, and the stationary distribution through the estimated model’s own fixed point.

The offer-composition margin. A companion condition allows entrants’ common capital choice to respond: the location of the offer distribution adjusts so that the F_A -weighted marginal value of capital on the viable set equals its calibrated marginal cost. Because the estimated offer distribution concentrates mass at the ends of its support, the location instrument is an exponential tilt of the fixed support rather than a translation. The tilt is smooth and symmetric, and it nests the estimated distribution exactly. In practice this margin is weakly determined away from the baseline: at counterfactual points the condition is flat within numerical precision over the economically sensible range, so the reported closure imposes free entry only, and the location margin is treated as a bounded sensitivity check (Appendix G). The shape of F_A beyond its location is held fixed throughout; endogenizing it requires primitive heterogeneity in firm scale, which the model excludes, and companion work takes up this margin.

The closure disciplines the counterfactuals in Section VI rather than amplifying them: at the estimated parameters the entry-surplus integral responds weakly to every policy lever we consider, tightness stays within a few percent of baseline, and partial-equilibrium results carry over nearly unchanged. We verify this property of the estimated economy directly rather than assuming it.

E. Empirical Details

E.1 Age profiles and employer-tenure comparisons

Figure 1 uses PNADC observations from 2012–2019 for persons aged 18–65. The three labor-market states are formal private employment with a signed work card, own-account self-employment, and unemployment under the ILO definition. The displayed age groups

¹⁶This is the convention of the bargaining literature closest to ours, in which the firm’s breakdown option is an empty vacancy with no value by free entry. Positive position values arise in fixed-firm-measure environments with sunk expansion costs; their structural analogue here is the liquidation value a bounding the firm’s loss.

are 20–24, 25–34, 35–44, 45–54, and 55–65. Stocks and transitions use PNADC sampling weights.

The self-employment share is the weighted number of own-account person-quarter observations divided by the weighted number of person-quarter observations in the three states within the same age group. It does not require observation in the following quarter. The entry rate is the weighted number of formal employees who report own-account work in the next quarter divided by the weighted number of formal employees observed in one of the three states in that next quarter. Age and weights are measured in the origin quarter. Transition pairs must be consecutive, with both interviews in 2012–2019 and both ages within 18–65. Informal employees, employers, other employment, and nonparticipation are outside these denominators.

Standard errors in Figure 1 use 200 bootstrap draws of individuals, retaining each sampled individual’s complete contribution to the weighted numerators and denominators. The figure reports point estimates plus or minus 1.96 bootstrap standard errors. The age-specific shares are observed cross-sectional shares; they are distinct from the aggregate stationary shares implied by the transition matrix that enter the estimation criterion.

Figure 2 applies the same entry definition and age groups to formal private employees whose origin jobs are in manufacturing divisions 10–33 and whose origin ages are 20–65. The sample contains 312,448 transitions from 141,267 workers, including 5,653 transitions into own-account work. The industry groups in panel (a) follow the PIA-Empresa ranking defined in Appendix E.9. Panel (b) groups establishments with 1–10 workers and those with at least 11 using origin-quarter PNADC variable V4018. Excluding 29,536 observations with unknown size leaves 282,912 transitions, including 5,052 entries into own-account work. Before restricting to manufacturing, the reconstructed entry rates reproduce all five cells in Figure 1 to numerical precision.

For each cell in Figure 2, standard errors are based on the weighted-ratio influence function and two-way clustering by the origin primary sampling unit (UPA) and two-digit industry. The variance adds the two cluster contributions and subtracts the contribution of their intersections, applying the finite-cluster correction to each. Intervals use Student- t critical values with 11 degrees of freedom for the investment cells, which each span 12 industries, and 23 for the size cells, which span all 24. This procedure allows dependence within survey sampling units and industries; it does not reproduce the full stratified survey variance estimator. The industry ranking is treated as fixed.

The increase in entry between ages 20–24 and 55–65 is 2.34 percentage points in establishments with 1–10 workers and 0.48 points in larger establishments. Their difference is 1.86 points, with a 95 percent confidence interval of [0.43, 3.30], retaining covariance across cells. Extending the same comparison to all sectors yields a difference of 1.08 points, with

an interval of $[0.10, 2.06]$. Within manufacturing, the weighted share with unknown size ranges from 9.30 to 9.84 percent across age groups. Restricting panel (a) to observations with known size changes each entry rate by at most 0.11 percentage points and preserves the difference in age profiles.

Tenure profiles. We examine how entry differs across the same employer groups at different durations of the current employment relationship. Previous work relates self-employment entry to current-job tenure (Evans and Leighton, 1989; Elfenbein, Hamilton, and Zenger, 2010). Figure 5 retains the formal-private manufacturing sample and destination definition of Figure 2, grouping current-employer tenure into less than three years, three to less than ten years, and ten or more years. There are no missing tenure values. Each rate pools weighted entries and weighted observations within its cell; confidence intervals use the same clustering and critical values as the age profiles above.

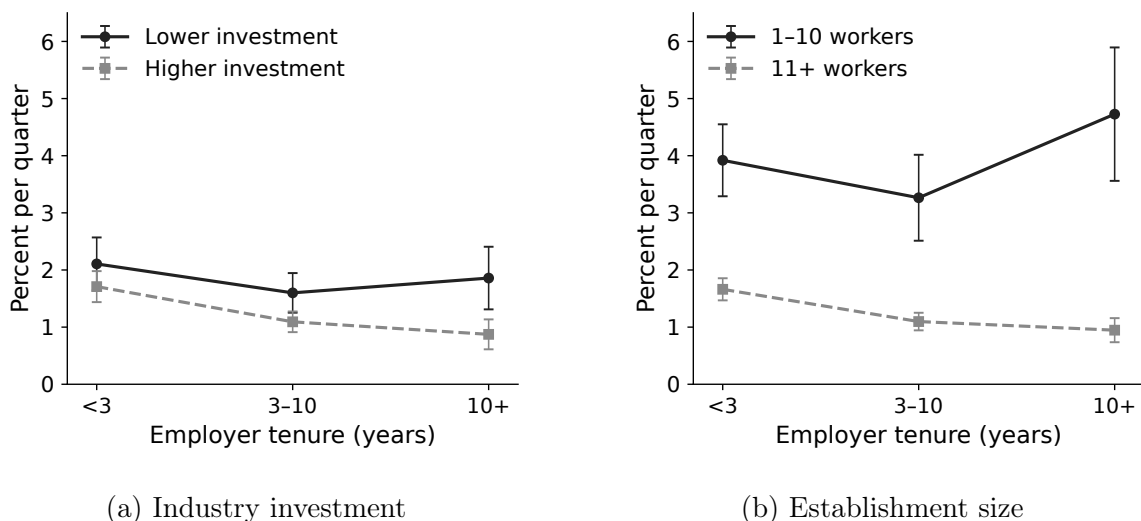


Figure 5: Entry into own-account work by tenure and employer characteristics

Notes: The figure reports sampling-weighted quarterly transitions from formal private employment in manufacturing to own-account work. Destinations are restricted to formal private employment, own-account work, or unemployment. Tenure, industry, and establishment size are measured in the origin interview. Tenure groups are less than three years, three to less than ten, and ten or more. Industry groups follow the PIA-Empresa investment ranking in Appendix E.9. Panel (a) contains 312,448 transitions; panel (b) retains 282,912 with reported establishment size. Both interviews lie in 2012–2019 and satisfy the age restriction of 18–65, with origin ages of 20–65. Vertical bars are pointwise 95 percent confidence intervals based on standard errors clustered by primary sampling unit and two-digit industry. Source: PNADC and PIA-Empresa, 2012–2019.

Panels (a) and (b) show that differences between employer groups are also larger at long tenure than at intermediate tenure. Comparing tenure of three to less than ten years with tenure of ten or more years, the entry-rate gap between lower- and higher-investment industries widens from 0.51 to 0.99 percentage points. At small establishments, entry rises from 3.26 to 4.73 percent; at larger establishments, it falls from 1.10 to 0.95 percent.

The change between the intermediate and longest tenure groups is 0.48 percentage points larger in lower-investment industries than in higher-investment industries, with a 95 percent confidence interval of [0.12, 0.83]. It is 1.61 points larger in small establishments than in larger establishments, with an interval of [1.00, 2.22]. Both comparisons retain covariance across cells and use 23 degrees of freedom, reflecting the 24 industries represented in each between-group contrast.

To account for differences in worker age, we estimate weighted linear probability models separately within each employer group, including the three tenure indicators, single-year age effects, origin-quarter effects, and two-digit-industry effects. Inference uses the same clustering convention and retains covariance between groups. The adjusted difference in tenure changes is 0.42 points for industry investment, with an interval of [0.08, 0.76], and 1.44 points for establishment size, with an interval of [0.85, 2.03]. Within lower-investment industries, the adjusted change is approximately zero (-0.05 points), compared with -0.47 points in higher-investment industries. The comparison therefore concerns divergence between employer groups, rather than an increase in entry within every group.

An alternative grouping uses less than five years, five to less than ten years, and ten or more years, retaining the same observations. The unadjusted differences in tenure changes are 0.48 points across investment groups, with an interval of [0.11, 0.85], and 1.73 points across size groups, with an interval of [0.82, 2.65].

The quantitative estimation uses the all-sector three-state sample underlying Figure 1.¹⁷ The additional employer comparisons described in Appendix E.4 use a broader population and do not condition on remaining in these three states.

E.2 Previous employment and startup finance in ECINF

Table 1 uses the proprietor records in IBGE’s 2003 ECINF. The primary population comprises own-account owner–business records, identified by V4203 equal to 1, without a restriction on the survey’s business-informality classification. The employment-history comparison retains owners who became proprietors of the current business less than five years before the survey date, October 31, 2003 (V4403). This is an ownership-duration restriction, not a restriction on the founding date of the business. The previous-job questions are asked only within this window. We select respondents who report leaving a

¹⁷Over 2012–2019 and ages 18–65, the three states cover 49 percent of the population; informal private employees are 7.5 percent, employers 2.8, other employment 14.0, and nonparticipation 26.8. The same destination conditioning is applied to the model, so the targets and their model counterparts are constructed identically. The conditioning matters most for unemployment origins: without it, the unemployment-to-self-employment and unemployment-to-formal rates are 38 percent smaller (0.086 against 0.138; 0.101 against 0.163), because 38 percent of destinations from unemployment lie outside the universe, mostly nonparticipation; the formal-to-self-employment rate changes by a tenth. The full six-state transition matrix is in the replication output.

job before entering the business (V4404 equal to 1), who were employees (V4406 equal to 2), and whose job was in a private enterprise (V4407 equal to 4). V4408 distinguishes jobs with and without a signed work card. V4410 equal to 2 identifies reported dismissal; it does not identify whether dismissal was with or without just cause. Other exits comprise all other valid reasons. Missing exit reasons exclude 19 former formal employees and eight former informal employees from the four-column comparison.

V4402 records the principal source of the capital needed to start the business. It does not report the amount raised, all financing sources used, or FGTS receipts separately, and does not establish that compensation came from the reported previous job. Table 1 excludes respondents reporting that no capital was needed (code 9). It preserves compensation received (code 1), previous saving or asset sales (code 3), and other own resources (code 4), and pools borrowing from relatives or friends, banks, and other lenders (codes 5–7). Other lenders include businesses or individuals advancing money or discounting postdated checks. The residual category combines inheritance, partner-supplied capital, other sources, and unreported responses. Partner-supplied capital remains in the denominator because capital was available to the business even though the respondent did not supply it. Owners can report more than one business, so the observational unit is an owner–business record. The main estimates use survey weights for business owners (PESOPES); alternative calculations restrict to owners reporting a single business or divide each owner’s survey weight across their reported own-account businesses.

Standard errors account for the survey’s stratification and clustering within census sectors, the geographic sampling units. We use Taylor linearization, which approximates each estimate’s sampling variation from the contributions of these units. Strata combine state, metropolitan status, and the published stratum code. The calculation retains all 2,499 sampled sectors in 185 strata, including sectors with zero contribution to a particular comparison, and applies no finite-population correction. The number of observed sectors in each stratum matches the number recorded in the survey design file. Standard errors for differences between shares retain their covariance by differencing the sector-level contributions.

Table 9 reports the distribution before excluding respondents who needed no startup capital. It also separates bank loans from borrowing from relatives or friends. The subsequent robustness and unadjusted tenure profiles apply the same exclusion as Table 1. The adjusted tenure comparisons report both denominators.

Table 9: Startup finance including businesses needing no capital

Financing source	Previous formal job		Previous informal job	
	Dismissed	Other exit	Dismissed	Other exit
Compensation received	24.4 (1.7)	16.8 (1.4)	4.7 (1.2)	5.5 (1.0)
Prior saving or asset sales	4.0 (0.8)	6.2 (0.8)	5.4 (1.7)	8.8 (1.1)
Other own resources	25.0 (1.6)	27.9 (1.6)	31.2 (3.3)	33.3 (1.9)
Loan from relatives or friends	5.6 (0.8)	7.9 (1.1)	12.7 (3.0)	13.9 (1.6)
Bank loan	1.2 (0.3)	2.4 (0.5)	1.3 (0.5)	2.1 (0.5)
No startup capital needed	35.2 (1.9)	31.4 (1.8)	37.4 (3.8)	28.0 (2.0)
Other sources or unreported	4.6 (0.6)	7.4 (1.1)	7.2 (1.5)	8.4 (1.3)
Observations	1,895	2,425	599	1,608

Notes: Entries are percentages, with standard errors in parentheses accounting for survey stratification and clustering within census sectors. The sample consists of own-account owner–business observations in ECINF 2003 whose owners became proprietors less than five years before the survey and previously held a private wage job. Estimates use survey weights and include respondents reporting that no startup capital was needed. Other sources or unreported includes inheritance, other lenders, partner-supplied capital, other sources, and unreported responses. Source: ECINF, 2003.

Table 10 compares the compensation-financing shares under alternative ownership-duration restrictions and weights. The single-business restriction excludes owners who report more than one business; person-normalized weights divide the owner’s survey weight by the number of reported own-account businesses. The business-weighted calculation uses the survey weight for businesses (PESOUNI). All rows other than the shorter-duration sample retain the less-than-five-year ownership window.

Table 10: Compensation financing: alternative samples and weights

Sample or weights	Formal, dismissed	Informal, dismissed	Difference (s.e.)
Under five years (baseline)	37.7	7.5	30.1 (3.0)
Under two years	36.8	9.6	27.3 (4.6)
Single-business owners	37.4	7.4	30.0 (3.0)
Person-normalized weights	37.5	7.5	30.1 (3.0)
Business weights	36.9	7.4	29.5 (3.0)

Notes: The first two columns report the percentage naming compensation received as the principal source of startup capital, among owners who report dismissal from a previous private wage job. Respondents reporting that no startup capital was needed are excluded. The final column is the formal-minus-informal difference in percentage points, with its standard error. Standard errors account for survey stratification and clustering within census sectors. Source: ECINF, 2003.

Compensation financing is 30.1 percentage points more common following dismissal from formal rather than informal private employment (standard error 3.0 points). The difference remains 27.3 points (standard error 4.6) when ownership duration is restricted to less than two years. The single-business restriction and alternative weights yield similar differences.

Table 11 divides each of the four previous-employment groups by tenure in the previous job, measured by V4409. The categories are less than one year, one to less than five years, and at least five years.

Table 11: Compensation financing and previous-job tenure: ECINF 2003

Previous-job tenure	Previous formal job		Previous informal job	
	Dismissed	Other exit	Dismissed	Other exit
Less than one year	16.1 (3.7)	11.1 (2.3)	6.7 (3.5)	5.3 (1.6)
Observations	183	268	117	287
One to less than five years	36.4 (3.2)	24.6 (3.1)	7.3 (2.4)	7.9 (1.8)
Observations	631	890	193	570
At least five years	45.6 (4.1)	30.5 (3.7)	9.9 (4.9)	9.5 (3.2)
Observations	493	586	64	319

Notes: Shares report the percentage naming compensation received as the principal source of startup capital. The sample comprises own-account owner–business observations with ownership duration below five years whose owners previously held a private wage job and report a reason for departure, excluding respondents reporting that no startup capital was needed. Formality and departure groups follow Table 1. Estimates use survey weights; standard errors in parentheses account for survey stratification and clustering within census sectors. N is the unweighted number of observations. Source: ECINF, 2003.

The increase in compensation financing with tenure is larger following formal employment, both after dismissal and after other departures. Among owners dismissed from formal jobs, the share rises from 16.1 to 45.6 percent; after other formal departures, it rises from 11.1 to 30.5 percent. The corresponding endpoint increases after informal employment are 3.2 and 4.2 percentage points. These comparisons are retrospective and conditional

on operating an own-account business at the survey date. They do not measure entry probabilities among all previously employed workers or identify the effect of compensation on entry.

Comparison with the 1997 survey. We repeat the financing comparison using the 1997 ECINF (IBGE, 1999). The earlier survey records previous employment for owners who entered the business less than five years before the survey, rather than those who became owners within five years. We retain own-account businesses whose owners previously worked as private employees and no longer held that job at the interview. Unlike the 2003 question, the 1997 question does not establish that the previous job ended before business entry. The two surveys therefore provide closely comparable employment-history samples, rather than identical cohorts of entrants.

Table 12 uses the same financing categories and signed-card distinction as Table 1. Business-entry duration is measured by V6408; V6409, V6411, and V6412 identify previous private employment, V6413 records signed-card status, V6414 identifies an ended job, and V6417 distinguishes dismissal from other exits. The financing-source variable is V6402. We exclude only reports that no capital was needed and use survey weights for business owners (PESOPES).

Standard errors follow the same survey-design adjustment as in 2003, retaining all 2,338 census sectors observed in the household file within the 155 survey strata, including sectors with no contribution to a particular comparison. The declared design contains 2,340 sectors; one sector is absent from each of two strata. A sensitivity calculation adds the absent sectors with zero contribution, without assuming that this resolves the discrepancy. Across the financing and tenure estimates and contrasts, this changes standard errors by less than 0.002 percentage points.

Table 12: Startup finance and previous employment: ECINF 1997

Financing source	Previous formal job		Previous informal job	
	Dismissed	Other exit	Dismissed	Other exit
Compensation received	41.4 (2.7)	28.6 (2.3)	10.6 (2.2)	8.7 (1.4)
Prior saving or asset sales	13.4 (1.9)	14.6 (2.0)	16.3 (3.0)	20.3 (2.6)
Other own resources	29.1 (2.4)	34.9 (2.7)	45.2 (5.8)	39.1 (2.5)
Borrowing	9.5 (1.3)	13.6 (1.4)	18.7 (3.7)	22.1 (2.4)
<i>of which:</i> Bank loans	0.5 (0.2)	1.1 (0.3)	1.8 (1.5)	0.9 (0.3)
<i>of which:</i> Relatives or friends	7.5 (1.2)	11.1 (1.3)	14.9 (3.4)	18.6 (2.4)
<i>of which:</i> Other lenders	1.6 (0.4)	1.5 (0.3)	2.0 (1.0)	2.6 (0.6)
Other sources or unreported	6.5 (1.2)	8.3 (1.3)	9.2 (2.4)	9.8 (1.5)
Observations	1,566	2,061	377	1,216

Notes: Percentages with standard errors in parentheses. The sample comprises own-account owner–business observations whose owners entered the business less than five years before the survey and previously held a private wage job that had ended by the interview. Respondents reporting no startup capital needed are excluded; their weighted shares of the original groups are 35.3, 33.3, 33.8, and 42.9 percent, respectively. Partner-supplied capital and unreported sources remain included. Entries describe principal financing sources, not shares of capital invested. Borrowing comprises the three indented categories; components may not sum after rounding. Formal jobs have a signed work card; reported dismissal does not distinguish just cause. Compensation does not identify FGTS receipts separately. Estimates use survey weights and standard errors accounting for survey stratification and clustering within census sectors. Source: Author’s calculations from ECINF 1997 (IBGE, 1999).

Among owners reporting dismissal from a previous formal job, 41.4 percent name compensation as their principal source of startup capital, compared with 28.6 percent among those reporting other exits from formal jobs. The difference, computed before rounding, is 12.9 percentage points, with a 95 percent confidence interval of [6.1, 19.6], compared with 13.2 points in 2003. The corresponding shares for previous informal jobs are 10.6 and 8.7 percent. Among dismissed workers, the formal–informal difference is 30.9 percentage points (standard error 3.7), close to the 30.1-point difference in 2003. Bank loans account

for only 0.5–1.8 percent across the four groups. The greater reliance on compensation among owners previously employed formally thus appears in both surveys.

Table 13 divides the four previous-employment groups by completed previous-job tenure, measured by V6416, using the same categories as the 2003 comparison. Two observations with missing or invalid tenure are excluded from the informal-other-departure group; neither names compensation as the principal startup source.

Table 13: Compensation financing and previous-job tenure: ECINF 1997

Previous-job tenure	Previous formal job		Previous informal job	
	Dismissed	Other exit	Dismissed	Other exit
Less than one year	39.3 (5.9)	19.2 (5.2)	8.2 (3.3)	5.8 (2.4)
Observations	205	259	99	296
One to less than five years	37.6 (4.6)	27.0 (3.3)	14.1 (3.4)	8.3 (1.7)
Observations	731	959	210	579
At least five years	47.1 (3.5)	32.4 (3.3)	3.5 (1.9)	12.3 (3.3)
Observations	630	843	68	339

Notes: Percentages naming compensation received as the principal source of startup capital, among owners with a previous private wage job and a reported reason for departure. The sample and inference follow Table 12, with missing tenure excluded. N is the unweighted number of owner–business observations. Source: Author’s calculations from ECINF 1997.

The formal-dismissal tenure profile is less pronounced than in 2003: compensation shares are 39.3, 37.6, and 47.1 percent across the three tenure groups. The difference between at least five years and less than one year is 7.7 percentage points, with a 95 percent confidence interval of $[-5.6, 21.1]$. After other formal departures, the corresponding shares are 19.2, 27.0, and 32.4 percent. The earlier survey reproduces the financing contrast by previous-job formality, but not the monotone formal-dismissal tenure profile observed in 2003.

Tenure comparisons adjusted for age. We next ask whether the tenure gradients differ between formal and informal employment after accounting for owners’ ages. The outcome is an indicator for compensation as the principal startup source. We compare previous-job tenure of at least five years with tenure below one year, retaining a separate indicator for the middle category. Table 14 reports these contrasts from survey-weighted linear probability regressions estimated separately by survey year and previous-job formality. Each regression includes five-year age indicators and an indicator for reported dismissal. Age effects and the dismissal coefficient may differ by formality; the tenure coefficients are common across dismissal and other departures within each formality group.

The table reports the denominator used in Table 1 and a comparison retaining owners who needed no startup capital. We also restrict ages to 25–54, where short- and long-tenure jobs

are better represented than at older ages. Standard errors apply the same survey-design adjustment to the regression estimates, using the contributions of census sectors within each survey stratum. Differences between coefficients retain their covariance across groups. The age variables are V2803 in the 2003 household-member file and IDADE in the 1997 proprietor file; no observation in the tenure samples is lost for missing age.

Table 14: Tenure gradients in compensation financing, adjusted for age and dismissal

Sample	Formal	Informal	Difference	<i>N</i>
<i>2003</i>				
Exclude no-capital reports, all ages	23.5 (3.8)	2.2 (3.0)	21.3 (4.8)	4,601
Exclude no-capital reports, ages 25–54	20.4 (4.1)	2.5 (3.6)	18.0 (5.3)	3,786
All startup responses, all ages	18.6 (2.8)	1.4 (2.2)	17.2 (3.5)	6,527
All startup responses, ages 25–54	17.0 (3.0)	1.5 (2.8)	15.4 (4.0)	5,250
<i>1997</i>				
Exclude no-capital reports, all ages	10.7 (4.7)	3.6 (3.1)	7.1 (5.7)	5,218
Exclude no-capital reports, ages 25–54	8.7 (5.5)	3.0 (3.5)	5.6 (6.7)	4,139
All startup responses, all ages	14.0 (3.1)	3.0 (2.0)	11.0 (3.8)	7,581
All startup responses, ages 25–54	13.3 (3.6)	2.8 (2.3)	10.5 (4.4)	5,882

Notes: Entries are percentage-point contrasts between previous-job tenure of at least five years and less than one year. Regressions include an indicator for tenure of one to less than five years, five-year age indicators, and a dismissal indicator, with all coefficients allowed to differ by previous-job formality. The difference column subtracts the informal contrast from the formal contrast. The capital-needed sample excludes only reports that no startup capital was needed; all owners retains those responses. Other cohort restrictions follow Tables 1 and 12. Standard errors in parentheses account for survey stratification and clustering within census sectors. *N* is the number of owner–business observations across both formality groups. Source: Author’s calculations from ECINF 2003 and 1997.

With the capital-needed restriction, the 2003 tenure contrast is 23.5 percentage points after formal employment and 2.2 after informal employment. Their difference is 21.3 points, with a 95 percent confidence interval of [11.9, 30.6]. The corresponding 1997 contrasts are 10.7 and 3.6 points, giving a difference of 7.1 points with an interval of [−4.2, 18.3]. Retaining owners who needed no startup capital gives formal–informal differences of 17.2 points in 2003 and 11.0 in 1997, with intervals of [10.3, 24.1] and [3.4, 18.5]. The 1997 conclusion therefore depends on the denominator. The restricted denominator describes principal-source composition after excluding no-capital reports; retaining those responses

describes the frequency of naming compensation as the principal startup source across the full employment-history sample.

Table 15 allows tenure gradients to differ across the four employment-history groups. Each group has its own tenure coefficients and five-year age indicators. The table reports the four tenure contrasts, the formal–informal differences by departure type, and the difference between dismissal and other departures within formal employment. The final row asks whether that dismissal–other difference is larger following formal than informal employment.

Table 15: Age-adjusted tenure gradients by previous employment and departure

	2003		1997	
	Capital needed	All owners	Capital needed	All owners
Formal, dismissed	25.5 (5.7)	20.3 (4.1)	5.7 (6.8)	12.8 (4.8)
Formal, other exits	19.3 (4.4)	15.8 (3.6)	13.9 (6.4)	14.7 (4.3)
Informal, dismissed	0.0 (7.2)	-0.3 (5.1)	-0.4 (4.7)	-0.6 (2.5)
Informal, other exits	2.4 (3.5)	1.7 (2.6)	4.9 (3.8)	4.1 (2.5)
Formal minus informal: dismissed	25.5 (9.1)	20.6 (6.6)	6.1 (8.2)	13.4 (5.4)
Formal minus informal: other exits	16.8 (5.6)	14.0 (4.4)	9.0 (7.4)	10.6 (4.9)
Dismissed minus other exits: formal	6.2 (6.9)	4.6 (5.5)	-8.2 (9.6)	-1.9 (6.6)
Triple interaction	8.7 (10.6)	6.6 (8.1)	-2.9 (11.2)	2.8 (7.4)

Notes: Entries are percentage-point contrasts between tenure of at least five years and less than one year, from separate survey-weighted regressions for each survey year and employment-history group. Each regression includes an indicator for the middle tenure category and five-year age indicators. Capital needed excludes only owners reporting no startup capital needed; all owners retains them. All valid ages are included. The final row is the formal-minus-informal difference between the dismissal and other-departure tenure contrasts. Standard errors in parentheses account for survey stratification and clustering within census sectors, preserving covariance between group estimates. Source: Author’s calculations from ECINF 2003 and 1997.

The 2003 capital-needed sample shows larger tenure gradients following formal employment after both dismissal and other departures. The additional gradient after dismissal within formal employment is 6.2 percentage points, with a 95 percent interval of $[-7.3, 19.7]$. In 1997 it is -8.2 points, with an interval of $[-26.9, 10.6]$. Retaining owners who needed no startup capital gives differences of 4.6 and -1.9 points. These comparisons do not establish an additional tenure gradient specific to dismissal in either wave.

A narrower departure comparison. The other-departure category includes fixed-term contracts, retirement, and several personal and job-related reasons. We restrict it to reports of personal or family reasons, not adapting to the job, low pay, or successful secondary work (codes 4, 5, 6, and 8 in both surveys). These reasons are consistent with voluntary departure. They are mutually exclusive alternatives to reported dismissal in the question asking for one principal reason for leaving; they do not independently verify the legal form of termination. The employment-history and financing restrictions otherwise remain unchanged.

Table 16 compares the formal-dismissal tenure gradient with the gradient in this narrower departure group. It also reports the difference between the dismissal–voluntary contrasts following formal and informal employment. Panel A retains all valid ages, panel B restricts ages to 25–54, and panel C excludes successful secondary work from the voluntary category because it describes a pre-existing activity that was already doing well. The all-age voluntary groups contain 1,427 formal and 1,067 informal observations in 2003 and 1,987 formal and 1,314 informal observations in 1997 before excluding owners who needed no startup capital.

Table 16: Tenure contrasts using reported reasons consistent with voluntary departure

	2003		1997	
	Capital needed	All owners	Capital needed	All owners
<i>A. Primary definition, all ages</i>				
Dismissed minus voluntary: formal	7.7 (7.8)	6.6 (6.0)	-14.3 (9.5)	-4.4 (6.5)
Triple interaction	9.7 (11.1)	8.2 (8.4)	-8.9 (11.5)	0.1 (7.5)
<i>B. Primary definition, ages 25–54</i>				
Dismissed minus voluntary: formal	9.7 (8.2)	8.3 (6.3)	-15.7 (10.5)	-4.9 (7.2)
Triple interaction	17.1 (11.4)	14.8 (8.6)	-14.2 (12.0)	-1.8 (7.9)
<i>C. Excluding successful secondary work, all ages</i>				
Dismissed minus voluntary: formal	8.4 (8.0)	8.2 (6.0)	-13.2 (9.5)	-3.1 (6.5)
Triple interaction	10.1 (11.2)	9.5 (8.3)	-4.9 (11.3)	3.0 (7.4)

Notes: Within each departure group, the tenure contrast compares at least five years with less than one year and adjusts for five-year age groups. Entries are differences between these contrasts, in percentage points. The primary voluntary category includes personal or family reasons, poor fit with the job, low pay, and successful secondary work; panel C excludes the last reason and retains all valid ages. The triple interaction subtracts the informal dismissal–voluntary contrast from the formal contrast. The two denominators follow Table 14. Standard errors in parentheses account for survey stratification and clustering within census sectors. Source: Author’s calculations from ECINF 2003 and 1997.

Among all owners, the formal-dismissal tenure contrast exceeds the voluntary-departure contrast by 6.6 percentage points in 2003, with a 95 percent interval of $[-5.1, 18.4]$. In 1997 the difference is -4.4 points, with an interval of $[-17.1, 8.4]$. Restricting ages or excluding successful secondary work leaves the same ordering in each year. Narrowing the departure category therefore does not establish the additional dismissal-related gradient. The tenure evidence connects the duration of formal employment to startup financing more clearly than it distinguishes dismissal from voluntary departure.

E.3 Prior wages and entry from formal employment

We examine whether entry into own-account work varies with the wage received before the transition. The sample follows Figure 1: origins are formal private employees aged 18–65, both observations are in consecutive quarters within 2012–2019, and the primary

denominator retains destinations in formal private employment, own-account work, or ILO unemployment. We additionally require positive habitual monthly earnings in the origin job. Wage groups are origin-weighted terciles within calendar year, with ties kept together. The same cutoffs are used for the alternative denominator that retains all observed destination states.

Table 17 reports the raw entry rates and probabilities adjusted for observed worker and job characteristics. Adjustment uses a weighted linear probability model with single-year age, tenure-band, education-year, two-digit occupation, two-digit industry, and calendar-quarter fixed effects, together with sex. Adjusted probabilities average predictions over the common sample’s covariate distribution. All raw and adjusted comparisons within a panel use the same observations; the change in the gradient does not arise from dropping observations with missing controls. Standard errors for the top-minus-bottom contrast are clustered by the origin primary sampling unit.

Table 17: Prior wages and quarterly entry into own-account work

	Bottom Entry probability	Middle (percent)	Top	Top – bottom (percentage points)	Observations
<i>A. Destinations: formal employment, own-account work, or unemployment</i>					
Unadjusted	2.24	2.27	2.42	+0.18 (0.05)	1,532,607
Adjusted	2.47	2.19	2.24	−0.23 (0.05)	1,532,607
<i>B. All observed destinations</i>					
Unadjusted	1.98	2.09	2.24	+0.26 (0.04)	1,703,723
Adjusted	2.20	2.01	2.07	−0.13 (0.05)	1,703,723

Notes: PNADC 2012–2019. Origins are private employees with a signed work card; both quarterly observations are at ages 18–65 and within 2012–2019. Entry denotes own-account work in the next quarter. Main-job habitual monthly earnings must be positive at the origin. Wage groups use origin-weighted terciles within calendar year, with common cutoffs across panels and ties kept together. Panel A conditions on the three labor-market states used in Figure 1; Panel B retains every observed destination. Adjusted probabilities are standardized predictions from a weighted linear probability model with single-year age, tenure-band, education-year, two-digit occupation, two-digit industry, and calendar-quarter fixed effects, plus sex. All rows within a panel use the same observations and origin sampling weights. Parentheses report standard errors clustered by the origin primary sampling unit (UPA) for the top-minus-bottom contrast; these are not the complete stratified survey variance estimator. The comparisons are descriptive and do not identify a causal wage effect or the prevalence of voluntary quits.

Within the three-state denominator, the raw top-minus-bottom difference is 0.18 percentage points. After adjustment, it is -0.23 points, with a 95 percent confidence interval of $[-0.33, -0.12]$. Retaining all observed destinations yields a raw difference of 0.26 points and an adjusted difference of -0.13 points, with a confidence interval of $[-0.23, -0.03]$. Prior wages therefore do not yield a robust positive conditional association with entry. These descriptive comparisons do not isolate the effect of accumulated wealth from earnings opportunities, nor do they identify whether a transition was voluntary.

E.4 Data and transition sample for employer comparisons

The PNADC is a nationally representative household survey conducted by Brazil’s national statistical agency, IBGE. We use quarterly observations from 2012 through 2019 and retain consecutive interviews for the same individual. Origin quarters run from 2012Q1 through 2019Q4; destinations for 2019Q4 origins are observed in the 2020Q1 interview. Regressions and weighted means use the origin-quarter PNADC sampling weight (V1028); results are invariant to the weight scale.

The additional employer comparisons below use a common sample and outcome. The origin population is formal and informal private wage employees aged 15–64, unlike the narrower sample behind Figures 1 and 2. A transition is retained when the same individual is observed in the immediately following quarter, irrespective of the destination state. Entry into self-employment means entry into own-account work; employers are excluded from this outcome. Age, current-job tenure, establishment size, industry, occupation, earnings, and the sampling weight are measured in the origin quarter. This construction yields 2,433,787 transitions and a sampling-weighted mean quarterly transition rate of 4.705 percent.

A transition to a new wage job requires wage employment in the following quarter and reported tenure of at most three months in that job. Nonparticipation denotes being outside the labor force. Origin-job tenure is constructed from reported months and years and is top-coded at 35 years. Establishments with 1–10 workers are classified as small; establishments with at least 11 workers form the comparison group.¹⁸ The age bins are 15–24, 25–34, 35–44, 45–54, and 55–64. The eleven tenure bins displayed in Appendix Figures 7 and 9 are less than one month, one to six months, six to twelve months, one to two years, two to three years, three to five years, five to seven years, seven to ten years, ten to fifteen years, fifteen to twenty years, and twenty or more years. The six-bin profiles in Figures 8 and 6 use less than one year, one to two years, two to three years, three to seven years, seven to ten years, and ten or more years.

¹⁸The PNADC questionnaire asks for the number of persons working at the business counting the respondent, so the establishment-size category labels follow the official coding of 1–5, 6–10, 11–50, and 51 or more persons.

The industry comparisons retain the 24 two-digit manufacturing divisions covered by IBGE’s Pesquisa Industrial Anual–Empresa (PIA–Empresa). Lower- and higher-investment industries are defined by a median split of average year-demeaned log fixed-asset acquisitions per worker over 2012–2019. Appendix E.9 gives the construction and division ranking. Establishment size and industry investment describe employer characteristics; neither directly measures firm assets or access to finance.

For tenure bin τ and destination d , the reported transition rate is

$$h_{d\tau} = \frac{\sum_i \omega_{it} \mathbf{1}\{D_{i,t+1} = d, \tau_{it} = \tau\}}{\sum_i \omega_{it} \mathbf{1}\{\tau_{it} = \tau\}}, \quad (\text{E1})$$

where ω_{it} is the origin-quarter sampling weight. Destination indicators are mutually exclusive by construction. The raw figures report 95 percent confidence intervals based on standard errors clustered by worker and two-digit origin industry. For the robustness specifications below, we recover the PNADC primary sampling unit (UPA) from the person identifier. We cluster by UPA and two-digit origin industry. A table note states any exception. The reobservation comparisons in Table 18 retain worker-by-industry clustering. Because workers are nested within UPAs, UPA-by-industry clustering also allows arbitrary serial correlation within worker. The full 2012–2019 extract does not retain the survey stratum, so the UPA-by-industry estimates do not reproduce the PNADC’s complete stratified survey variance estimator.

E.5 Scheduled follow-up and reobservation

The employer-comparison sample conditions on observation in the next quarter. We assess selection from this condition using the PNADC rotation schedule. We infer each household’s interview cohort from its first observed quarter, retain the first four scheduled interviews, and exclude households already present in the first quarter of 2012 because their position in the rotation cannot be recovered. This construction yields 2,180,318 wage-employee origin observations with a scheduled subsequent interview. The sampling-weighted follow-up rate is 86.3 percent.

We estimate the probability of follow-up with a linear probability model in age, tenure, employer size, sex, schooling, log wages, formal status, full-time status, industry fixed effects, and state-by-quarter fixed effects. Employer size enters as three category indicators plus an indicator for missing size. Standard errors are clustered by UPA. Because the linear model can produce fitted values outside the unit interval, fitted probabilities are restricted to the interval $[0.05, 0.995]$ before constructing inverse probability weights; 0.43 percent of observations are affected by this restriction. Table 18 compares the principal contrasts with and without the inverse probability adjustment.

Table 18: Robustness to next-quarter reobservation

	PNADC weights	Follow-up IPW
Tenure: 7+ minus 3–7 years	0.194 (0.216)	0.197 (0.218)
Small-minus-large long-tenure change	2.819 (0.551)	2.857 (0.553)
Low-minus-high investment age slope	0.0354 (0.0175)	0.0351 (0.0176)

Notes: Entries are percentage-point differences in quarterly transition probabilities. The tenure row compares at least seven years with three to seven years; the size row compares at least ten years with three to seven years; the third row is an annual age-slope difference. Standard errors in parentheses are clustered by worker and two-digit origin industry. The tenure, size, and industry samples contain 1,867,176, 1,676,773, and 350,982 scheduled-follow-up observations, respectively. Both columns use PNADC sampling weights; the second also uses the inverse estimated probability of next-quarter observation.

E.6 Age and tenure comparisons across employer groups

Table 19 reports how age and tenure patterns differ across employer groups. The four comparisons pair worker age and current-job tenure with establishment size and industry investment. These are descriptive comparisons of employer heterogeneity, rather than direct measures of the effects of financing constraints.

Let $Y_{i,t+1}$ indicate entry into own-account work in the following quarter, X_{it} denote observed controls, μ_{it} collect the fixed effects specified below, and ε_{it} denote the regression residual. Let C_{it} indicate the origin-employer group. In the establishment-size specification, $C_{it} = 1$ for establishments with 1–10 workers and zero for establishments with at least 11 workers. In the industry specification, $C_{it} = 1$ for manufacturing divisions below the median of average year-demeaned log fixed-asset acquisitions per worker over 2012–2019. Appendix E.9 gives the construction and complete division ranking. For the age cells, we estimate

$$Y_{i,t+1} = \beta_a \text{Age}_{it}^c + \delta C_{it} + \gamma \text{Age}_{it}^c C_{it} + X_{it}'\theta + \mu_{it} + \varepsilon_{it}, \quad (\text{E2})$$

where age is centered at 40. The employer-group main effect is absorbed by industry fixed effects in the PIA specification. The reported coefficient is 10γ , expressed as a percentage-point difference in the age gradient per decade. We also replace the linear age terms with five age-bin indicators and their interactions with C_{it} and test the interaction terms jointly.

For the tenure cells, let T_{it}^j denote the six tenure groups used in Figure 8, with three to seven years omitted. We estimate

$$Y_{i,t+1} = \sum_{j \neq 4} \delta_j T_{it}^j + \psi C_{it} + \sum_{j \neq 4} \kappa_j C_{it} T_{it}^j + X_{it}'\theta + \mu_{it} + \varepsilon_{it}. \quad (\text{E3})$$

The reported coefficient is the interaction for tenure of ten or more years, κ_{10+} , which measures the difference between employer groups in the change from three to seven years to ten or more years. It subtracts the change at larger establishments from that at smaller establishments, or the change in higher-investment industries from that in lower-investment industries. We also test all employer-group-by-tenure interactions jointly.

The baseline specifications include sex, schooling, log origin wages, and the other worker-side variable, together with origin-industry, origin-occupation, and origin-quarter fixed effects. The rich specifications add origin-job formality and full-time status, household composition and other household labor income, and state-by-quarter, sector-by-year, and occupation-by-year fixed effects. All regressions use PNADC sampling weights. Standard errors are clustered by UPA and two-digit origin industry. Table 19 reports all four pairings under both specifications.

Table 19: Entry patterns across employer groups: Baseline and rich specifications

Worker variable	Employer measure	Baseline	Joint p	Rich controls	Joint p
Age	Establishment size	1.319 (0.207)	< 0.001	1.015 (0.171)	< 0.001
Age	Industry investment	0.476 (0.162)	0.005	0.394 (0.156)	0.032
Tenure	Establishment size	3.281 (0.600)	< 0.001	2.492 (0.463)	< 0.001
Tenure	Industry investment	1.041 (0.401)	0.022	0.975 (0.409)	0.037

Notes: Age entries are differences in the quarterly-entry gradient per decade; tenure entries are differences in the change from three to seven years to ten or more years. Differences subtract larger establishments from smaller establishments, or higher-investment industries from lower-investment industries. Standard errors are in parentheses. Joint p -values test the complete set of employer-group-by-age-bin or employer-group-by-tenure-bin interactions. The baseline size and investment samples contain 2,158,109 and 417,862 observations; the rich-control samples contain 1,823,611 and 382,253 observations. Relative to the common transition sample, missing occupation codes and wages account for the reduction to the baseline regression samples (2,383,177 tenure and 2,158,109 size observations), and missing full-time status accounts for the further reduction to the rich-control samples (2,014,014 and 1,823,611).

All four comparisons remain positive with the rich controls. The age gradient is 1.01 percentage points per decade steeper at small establishments and 0.39 percentage points steeper in lower-investment industries. The increase from three to seven years of tenure to ten or more years is respectively 2.49 and 0.98 percentage points larger in these groups. Figures 8 and 10 display the underlying tenure-by-size and age-by-industry profiles. Appendix E.12 assesses the corresponding employer differences in the model.

Table 20 replaces the binary employer measures with their more disaggregated counterparts.

Table 20: Alternative measures of origin-employer characteristics

Employer comparison	Age gradient	Long-tenure change
One-SD increase in PIA investment	-0.297 (0.081)	-0.755 (0.217)
1–5 workers relative to 51+	1.435 (0.259)	
6–10 workers relative to 51+	0.604 (0.099)	
11–50 workers relative to 51+	0.214 (0.078)	

Notes: The age column reports percentage-point differences in the quarterly-entry gradient per decade. The tenure column reports the change in the ten-or-more-years contrast associated with a one-standard-deviation increase in PIA investment. The PIA age and tenure coefficients have p -values of 0.001 and 0.002; the PIA tenure-bin interactions are jointly significant with $p < 0.001$. The size-by-age interactions are jointly significant with $p < 0.001$. All specifications use the rich controls and inference described above.

A one-standard-deviation increase in PIA investment is associated with smaller differences along both worker-side measures. Across the four establishment-size categories, the age-gradient difference declines monotonically as establishment size rises. Appendix E.8 reports the corresponding tenure contrasts for all four size categories.

Requiring the destination to persist leaves the pattern in place. When entry is redefined as own-account work in both of the next two quarters, all four interactions remain positive, all four joint tests remain significant at the five percent level, and the coefficients retain one half to three quarters of their magnitudes. The two establishment-size interactions remain individually significant.¹⁹

E.7 Pooled tenure profiles and conditional estimates

We compare entry into own-account work with departures to a new wage job, unemployment, and nonparticipation to examine whether its tenure profile resembles that of other exits. Figure 6 reports sampling-weighted quarterly transition rates across six tenure groups. Each point is a nonparametric bin mean; no duration profile is imposed.

¹⁹The persistent outcome is observable for the two thirds of the sample whose rotation includes a third interview. On that sample, the industry-investment interaction is significant at the ten percent level for age and imprecise for tenure. The tenure cell's joint test is insignificant there under the original outcome and significant under the persistent one, so the lost precision reflects the smaller sample rather than transient spells.

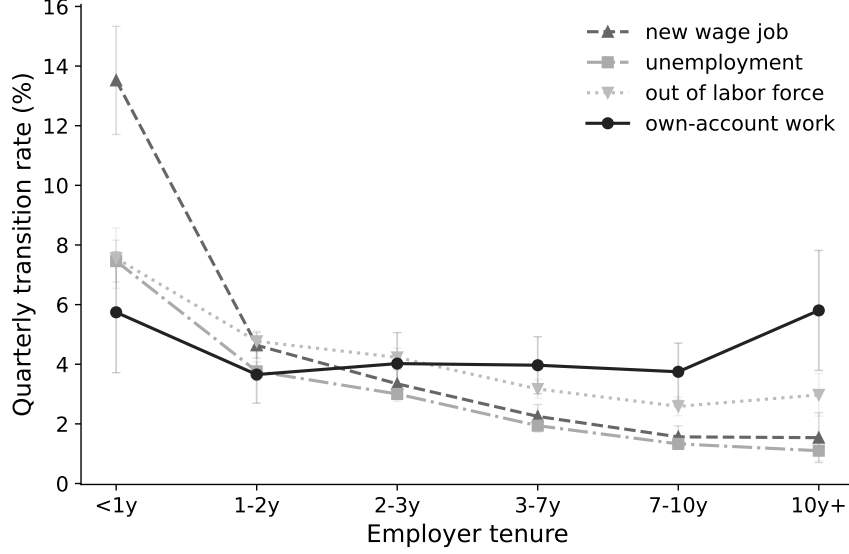


Figure 6: Destination-specific transitions by employer tenure

Notes: The sample contains 2,433,787 transitions among private wage employees aged 15–64 who are observed in the following quarter. Tenure is measured in the origin job. Each series is a sampling-weighted quarterly transition rate. Entry into own-account work excludes employers. Tenure groups are less than one year, one to two years, two to three years, three to seven years, seven to ten years, and ten or more years. Vertical bars are 95 percent confidence intervals based on standard errors clustered by worker and two-digit origin industry. Source: PNADC, 2012–2019.

Entry into own-account work falls from 5.7 percent below one year to 3.7 percent at one to two years, remains near 4 percent through ten years, and rises to 5.8 percent at ten or more years. The high initial rate is concentrated among jobs lasting less than one year. Equal two-year bins retain the initial decline: entry is 5.07 percent at zero to two years, 3.97 at two to four years, and 5.79 at ten to twelve years. Transitions to a new wage job and unemployment decline sharply overall. Nonparticipation falls initially and rises modestly at long tenure. These pooled profiles do not condition on the selected survival of the origin match.

The contrasting exit profiles are compatible with both financial accumulation and entrepreneurial learning. Skills and contacts useful for starting a business can accumulate alongside skills specific to the current employer, which raise the value of remaining in the job.

To examine whether age and current-job tenure contain distinct information, we estimate the grouped specification

$$Y_{i,t+1} = \sum_{g=2}^5 \alpha_g A_{it}^g + \sum_{q=2}^4 \delta_q T_{it}^q + X'_{it} \theta + \mu_{s(i,t)} + \mu_{o(i,t)} + \lambda_t + \varepsilon_{it}, \quad (\text{E4})$$

where $Y_{i,t+1}$ indicates entry into own-account work. The age indicators A^g use ages 15–24 as the omitted group. The tenure indicators T^q group tenure into less than one year, one to three years, three to seven years, and seven or more years, with less than one year omitted. The vector X_{it} contains a female indicator, years of schooling, and log origin wages. The terms μ_s , μ_o , and λ_t denote two-digit origin-industry, two-digit origin-occupation, and origin-quarter fixed effects. We estimate weighted linear probability models and cluster standard errors by worker and two-digit origin industry.

In the full regression sample, the conditional rate at seven or more years is 0.554 percentage points higher than at three to seven years (standard error 0.304; $p = 0.072$; Table 21). This estimate is sensitive to sample composition and additional controls. On the common sample with complete job and household covariates, the baseline contrast falls to 0.206 percentage points (standard error 0.214). Adding formal status, full-time status, household composition, and other household income reduces it to 0.099 percentage points (standard error 0.157). State-by-quarter, sector-by-year, and occupation-by-year fixed effects leave the latter estimate nearly unchanged, as does UPA-by-industry clustering. Table 22 collects these contrasts.

A complementary worker fixed-effect specification compares observations on either side of seven years of current-job tenure. The within-worker contrast is -0.387 percentage points (standard error 0.135) in 148,793 observations. This specification uses a selected set of long matches and does not estimate a general duration effect. It nevertheless provides no support for interpreting the raw recovery as accumulation within the current job. Current tenure conditions on endogenous match survival and may proxy for worker and employer heterogeneity, wage growth, and household circumstances.

The regression tenure groups (less than one year, one to three years, three to seven years, and seven or more years) pool the figure’s one-to-two and two-to-three-year bins, and its seven-to-ten and ten-or-more-year bins. The reported long-tenure contrast therefore uses a different grouping from the figure.

Taken together, these checks imply that the standalone tenure profile should be treated as descriptive. The more stable empirical result is the interaction between long tenure and employer resources documented in Appendix E.8.

Table 21: Joint age and tenure profiles

	Entry into own-account work
Age 25–34	1.841 (0.155)
Age 35–44	2.738 (0.218)
Age 45–54	3.214 (0.289)
Age 55–64	3.724 (0.337)
Tenure 1–3 years	-0.785 (0.091)
Tenure 3–7 years	-0.672 (0.104)
Tenure 7 or more years	-0.118 (0.368)
Tenure 7 or more minus 3–7 years	0.554 (0.304)
Observations	2,383,177
Age indicators jointly zero: p -value	< 0.001
Tenure indicators jointly zero: p -value	< 0.001

Notes: Coefficients and standard errors are multiplied by 100 and hence are percentage-point changes in the quarterly transition probability. Omitted categories are ages 15–24 and tenure below one year. All specifications include sex, schooling, log origin wages, and origin-industry, origin-occupation, and origin-quarter fixed effects. Standard errors in parentheses are clustered by worker and two-digit origin industry. The final tenure row is a linear contrast; its p -value is 0.072.

Table 22: Robustness of the long-tenure contrast

Specification	Contrast	Observations
Full baseline sample	0.554 (0.304)	2,383,177
Common covariate sample	0.206 (0.214)	2,014,014
Job controls	0.099 (0.157)	2,014,014
Household controls	0.099 (0.157)	2,014,014
Rich time interactions	0.099 (0.158)	2,014,014
Rich specification, UPA inference	0.099 (0.158)	2,014,014
Within-worker crossing of seven years	-0.387 (0.135)	148,793

Notes: Entries are percentage-point contrasts between tenure of at least seven years and tenure of three to seven years. The job controls add formal and full-time status. Household controls add the number of household members aged 14 or older, the number of other employed household members, and log other household labor income. Rich time interactions add state-by-quarter, sector-by-year, and occupation-by-year fixed effects. Standard errors in parentheses are clustered by worker and two-digit origin industry except in the row labeled UPA inference and the worker fixed-effect row, which cluster by UPA and industry.

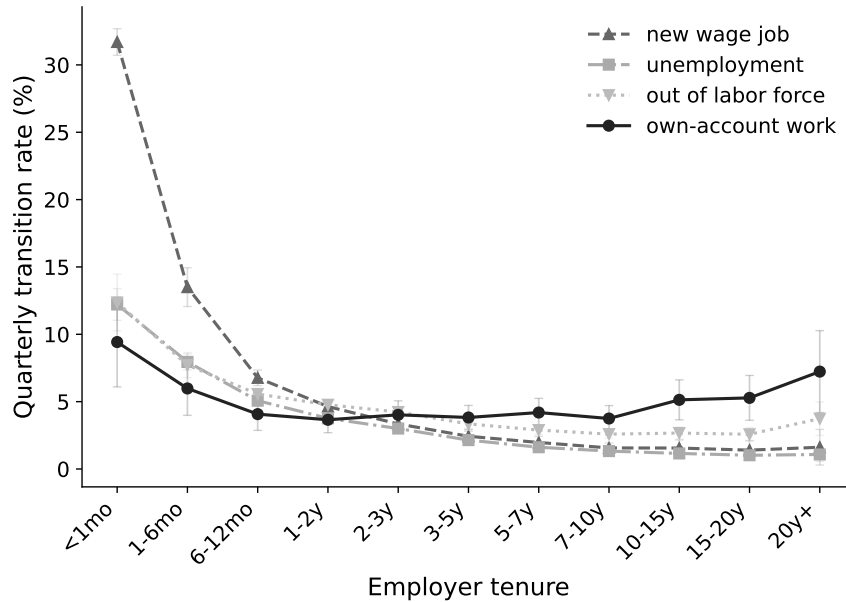


Figure 7: Disaggregated destination-specific transitions by employer tenure

Notes: The figure reports the same sample and outcomes as Figure 6 using eleven tenure bins. Figure 6 pools the three bins below one year, separates tenure of one to two years from tenure of two to three years, and groups the later bins into three to seven, seven to ten, and ten or more years. Vertical bars are 95 percent confidence intervals based on standard errors clustered by worker and two-digit origin industry. Source: PNADC, 2012–2019.

E.8 Employer-size interactions

Figure 8 reports quarterly entry rates across the six tenure groups defined in Appendix E.4. Rates are calculated separately for establishments with 1–10 workers and those with at least 11 workers. Each point is a weighted group mean; no duration profile is imposed.



Figure 8: Entry into own-account work by employer size

Notes: The sample contains 2,204,334 transitions among private wage employees aged 15–64 who are observed in the following quarter and report employer size and tenure. Employer size and tenure are measured in the origin job. Small establishments have 1–10 workers; larger establishments have at least 11. Tenure groups are less than one year, one to two years, two to three years, three to seven years, seven to ten years, and ten or more years. Each series is a sampling-weighted quarterly transition rate into own-account work. Vertical bars are 95 percent confidence intervals based on standard errors clustered by worker and two-digit origin industry. Source: PNADC, 2012–2019.

Figure 8 shows a larger increase in entry at long tenure among employees of small establishments. Entry rates are 7.4 percent at tenure of three to seven years and 12.0 percent at ten or more years in establishments with 1–10 workers. The corresponding rates in larger establishments are 2.3 and 2.7 percent. These are differences across observed employment relationships, whose survival to long tenure may depend on worker and employer characteristics.

To test the size comparison directly, let T_{it}^j denote indicators for tenure of less than one year, one to two years, two to three years, three to seven years, seven to ten years, and ten or more years. We omit three to seven years. Let S_{it} indicate an establishment with 1–10 workers. We estimate

$$Y_{i,t+1} = \sum_{j \neq 4} \delta_j T_{it}^j + \psi S_{it} + \sum_{j \neq 4} \kappa_j S_{it} T_{it}^j + \sum_g \alpha_g A_{it}^g + X_{it}' \theta + \mu_{s(i,t)} + \mu_{o(i,t)} + \lambda_t + \varepsilon_{it}. \quad (\text{E5})$$

The controls and inference follow equation (E4). The sample contains 2,158,109 transitions with reported establishment size. The size-by-tenure interactions are jointly significant ($F = 9.93$, $p < 0.001$). Between tenure three to seven years and tenure of ten or more years, the increase in the transition probability is 3.281 percentage points larger at small establishments (standard error 0.601). The two-way clustered p -value is below 0.001; a restricted wild-cluster test at the two-digit-industry level gives $p = 0.001$.

The binary cutoff does not generate the result. Table 23 reports the change from three to seven years to ten or more years separately for all four PNADC size categories. The increase is concentrated among establishments with 1–5 workers and is smaller among those with 6–10 workers; the estimates are nonpositive for both larger categories. In a binary specification that adds job and household controls and state-by-quarter, sector-by-year, and occupation-by-year fixed effects, the small-minus-large contrast is 2.492 percentage points (standard error 0.463) under UPA-by-industry clustering. Table 18 shows that inverse probability weighting for scheduled follow-up changes the corresponding estimate from 2.819 to 2.857 percentage points. The reobservation comparison re-estimates the baseline specification on the scheduled-follow-up cohort, which is why its unweighted estimate (2.819) differs from the rich-control estimate (2.492).

Table 23: Long-tenure change by establishment size

Origin establishment size	10+ minus 3–7 years	Standard error
1–5 workers	4.009	0.640
6–10 workers	0.844	0.402
11–50 workers	-0.301	0.197
51+ workers	-0.395	0.106
Observations	2,158,109	

Notes: Entries are percentage-point changes in the quarterly probability of entry into own-account work between tenure of three to seven years and tenure of ten or more years. The specification includes the controls and fixed effects in equation (E5). Standard errors are clustered by worker and two-digit origin industry.

Figure 9 reports the eleven-bin profile. The three separate points below one year display the sub-year detail behind the pooled first bin. Figure 8 pools them and reports separate groups for one to two and two to three years.

E.9 Industry investment specifications

We use IBGE’s Pesquisa Industrial Anual–Empresa (PIA–Empresa) to construct an external industry investment measure (IBGE, 2012–2019b). Let I_{st} denote fixed-asset acquisitions and L_{st} average employment in two-digit manufacturing division s and year t .

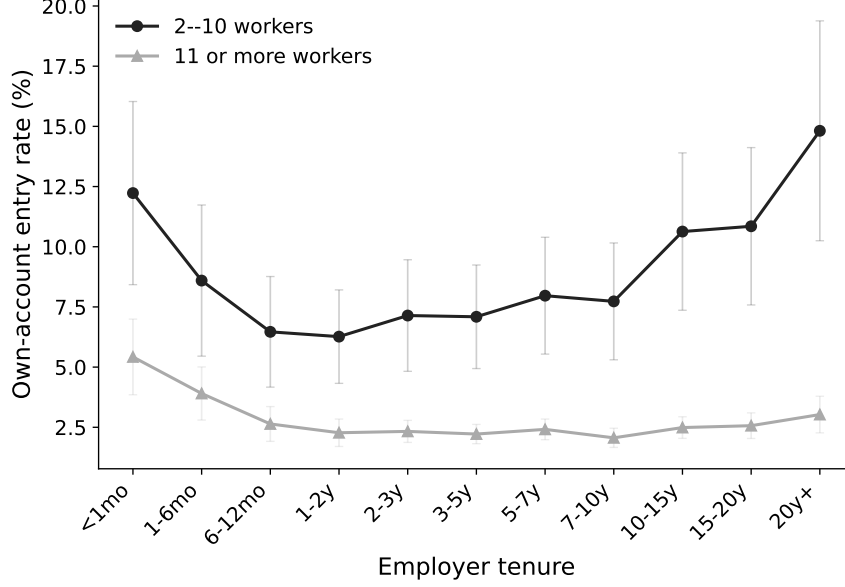


Figure 9: Disaggregated entry into own-account work by employer size

Notes: The figure reports the same sample and outcome as Figure 8 using eleven tenure bins. The six-bin figure pools the three bins below one year and groups later observations to match the specification in equation (E5). Vertical bars are 95 percent confidence intervals based on standard errors clustered by worker and two-digit origin industry. Source: PNADC, 2012–2019.

We define

$$\tilde{k}_s = \text{standardize}_s \left\{ \frac{1}{8} \sum_{t=2012}^{2019} \left[\log \left(\frac{I_{st}}{L_{st}} \right) - \frac{1}{24} \sum_{s'=10}^{33} \log \left(\frac{I_{s't}}{L_{s't}} \right) \right] \right\}. \quad (\text{E6})$$

Year demeaning removes common nominal trends before the division-level measure is standardized. The measure is an investment flow per worker, not a capital stock.

The grouped comparisons in panel (a) of Figures 2 and 5, and in Figure 10 divide the 24 divisions at the median of $\tilde{k}_s$. The lower-investment group comprises divisions 13–16, 18, 25–28, and 31–33; examples include textiles and apparel, leather and footwear, wood products, fabricated metal products, and furniture. The higher-investment group comprises divisions 10–12, 17, 19–24, and 29–30; examples include beverages, paper, petroleum refining, chemicals, basic metals, and motor vehicles. Thus, the binary grouping is a coarsening of the same division-level PIA measure used in the continuous specification rather than a cutoff in the CNAE codes.

Figure 10 extends the main-text industry comparison to the broader sample defined in Appendix E.4: formal and informal private employees aged 15–64, retaining all observed next-quarter destinations. The remaining specifications in this subsection use that broader sample.

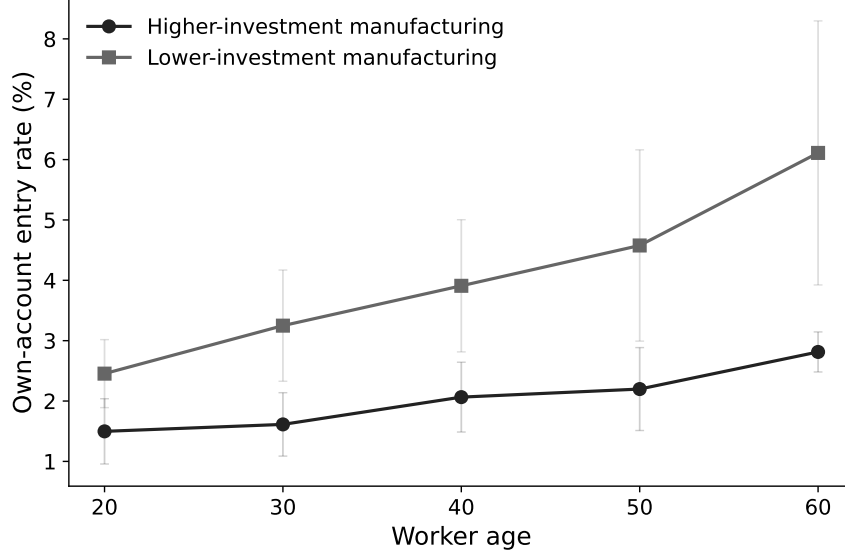


Figure 10: Entry into own-account work by worker age and origin industry

Notes: The figure reports sampling-weighted quarterly transition rates from private wage employment to own-account work by ten-year age bin. Age and industry are measured in the origin wage job. Manufacturing divisions are divided at the median of the PIA-Empresa investment measure defined in Appendix E.9. The sample contains 423,311 transitions among workers aged 15–64. Source: PNADC and PIA-Empresa, 2012–2019. Vertical bars are 95 percent confidence intervals based on standard errors clustered by worker and two-digit origin industry.

Figure 10 shows a larger difference in entry rates between younger and older workers in lower-investment manufacturing industries. Quarterly entry is 2.5 percent among workers aged 15–24 and 6.1 percent among those aged 55–64 in this group. The corresponding rates in higher-investment industries are 1.5 and 2.8 percent. These age profiles combine differences in accumulated resources, cohort composition, labor-market histories, and selection into wage employment.

As a check on the grouping, employment-weighted fixed-asset acquisitions per worker are higher in the above-median group in every sample year. The annual ratio ranges from 5.29 to 6.54, with a median of 5.76. Across divisions, pooled acquisitions per worker have medians of R\$5,948 below the cutoff and R\$28,216 above it; a one-sided rank-sum test gives $p < 0.001$.

On the manufacturing sample, the binary and continuous equations are

$$Y_{i,t+1} = \beta_a \text{Age}_{it}^c + \beta_L \text{Age}_{it}^c L_{it} + X'_{it} \theta + \mu_{s(i,t)} + \mu_{o(i,t)} + \lambda_t + \varepsilon_{it}, \quad (\text{E7})$$

$$Y_{i,t+1} = \gamma_a \text{Age}_{it}^c + \gamma_k \text{Age}_{it}^c \tilde{k}_{s(i,t)} + X'_{it} \pi + \mu_{s(i,t)} + \mu_{o(i,t)} + \lambda_t + \nu_{it}, \quad (\text{E8})$$

where age is centered at 40 and L_{it} indicates a division below the median of $\tilde{k}_s$. In addition to sex, schooling, and log origin wages, X_{it} includes grouped current-job-tenure indicators.

Both equations include origin-industry, origin-occupation, and origin-quarter fixed effects and use sampling weights. Standard errors are clustered by worker and two-digit industry.

Table 24: Age gradients and origin-industry investment

	PIA median split	Continuous PIA measure
Age	0.000595 (0.000044)	0.000761 (0.000053)
Age $\times$ below-median investment	0.000476 (0.000163)	
Age $\times$ PIA investment measure		-0.000322 (0.000083)
Observations	417,862	417,862
Wild-cluster p -value for interaction	0.008	0.002

Notes: The outcome is entry into own-account work in the following quarter. The median-split specification's age coefficient is the slope for above-median divisions; adding the interaction gives a slope of 0.001072 for below-median divisions. The PIA measure is defined in equation (E6). Standard errors in parentheses are two-way clustered by worker and two-digit industry. Wild-cluster tests are restricted tests at the two-digit-industry level with 4,999 replications.

The linear median-split interaction is positive (Table 24): the annual age slope is 0.000476 larger below the PIA median. Its two-way clustered and wild-cluster p -values are both 0.008. The continuous interaction is negative. A one-standard-deviation increase in the PIA measure reduces the annual age slope by 0.000322; the two-way clustered p -value is 0.001 and the wild-cluster p -value is 0.002.

We relax the linear age restriction by replacing centered age in equations (E7) and (E8) with indicators for age 20, 30, 40, 50, and 60 and their interactions with the industry measure. The median-split interactions are jointly significant ($F = 4.66$, $p = 0.007$), as are the continuous interactions ($F = 4.88$, $p = 0.005$). Relative to age 20, the low-minus-high investment difference rises with age, as reported in Table 25. Figure 11 displays these adjusted profiles.

The result also survives richer controls and alternative constructions. Adding formal and full-time status, household composition and income, and state-by-quarter, sector-by-year, and occupation-by-year fixed effects gives a low-minus-high annual age slope of 0.000394 (standard error 0.000156) under UPA-by-industry clustering. Using pooled acquisitions per worker rather than the year-demeaned measure gives a continuous interaction of -0.000321 (standard error 0.000085). The continuous interaction is negative in all 24 specifications that omit one two-digit industry at a time, ranging from -0.000358 to -0.000200.

Table 26 shows whether lower-investment industries have steeper age gradients for departures generally. The positive differential is specific to own-account entry. The estimates for a new wage job, unemployment, and nonparticipation are nonpositive or close to zero.

Table 25: Flexible age profiles and industry investment

	Low-minus-high	One-SD PIA increase
Age 30 relative to age 20	0.774 (0.191)	-0.425 (0.100)
Age 40 relative to age 20	0.856 (0.264)	-0.567 (0.140)
Age 50 relative to age 20	1.391 (0.511)	-0.948 (0.237)
Age 60 relative to age 20	2.198 (0.752)	-1.371 (0.421)
Observations	417,862	417,862
Joint p -value	0.007	0.005

Notes: Entries are percentage-point differences in quarterly entry into own-account work, normalized at age 20. The first column compares divisions below and above the PIA median. The second reports the effect of a one-standard-deviation increase in the continuous PIA measure. Standard errors in parentheses are clustered by worker and two-digit origin industry. Joint tests cover all age-bin interactions.

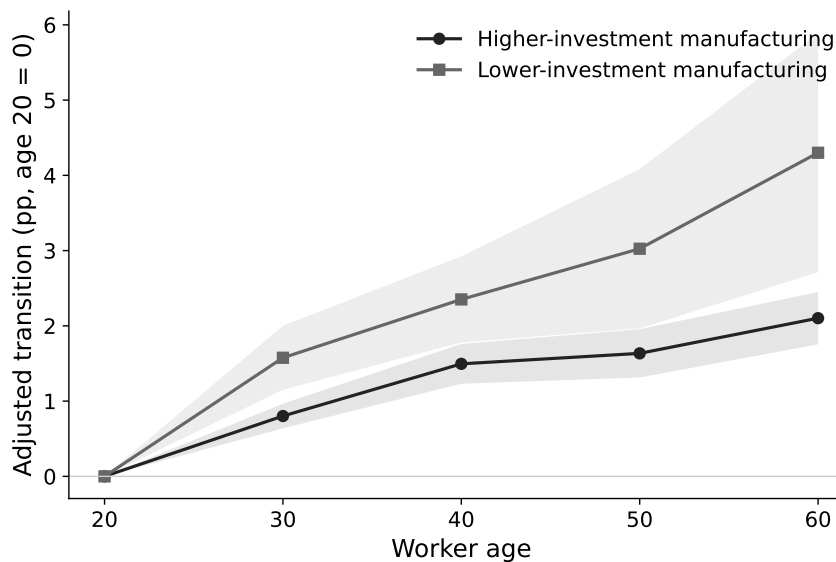


Figure 11: Adjusted age profiles by the PIA investment median

Notes: The figure plots adjusted age-bin coefficients from a specification that replaces the linear age terms in equation (E7) with unrestricted age-bin indicators and interactions. Profiles are normalized to zero at age 20. The regression contains 417,862 transitions. Shaded areas are 95 percent confidence bands based on standard errors clustered by worker and two-digit origin industry. Source: PNADC and PIA-Empresa, 2012–2019.

Table 26: Industry investment and alternative destinations

Next-quarter destination	Low-minus-high age slope	Standard error
Own-account work	0.0476	0.0163
New wage job	-0.0264	0.0152
Unemployment	0.0010	0.0067
Nonparticipation	-0.0156	0.0147
Observations	417,862	

Notes: Entries are percentage-point differences in the annual age slope between manufacturing divisions below and above the PIA investment median. Each row is a separate weighted regression with the controls and fixed effects in equation (E7). Standard errors are clustered by worker and two-digit origin industry.

E.10 Outcome definition

The primary outcome is entry into own-account work because it is the closest observed counterpart to the model’s nonemployer self-employment state. Table 27 reports the principal contrasts when entry as an employer is either added to or separated from own-account entry. Both empirical patterns strengthen under the broader self-employment definition. Employer entry alone also exhibits the same signs, although it is a less direct counterpart to the model.

Table 27: Alternative definitions of self-employment entry

Outcome	Long-tenure contrast	PIA age-slope difference
Own-account entry	0.554 (0.304)	0.0476 (0.0163)
All self-employment entry	0.964 (0.302)	0.0619 (0.0159)
Employer entry	0.410 (0.084)	0.0143 (0.0040)

Notes: Entries are percentage-point differences in quarterly transition probabilities. The long-tenure contrast compares tenure of at least seven years with three to seven years and uses 2,383,177 observations. The PIA contrast is the low-minus-high investment difference in the annual age slope and uses 417,862 manufacturing observations. Standard errors in parentheses are clustered by worker and two-digit origin industry.

E.11 Separate formality comparison

The formality exercise is separate from the common employer-comparison sample above. It excludes employers from the origin state and retains complete observations for age, sex, schooling, sampling weight, prior log wage, industry, occupation, and formal status. The

specification is

$$Y_{it} = \theta_0 + \theta_a \text{Age}_{it} + \theta_F \text{Formal}_{it} + \theta_{aF} \text{Age}_{it} \times \text{Formal}_{it} + X'_{it}\delta + \mu_{s(i,t)} + \mu_{o(i,t)} + \epsilon_{it}, \quad (\text{E9})$$

where X_{it} contains sex, schooling, and prior log wages. The regression has 1,223,779 observations and is estimated by weighted least squares. The estimated informal age slope is 0.001012. The age-by-formality interaction is -0.000692 (standard error 0.000621; $p = 0.269$), implying a formal-sector slope of 0.000321. Figure 12 reports the corresponding raw weighted age profiles.

Formal employment provides access to employment-linked severance accounts. Formal status also reflects regulation, employer characteristics, benefits, and worker selection. The comparison is therefore reduced-form evidence, and equation (E9) does not isolate savings access from these other channels. Moreover, the age-slope difference is imprecise under UPA-by-industry clustering. We therefore report the comparison in this appendix and do not use it to motivate the model.

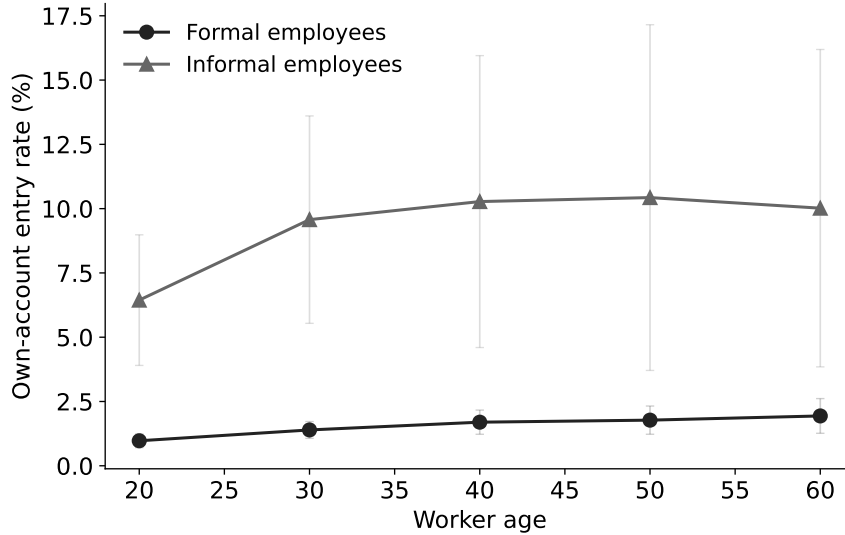


Figure 12: Entry into own-account work by age and formality

Notes: The figure reports sampling-weighted quarterly transition rates for wage employees in consecutive interviews. Formal status is measured in the origin job. The corresponding regression sample contains 1,223,779 employee-quarter observations. Vertical bars are 95 percent confidence intervals based on standard errors clustered by UPA and two-digit origin industry. Source: PNADC, 2012–2019.

E.12 Employer heterogeneity and the model

We examine whether differences in firm assets in the estimated model generate tenure patterns resembling the employer comparisons in Table 19. Because assets converge to a common target within matches, matches are split at the median of assets at formation, and each group’s entry profile is followed from formation. Initial heterogeneity is

large. Mean initial assets in the high group are more than four times those in the low group, and 74 percent of matches form below the target. Aggregating the two groups reproduces the model’s unconditional tenure profile exactly. Model firm assets and the observed establishment-size and industry-investment groups are different measures, so this comparison assesses the quantitative scope of the mechanism rather than a directly matched empirical moment.

The two profiles are nearly identical. Quarterly entry into own-account work rises from 1.6 percent in the first two years to 2.6 percent at ten or more years in both groups. The low-minus-high difference in the tenure change from three-to-seven years to ten or more is 0.01 percentage points, against 2.49 across establishment size and 0.98 across industry investment in the data.

Two facts explain the small model difference. The model’s firm-asset heterogeneity is short-lived: the median low-asset match reaches the common target within seven quarters, and by the three-to-seven-year band 99.6 percent of surviving low-asset matches sit at the target, so almost no heterogeneity remains where the data’s profiles diverge. While the heterogeneity persists, low initial assets barely affect the worker’s exit margin: they lower the wage by 3 percent, lower the dissolution hazard slightly, and leave entry unchanged, because entry flows through separations of eligible workers at a rate that depends on the worker’s wealth rather than the firm’s assets.

The estimated model therefore does not account for the observed differences across employer groups. Firm assets in the model are transitory, whereas establishment size and industry investment can reflect more persistent differences in technology, organization, and worker composition as well as financing. The empirical comparisons do not identify which of these differences produces the entry patterns. They remain a limitation of the quantitative account, rather than evidence that its firm-financing channel has been identified.

F. Stationary Equilibrium and Estimation

F.1 Stationary accounting

Let $g_U(b)$, $g_{SE}(b)$, and $g_F(a, b)$ denote the stationary densities of unemployed workers, self-employed workers, and formal matches. These densities solve the Kolmogorov forward equations implied by the policy drifts and transition intensities in Section III. Turnover removes mass from every state at rate ν and replaces it with unemployed entrants drawn from $\psi(b)$. In the quantitative model, ψ is a truncated exponential on the wealth grid with scale parameter $\bar{b}_0$ (the pre-truncation mean).

An unemployed worker enters self-employment at intensity

$$\rho I_{\text{SE}}(b) \mathbf{1}\{V^{\text{SE}}(b) > U(b)\} \quad (\text{F1})$$

and formal employment at intensity

$$p \int \mathbf{1}\{a' \in \mathcal{A}(b)\} dF_A(a'). \quad (\text{F2})$$

Self-employed workers return to unemployment at rate ξ_{SE} . Because the startup requirement applies to entry only, the stationary self-employed density can in principle have support below b_{SE} : incumbents who dissave continue operating and retain the ξ_{SE} failure hazard, and after failure below the threshold they are ineligible to re-enter until wealth again reaches b_{SE} . At the estimated parameters this channel is quantitatively inert: self-employed wealth drifts upward at every eligible wealth level, so the below-threshold self-employed mass is negligible. Formal matches separate exogenously at rate s and end endogenously when their state exits $\mathcal{A}$. At an *exogenous* separation, the routing probability λ assigns the worker directly to self-employment only where $I_{\text{SE}}(b) = 1$ and $V^{\text{SE}}(b) > U(b)$; all other separation mass enters unemployment. This reallocation enters the worker's match value, equation (B3). *All* endogenous dissolutions enter unemployment, from which the ρ process converts wealthy workers to self-employment with its equilibrium delay. The baseline sets $p_{\text{SE}} = 0$, so self-employed workers do not receive direct formal offers.

The continuous-time generator combines these transitions with the endogenous asset drifts $\dot{b}_U$, $\dot{b}_{\text{SE}}$, $\dot{a}$, and $\dot{b}$. Two related operators are used. Let $Q_0(\theta)$ denote the discretized individual-level generator, which includes exit at rate ν but excludes entrant replacement. The stationary distribution $g(\theta)$ solves the generator-plus-births system

$$Q_0(\theta)'g(\theta) + \nu \psi_U = 0, \quad \mathbf{1}'g(\theta) = 1, \quad (\text{F3})$$

where ψ_U places the entrant wealth density ψ in the unemployment block. Quarterly transition probabilities are computed from the transition semigroup $\exp[Q_0(\theta)/4]$, approximated by 64 implicit-Euler substeps (an L-stable, probability-preserving scheme); the semigroup allows multiple transitions within a quarter. Because Q_0 excludes the replacement kernel, these probabilities follow the original individuals through the quarter. The mass exiting at rate ν is removed by the conditioning on remaining in $\{F, \text{SE}, U\}$ described below. The conservative upwind discretization accommodates a possible unemployed atom at b_{SE} as mass concentration in the threshold cell. Appendix A.1 derives the one-sided value derivatives and wealth drifts that determine whether such an atom forms.

F.2 Moment definitions

The empirical states are formal private employment F , own-account self-employment SE , and ILO unemployment U . The estimation sample covers workers aged 18–65. Transition probabilities condition on the destination remaining in $\{F, SE, U\}$, matching the closed state space of the quantitative model.

The 17 targets are listed in Table 2 in the main text. The alternative earnings ratio based on within-person-averaged earnings is 0.741.

The estimator minimizes equation (19). For empirical target t_k , model moment $m_k(\theta)$, and bootstrap standard error $\hat{\sigma}_k$, define the scaled deviation

$$e_k(\theta) = \frac{m_k(\theta) - t_k}{s_k}, \quad s_k = \max\{\hat{\sigma}_k, 0.05 |t_k|\}. \quad (\text{F4})$$

The floor binds for all seventeen targets in the sample of roughly 11.5 million person-quarters, so one scale unit equals five percent of the target’s value. The momentwise loss is

$$h(e) = \begin{cases} e^2, & |e| \leq 20, \\ 20(2|e| - 20), & |e| > 20. \end{cases} \quad (\text{F5})$$

This is twice the conventional normalization of the [Huber \(1964\)](#) loss, with threshold 20; the common factor leaves the minimizer unchanged. Since the five-percent scale floor binds for all targets, the linear region begins at an absolute error exceeding 100 percent of the target’s value. Each moment error at the estimate falls inside the quadratic region. The diagonal weighting has an applied precedent in [Blundell, Pistaferri, and Preston \(2008\)](#); the five-percent floor and the threshold of 20 specify this paper’s emphasis on proportional fit and its treatment of large errors during the search.

In addition to the moment terms, the implemented loss carries numerical-quality guards: an evaluation whose equilibrium solve fails its residual or complementarity check receives a large additive penalty. A diagnostic flag records whether the viable band remains open at the top of the computational wealth grid; the flag carries no penalty and is reported with every evaluation. Final admission additionally requires agreement across the examined equilibrium initializations and stability under the wealth-window audit reported in Appendix F.3.

The jointly searched parameter block is

$$\theta = (z_{SE}, \alpha_{SE}, z_u, \bar{b}_0, \beta_1, \beta_2, \eta, \lambda, b_{SE}, s, \rho). \quad (\text{F6})$$

The startup threshold b_{SE} and the entrant-wealth scale $\bar{b}_0$ jointly shape self-employment among young workers;²⁰ $\bar{b}_0$ is the mean of the underlying exponential before truncation, and (β_1, β_2) govern the formal-offer distribution. The working-capital duration is calibrated at $\chi = 0.25$ and the collateral multiplier at $\phi = 2.6$; Appendix A describes the financing calibrations and their sensitivity ranges. The rates p and ξ_{SE} are recovered from unemployment-to-formal and self-employment-to-unemployment transitions, respectively, by conditional flow inversion within each objective evaluation. The opportunity-arrival rate ρ is searched jointly, because its own-moment inversion conflicts with the self-employment stock moments once the startup requirement thins the eligible unemployed pool: exactly matching the unemployment-to-self-employment flow overshoots the stock, so the joint loss balances these targets. The objective is evaluated directly from the discretized distributional equations. A step-refinement audit measures its deterministic approximation sensitivity: refining the quarterly transition operator and the cohort marches shifts no targeted moment by more than 0.25 scale units, with the largest effect on the youngest age band. Widening the upper bound of the wealth grid sevenfold at unchanged resolution gives a maximum shift of 0.52 scale units on the audited cells.

Point estimates are reported in Table 3 in the main text.

The design has 17 targeted moments and 13 adjusted structural parameters: eleven are searched jointly and two are recovered by conditional inversion. The four excess targets provide a nominal count of overidentifying restrictions. The criterion, however, uses percentage-error weights and is subject to binding parameter and numerical-admission restrictions, so its reported value has no conventional chi-squared interpretation. We assess specification through the targeted fit and untargeted predictions, and report numerical reproducibility separately as a check on the solution.

²⁰The threshold enters the solved model through comparisons at wealth-grid nodes, so the objective is piecewise constant in b_{SE} between adjacent nodes. The reported threshold therefore carries a grid-cell qualification.

Table 28: Model fit: targeted moments and specification diagnostics

	Data	Model	Scaled dev.
<i>Targeted</i>			
Self-employment share	0.316	0.409	+5.9
Unemployment rate	0.101	0.105	+0.8
Unemployment to self-employment	0.138	0.105	−4.8
Unemployment to formal	0.163	0.163	+0.0
Self-employment to unemployment	0.042	0.042	−0.2
Formal to unemployment, tenure 10+	0.0085	0.0082	−0.6
Formal to self-employment, aggregate	0.0232	0.0205	−2.3
Formal to self-employment, tenure 10+	0.0236	0.0259	+1.9
Formal to self-employment, age 35–44	0.0258	0.0231	−2.2
Formal to self-employment, age 45–54	0.0278	0.0267	−0.8
Formal to self-employment, age 55–65	0.0322	0.0274	−3.0
Self-employment share, age 20–24	0.150	0.146	−0.6
Self-employment share, age 25–34	0.241	0.245	+0.3
Self-employment share, age 35–44	0.344	0.401	+3.3
Self-employment share, age 45–54	0.452	0.490	+1.7
Self-employment share, age 55–65	0.591	0.519	−2.5
Median self-employment/formal earnings	0.769	0.856	+2.3
<i>Non-targeted diagnostics</i>			
Formal to unemployment, aggregate	0.0291	0.0138	—
Formal to unemployment, tenure < 2	0.0520	0.0181	—
Formal to self-employment, tenure < 2	0.0270	0.0164	—
Formal to self-employment, age 20–24	0.0163	0.0077	—
Self-employment to formal	0.0447	0.0044	—
p75/p25 earnings, formal	2.00	1.26	—
p75/p25 earnings, self-employment	3.24	1.01	—
Own-account to employer, quarterly	0.035	0	—
Employers, share of the employed	0.042	0	—

Notes: Model moments at the estimates of Table 3. Scaled deviations divide by $\max\{\hat{\sigma}_k, 0.05|t_k|\}$; the floor binds for every targeted moment. The non-targeted rows are the specification diagnostics discussed in the text: the short-tenure separation shakeout and the direct self-employment-to-formal margin are absent from the model by design ($p_{SE} = 0$), and within-state earnings dispersion is limited by the single-type wealth channel. The employer state is absent by construction. In the data, own-account workers become employers at 3.5 percent per quarter. Each quarter, 19 percent of employers move to own-account work.

F.3 Identification and global search

Local sensitivity. At the estimates we compute the Jacobian of the seventeen scaled moment deviations with respect to the eleven jointly searched parameters, by central differences through the full loss, including the conditional flow inversion. The Jacobian has full rank, with condition number 1,162 and minimum singular value 0.97, so even the least-informed direction moves the moment vector by roughly one scale unit (five percent of a moment’s level) per unit of parameter movement. This least-identified direction loads equally on the startup threshold b_{SE} and the entrant-wealth scale $\bar{b}_0$: the young-worker self-employment margin admits a partial trade-off between a higher threshold and wealthier entrants, and because the threshold acts through wealth-grid nodes, the loss is locally flat within a grid cell along this pair. The second-weakest direction mixes the offer-distribution shapes with the bargaining weight and the self-employment curvature. The conditioning reflects the choice of searched parameters: releasing the externally calibrated financing parameters into the search reintroduces a near-null direction (the loss is nearly flat in ϕ alone). Table 29 reports the three most responsive moments for each parameter. Table 30 reports the full matrix.

Table 29: The most responsive moments, by parameter

Parameter	Largest responses of the scaled deviations		
z_{SE}	SE share, 35–44 (+2.3)	SE share, 25–34 (+2.0)	F→SE, 35–44 (+2.0)
α_{SE}	F→U, ten. 10+ (−5.8)	SE/F earnings (−2.7)	F→SE, ten. 10+ (+2.2)
z_u	SE share, 35–44 (+0.6)	F→SE, 35–44 (+0.5)	SE/F earnings (−0.5)
$\bar{b}_0$	SE share, 20–24 (+1.8)	SE share, 25–34 (+1.2)	SE/F earnings (−0.5)
β_1	F→U, ten. 10+ (+16.1)	F→SE, ten. 10+ (−4.8)	U→SE (−4.4)
β_2	F→U, ten. 10+ (−16.0)	Unemployment rate (−5.3)	F→SE, ten. 10+ (+4.8)
η	F→U, ten. 10+ (+13.5)	F→SE, ten. 10+ (−3.8)	SE/F earnings (−3.4)
λ	F→U, ten. 10+ (−5.2)	F→SE, ten. 10+ (+2.0)	F→SE, 45–54 (+1.7)
ρ	F→U, ten. 10+ (+12.8)	F→SE, ten. 10+ (−4.3)	F→SE (−2.6)
b_{SE}	SE share, 25–34 (−2.6)	SE share, 20–24 (−2.5)	SE share, 35–44 (−1.9)
s	F→U, ten. 10+ (−9.0)	F→SE, ten. 10+ (+5.6)	F→SE (+3.7)

Notes: Each entry is the change in a scaled moment deviation e_k in response to a 10 percent increase in the row parameter, holding the remaining parameters fixed (central differences through the full loss, including the conditional flow inversion). The three largest-magnitude responses are shown per parameter. The long-tenure separation margin responds strongly throughout because its empirical level is small, so a given absolute movement is many scaled units.

Table 30: The moment Jacobian at the estimates

Moment	z_{SE}	α_{SE}	z_u	$\bar{b}_0$	β_1	β_2	η	λ	b_{SE}	s	ρ
Self-employment share	+46.29	+38.59	+0.03	+4.40	-65.55	+54.32	-38.52	+6.96	-13.28	+197.50	-12.61
Unemployment rate	-35.37	-25.94	+37.17	-0.98	+86.83	-83.26	+56.05	-4.23	+6.84	-63.99	+12.85
U→SE	+58.14	+43.39	-24.62	+5.16	-87.20	+68.17	-56.48	-0.54	-15.58	+142.45	-9.89
U→F	-4.24	-3.44	+0.43	-0.47	+11.16	-2.71	+3.48	+0.06	+1.30	-12.45	+0.61
SE→U	-0.05	-0.05	-0.02	-0.00	-0.63	-0.07	-0.00	-0.02	+0.02	+0.02	-2.29
F→U, ten. 10+ ^a	-60.78	-149.82	+20.33	+6.15	+318.15	-251.89	+265.70	-60.42	+4.17	-648.16	+128.19
F→SE	+45.77	+41.89	+3.65	+2.51	-70.50	+56.63	-48.10	+17.90	-11.59	+265.46	-26.46
F→SE, ten. 10+	+30.52	+56.39	+8.69	-1.17	-93.94	+75.19	-74.84	+23.53	-2.29	+404.95	-43.62
F→SE, age 35-44	+63.17	+31.55	+54.95	+1.08	-14.34	+15.09	+4.38	+17.65	-18.97	+144.79	-8.72
F→SE, age 45-54	+38.20	+32.48	+20.60	-1.91	-35.71	+34.50	-17.39	+19.55	-7.11	+199.28	-13.49
F→SE, age 55-65	+24.78	+32.14	-5.12	-2.40	-56.28	+50.63	-37.11	+17.66	-1.65	+227.34	-18.70
SE share, age 20-24	+29.93	+33.13	+11.63	+30.61	-27.83	+20.72	-12.41	+2.49	-35.86	+94.24	-4.98
SE share, age 25-34	+63.87	+33.60	+42.00	+19.64	-14.11	+11.99	+5.89	+6.07	-37.28	+101.53	-3.27
SE share, age 35-44	+71.71	+33.27	+56.50	+5.93	-10.87	+10.45	+9.84	+7.16	-26.58	+100.05	-0.82
SE share, age 45-54	+38.91	+24.84	+25.59	+0.27	-17.61	+16.11	-3.08	+5.69	-10.52	+95.99	-0.89
SE share, age 55-65	+19.97	+17.78	+3.05	-0.91	-24.06	+21.87	-12.24	+4.20	-3.27	+90.31	-2.33
SE/F median earnings	+3.71	-68.86	-47.43	-8.75	-26.21	+16.11	-66.49	+3.36	+3.46	+203.84	-13.06

Notes: Each entry is the derivative of the scaled moment deviation with respect to the column parameter at the estimates, computed by central differences through the full loss, including the conditional flow inversion. The startup-threshold column is node-conditional: the loss is piecewise constant in b_{SE} between wealth-grid nodes, and the probe is one grid cell per side. The z_u column is one-sided at the declared plausibility floor. The Jacobian has full rank, with condition number 1,162 and minimum singular value 0.97.^a Entries in this row are local slopes at the documented probe steps; halving the step moves them substantially, a curvature property of this small cell rather than solver noise.

Standard errors. At the estimates every scaled deviation lies inside the quadratic region of the robust loss, so the estimator is locally minimum distance with weighting matrix $W = \text{diag}(1/s_k^2)$. The reported delta-method covariance uses the sandwich form

$$V = (G'WG)^{-1}G'W\Omega WG(G'WG)^{-1}, \quad (\text{F7})$$

where $G = \partial m / \partial \theta'$ is the Jacobian of the *unscaled* model moments and Ω is the covariance of the empirical moments. Thus row k of G equals s_k times row k of the scaled-deviation Jacobian in Table 30. Model moments are computed directly, so the covariance has no Monte Carlo simulation term. We estimate Ω by a joint individual-level bootstrap ($B = 200$ draws, each resampling individuals and recomputing all seventeen moments). Cross-moment covariances enter inference through Ω , while the diagonal fitting weights remain those of equation (F4).

The unemployment income z_u is estimated at the plausibility floor imposed below, so its sampling distribution is nonstandard and its standard-error entry is left blank. The startup threshold b_{SE} acts through wealth-grid nodes. Its reported standard error holds the grid fixed and uses a one-cell finite-difference probe; it exceeds the cell width of 0.058. It should therefore be read as a grid-conditional local approximation. The covariance reproduces the least-identified direction: $\text{corr}(\bar{b}_0, b_{\text{SE}}) = 1.00$, with individual standard errors of 0.19, while the difference $b_{\text{SE}} - \bar{b}_0$ is estimated at 0.105 with a standard error of 0.005. Ignoring cross-moment correlations (a diagonal Ω) changes most standard errors by less than ten percent, with the largest changes for the routing and arrival rates (λ : 0.033 to 0.025; ρ : 0.022 to 0.016).

Search across starting values. Pattern searches from more than a dozen parameter starting points locate several local minima. Every reported point reproduces across the examined from-scratch equilibrium solutions. Among the other local minima located, the closest loss lies 29 percent above the estimate and the remaining losses lie at least 47 percent above it. These points describe a rival economy with more productive but less accessible self-employment and higher unemployment income.

Income restriction and wealth-domain sensitivity. The reported estimate has unemployment income at the maintained lower bound, $z_u = 0.10$, and a loss of 107.5. This income is about 40 percent of the model's average formal wage. The comparison expresses the bound in earnings units; the lower bound is an additional assumption restricting outside options and saving incentives. Adding the aggregate formal-to-unemployment rate and the formal 75/25 earnings ratio leaves substantial tension among the targets and pushes estimates to parameter bounds; these moments are retained as specification diagnostics.

Table 31 reports lower-loss parameter vectors with higher entrant wealth and startup requirements. Unemployment income is above its lower bound at each of these points, so the income restriction and the wealth-domain check impose distinct restrictions on the reported estimate. The lower-loss points fail the domain check: widening the wealth window changes their audited moments by 1.03 to 4.60 scale units, compared with 0.52 at the estimate. The share of separations at the upper grid edge also rises. The table reports the parameter values, losses, and moment shifts underlying their exclusion.

Table 31: The rejected descent along the least-identified direction

	Estimate	Descent 1	Descent 2	Descent 3
Loss (certified)	107.5115	105.7780	91.8278	91.4987
z_{SE}	0.3160	0.3703	0.3334	0.3274
α_{SE}	0.3858	0.3655	0.4500	0.4753
z_u	0.1000	0.1203	0.1137	0.1250
$\bar{b}_0$	0.5938	0.7832	0.8519	0.8509
b_{SE}	0.6986	0.8895	0.9270	0.9037
Verification	pass	pass ^a	pass ^a	pass
Extended-window shift (audited cells)	0.52	1.03	3.75 ^a	4.60
Top-edge flux (% of separations)	4.1	4.1 ^a	10.6 ^a	10.5
Window audit (shift ≤ 0.75)	pass	fail	fail	fail
Status	admitted	excluded	excluded	excluded

Notes: Verified stationary points of the descent along the loss’s least-identified direction, against the estimate. All four points verify: two independent from-scratch solutions agree exactly, and a 10^{-5} perturbation of unemployment income moves the loss negligibly. Verification cannot see the wealth-window axis. The window audit re-solves each point with the wealth window extended at unchanged resolution and reports the largest shift, in scale units, on the audited cells (shares, entry flow, direct transition, earnings ratio); the pre-stated standard is 0.75. Top-edge flux is the share of separations occurring at the upper edge of the estimation window. Every descent point fails the audit while improving the in-window loss: the apparent improvement is wealth mass truncated at the window edge and recycled through separation into the targeted entry cells. Descent 2 sits at the routing rate’s declared upper bound ($\lambda = 0.95$); a fourth point (89.71) pinned the routing rate at 0.99 in a design test and is not a candidate. ^a Recomputed for this table; all other entries are from the archived records.

F.4 Non-targeted diagnostics

The formal-to-unemployment rates at tenure below two years and in the aggregate are reported but not targeted. The model has no match-quality shock (Jovanovic, 1979) capable of generating the steep decline in separations observed at short tenure. Targeting that decline would therefore require s or the boundary parameters to absorb a friction absent from the model. We use the long-tenure formal-to-unemployment rate to discipline exogenous separation and reserve the short-tenure profile as a specification diagnostic.

The young formal-to-self-employment cells (age 25–34 and tenure below two years) are likewise reported but not targeted. The model generates short-tenure formal-to-self-employment flow only through λ -routing at exogenous separations and through within-

quarter passage into unemployment and back out through the ρ process; at any λ this flow remains several times smaller than the data cells. Those cells reflect the same short-tenure shakeout that the model omits, and targeting them would force the routing parameters to absorb it; they are treated as specification diagnostics.

We also report, but do not target, the p75/p25 earnings ratios within formal employment and self-employment. The corresponding data moments are 2.000 and 3.240. In the model, wealth generates all earnings dispersion because workers do not differ in skill or sector. The gap between model and data therefore measures dispersion that the maintained heterogeneity cannot reproduce; it is not used to move parameters governing the wealth channel.

F.5 Do workers quit into self-employment?

The baseline model allows an eligible worker to enter self-employment when a match ends. An extension also allows opportunities to enter self-employment to arrive during employment at rate ρ_E . A worker accepts when the value of self-employment exceeds the value of the current match. On that set of states, the hazard enters both value functions, the bargained wage, and the stationary flows.

At the reported estimate, employed workers prefer the bargained match to self-employment, so this acceptance set is empty. When no worker accepts an opportunity, changes in its arrival rate generate no quit flow and provide no local identification of ρ_E . Exploratory estimates of an earlier specification also produced an inactive quit margin at the preferred parameters. Those exercises do not identify the prevalence of voluntary quits in the data.

The restriction that the self-employment value is a deterministic function of wealth limits the extension’s ability to represent different decisions by workers with similar resources and jobs. Heterogeneity in business opportunities could generate voluntary entry even when the common opportunity is less valuable than continued employment. Neither a positive association between prior wages and entry nor a comparison of realized earnings changes identifies that heterogeneity. Workers choose between expected lifetime values, and voluntary entrants can earn less initially. We therefore interpret retention as a feature of the estimated model, without treating the extension or the earnings comparison as a bound on the empirical quit share.

G. Counterfactual Conventions and Details

G.1 Conventions

All experiments are evaluated at the estimates of Table 3, with every parameter other than the policy lever held at its estimated value; the rates recovered by conditional inversion

during estimation are treated as structural and are not re-solved. Levers move along graduated paths. Each equilibrium is solved by continuation from the preceding point. Every reported equilibrium is certified: two independent from-scratch solutions must agree to numerical precision (they agree exactly at every reported point), and a 10^{-5} perturbation of unemployment income must move reported shares by less than 5×10^{-3} (realized perturbation responses are 10^{-4} or smaller). Points failing certification would be reported as equilibrium-ambiguous; none did. Along the threshold-removal path of Table 4, the two independent solutions agree exactly at every computed point, so the experiment does not enter the multiplicity region documented in Appendix A.5. At every computed equilibrium the on-the-job quit region of Section F.5 is checked. It is empty at every reported point, including the lowest-threshold entries of the entry experiment; results are therefore exactly invariant to the unidentified quit-arrival rate.

Entrant welfare. The welfare metric is the expected value of labor-market entry, $\mathbb{E}_\psi[U(b)]$, with ψ the entrant wealth distribution, converted to a consumption-equivalent level: with CRRA utility at $\gamma = 2$, a value V corresponds to the constant consumption $\bar{c} = -1/((r + \nu)V)$, and welfare changes are reported as percent changes in $\bar{c}$.

G.2 The wealth-accumulation boundary in the savings-access experiment

Workers discount at $r + \nu$. As the savings return r_k approaches this bound, the target wealth of an impatient saver diverges, and at $r_k \geq r + \nu$ no stationary wealth distribution exists: the experiment of Section VI.C is structurally confined to the corridor between the market return and the discount rate (2.2 percentage points at the calibration). Within the corridor the divergence is quantitatively rapid. The audit statistic is the share of separations occurring at the upper edge of the numerical wealth window. These edge events are a numerical artifact rather than an economic margin, because the interior joint-surplus band remains open through twice the estimation window at every computed point. At baseline, the share is 4.1 percent on the estimation window and 2.3 percent on the quadrupled window (the extended window of the wealth-grid audit in Appendix F). The +25bp equilibrium is solved on the quadrupled window. It carries an edge share of 4.0 percent. This is well above the 0.5 percent trigger for disclosure. The savings-access experiment is accordingly read from the difference between two equilibria solved on the same window and carrying the artifact at comparable levels, and the direction of the residual bias strengthens the text’s conclusion: edge events route wealthy, eligible workers through unemployment, where self-employment is preferred, so the artifact inflates the self-employment share and the reported decline is a conservative bound. Points deeper in the corridor fail the audit at any feasible window and are reported only through value-side objects (Table 6), which involve no stationary distribution.

G.3 The entry closure and the location margin

The general-equilibrium column uses the free-entry closure of Appendix D, calibrated so that both entry conditions hold exactly at the estimated equilibrium, and re-calibrated whenever an experiment requires an extended wealth window (the calibration must be window-consistent or baseline invariance fails). At counterfactual points the marginal-capital (offer-location) residual proves flat within its numerical noise floor across the economically sensible range of the location instrument. Imposing the marginal-capital condition would fit numerical noise and select degenerate locations. The reported closure therefore imposes free entry only. Tightness adjusts, and the offer distribution’s location is held at its estimated position. The location margin is reported as a bounded sensitivity. Tilting the offer distribution by ± 0.6 (in the exponential-tilt parameterization of Appendix D) moves the self-employment share under full threshold removal across $[0.37, 0.56]$ around the reported 0.52. Tightness responses under every lever at estimated matching efficiency lie within three percent of baseline, and partial-equilibrium and entry-closure shares differ by at most +0.006: the entry margin is quantitatively passive at the estimate. The full interaction grid is in the replication archive. It comprises fifteen converged tightness equilibria, including the unemployment and welfare entries omitted from Table 7. The archive also holds the window-consistent baseline equilibria from which the savings-access experiment is differenced.

G.4 The routing decomposition

Setting the reallocation-routing rate λ to zero at the estimates reduces the quarterly formal-to-self-employment transition probability from 0.0205 to 0.0025, a decline of 87.6 percent. The self-employment share falls from 0.409 to 0.294. These are counterfactual changes in the model, not a decomposition of observed transitions into dismissals and quits. A positive quarterly transition remains possible without direct routing because a worker can pass through unemployment and enter self-employment between interviews. The exercise measures the importance of direct access to self-employment at separation under the model’s maintained transition structure. Both equilibria are verified under the conventions above.

G.5 The firm-side grant

The firm-side comparison of Section VI is computed under the transfer conventions of the targeted eligibility grant: the value layer is unchanged, and the stationary layer is re-solved with the formation kernel relocating initial assets to the common target $a^* = 0.421$, well inside the asset grid (2.1 below its top). For the targeted grant, the value-layer freeze is exact. Its entrant density enters no Bellman equation. For the firm-side grant, the freeze is a convention. In a consistent partial-equilibrium solution, the relocated offer lottery

would raise the unemployed value through the offer-surplus term. The omitted term is small because match values depend only weakly on initial firm assets. At the estimate, the within-wealth wage gradient times the interdecile asset range is half a percent of the mean wage. No dissolution occurs at the asset boundary. The fiscal flows match to machine precision (0.0059 units of wealth per year, difference 4×10^{-17}), and the worker-side leg reproduces the targeted-grant experiment of Table 5 exactly (welfare gain 19.809 percent). The targeted grant’s combined earnings gain of +0.0046 decomposes into a within-state term of +0.0081, a between-state shift into self-employment of -0.0024 , and an interaction of -0.0012 ; within employment, the mean wage rises by +0.0151, of which composition along worker wealth contributes +0.0151 and along firm assets +0.0013, with a joint term of -0.0014 ; the firm-side grant’s whole effect on the mean wage is +0.0007, of which the along-assets composition is +0.0007. The full decomposition is in the replication archive. Three disclosures apply. The offer distribution’s zero-asset node, about six percent of the true offer probability, never generates an accepted match in the estimated economy and lies outside the instrument: its exclusion follows from the lattice offer-measure convention, whose clipped edge-singularity rectangle mass (5.19 against a true node probability of 0.063) cannot be relocated without fabricating offer measure. The exclusion does not reflect an economic judgment that zero-asset offers have no value. Entrant welfare is computed from worker states, so the firm-side grant leaves it unchanged by construction under the fixed value layer. The comparison does not evaluate gains to firm owners, which comprise the continuation value and the additional output net of capital costs that higher match capitalization produces and that the fixed wage schedule leaves with the firm as retained earnings. The earnings comparison is the informative worker-side object. The comparison is stationary and partial-equilibrium. It evaluates the level trade-off between capitalizing the two sides at the estimate rather than the dynamic sequencing that the optimal-policy analysis of [Itskhoki and Moll \(2019\)](#) concerns.

H. The Statutory Severance Account

This appendix describes the extended model behind Section [VI.B](#), its estimation, and the transfer comparison.

H.1 Design

A match carries the state (a, b, F) : firm assets, worker liquid wealth, and the account balance. The balance accrues out of bargained compensation, $\dot{F} = \theta w$ with $\theta = 0.08/1.08$, so the cash wage is $(1 - \theta)w$; the account earns no real return, matching its statutory remuneration of about zero in real terms over the sample period. At a dismissal the worker receives $b^+ = b + (1 + \varphi)F$ with $\varphi = 0.40$, and the separation decision is evaluated at b^+ ; the firm pays the fine φF . A firm-initiated dissolution is a dismissal on the same terms; a

quit releases nothing. A match continues only while the firm can cover the fine ($a \geq \varphi F$); a firm whose liability exceeds its assets dismisses the worker and pays out its assets. The statute also allows withdrawals on grounds other than dismissal, including housing, retirement, and illness (art. 20). In CAIXA’s 2018 administrative report, dismissal without just cause accounted for 60.9 percent of withdrawal value. Other channels accounted for the remaining 39.1 percent, including 15.8 percent for housing and 13.5 percent for retirement (Caixa Econômica Federal, 2019).²¹ We treat the account as locked until dismissal. Partial liquidity would bring the balance closer to private wealth, and we expect it to strengthen the displacement of private saving. No statutory withdrawal channel finances business entry directly.

The threat points of the bargain are the dissolution outcomes: $W - U(b^+) = \kappa(J - a + \varphi F)$, and the wage equation adds the accrual’s value to the worker and the fine’s cost to the firm. The bargained wage remains unique where it exists. Two implications follow. In undiscounted terms, a unit of compensation delivers $1 + \varphi\theta$ to the worker, since the deposited share returns with the fine at dismissal. The fine also lowers the firm’s outside option as the balance grows, so wages rise with tenure within a match. At the baseline estimate, the rise at the mean balance is about four percent of the mean wage.

H.2 Estimation

The balance is discretized on a twelve-cell geometric grid, and the wealth window is extended so that no release leaves the grid. At $\theta = \varphi = 0$ the extended model reproduces the baseline solution and loss exactly. We estimate it following Appendix F, with the same seventeen moments and formal earnings measured as the cash wage. Table 32 reports the estimates, with standard errors from the sandwich formula of Appendix F.3.

No parameter differs from the baseline estimate by more than one standard error of the extended model, or by more than 1.2 of the baseline’s. The identified combination, the startup threshold net of mean entrant wealth, is 0.114 (standard error 0.010) against 0.105 (0.005). The institution widens the standard errors but barely moves the point estimates. The routing probability and the opportunity-arrival rate are two routes to the same entry flows. Their estimates become correlated at 0.94, and the arrival rate’s standard error triples. Fit is somewhat poorer than the baseline’s (121.2 against 112.4 on the same window), with the deterioration concentrated in the earnings ratio, whose model counterpart now nets out the deposit, and in the self-employment share. Like the baseline loss, the extended loss has a second local minimum (125.1), which trades

²¹The report records R\$111.4 billion in withdrawals, of which R\$67.9 billion followed dismissal without just cause. These figures exclude inactive-account payments under Law 13,446/2017; see the administrative withdrawal table in Caixa Econômica Federal (2019, p. 45).

Table 32: Estimates of the extended model and standard errors

	Extended model		Baseline (Table 3)	
	estimate	s.e.	estimate	s.e.
z_{SE}	0.318	(0.051)	0.316	(0.055)
α_{SE}	0.390	(0.024)	0.386	(0.022)
z_u	0.100	(—)	0.100	(—)
$\bar{b}_0$	0.592	(0.160)	0.594	(0.189)
β_1	0.501	(0.021)	0.507	(0.020)
β_2	0.626	(0.021)	0.637	(0.013)
η	0.509	(0.026)	0.510	(0.020)
λ	0.887	(0.037)	0.860	(0.025)
ρ	1.014	(0.054)	0.995	(0.016)
b_{SE}	0.706	(0.161)	0.686	(0.191)
s	0.140	(0.002)	0.139	(0.003)
$b_{SE} - \bar{b}_0$	0.114	(0.010)	0.105	(0.005)

Notes: Sandwich standard errors in scaled units with the bootstrap moment covariance of Appendix F.3. No standard error is reported for z_u , estimated at its declared floor in both designs. The wealth thresholds are identified up to a grid cell: the baseline column reports the grid-node value (0.686), and the extended optimum (0.706) lies at the same node. The extended column conditions on the statutory objects (θ , φ , the account's return).

unemployment income against the bargaining weight along the ridge of Appendix F.3; the transfer comparison below reaches the same conclusions there.

H.3 The transfer comparison

The comparison holds the extended model at its estimate and solves three economies on one window: the estimated economy, the same economy without the account and fine, and the latter with an eligibility grant generalizing Section VI.A. The statutory release flow is 0.9 percent of workers' liquid wealth per year; per separation it is about seven months of the average formal wage, five of balance and two of fine. The grant places the 66 percent of entrants who arrive below the startup threshold at a common target 42 percent above it, chosen so that the grant's flow equals the statutory flow exactly. Relative to the no-account economy, the statutory institution raises the self-employment share by an imprecisely estimated 0.45 percentage points and lowers entrant welfare by 0.02 percent; the grant raises the share by 9.8 points and entrant welfare by 28.7 percent, roughly twenty-two times the entry per unit of flow. Delta-method standard errors are 0.45 (1.5) and 9.8 (0.2) percentage points on the entry effects, and -0.02 (0.005) and 28.7 (2.1) percent on welfare; the ratio is not separately precise. The threshold-completing grant of Table 5, run in the same economy, uses 58 percent of the flow and raises entrant

welfare by 20.0 percent, in line with the 19.8 percent of Section VI; lifting entrants strictly above the threshold raises the gain to 28.7 percent.